

On Optimal Information-Theoretic Security in Symmetric Encryption under Low-Entropy Keys

Abstract—We study the *achievable* level of information-theoretic security for symmetric encryption *under low-entropy keys* (e.g., passwords and biometrics), where classical notions such as perfect secrecy and entropic security are usually unattainable. We consider a model in which messages M and keys K are drawn independently from distributions (p_m, p_k) . Prior work on homophonic ciphers (HC) and honey encryption (HE) suggests that randomized encryption tailored to p_m can improve security. We ask what the *optimal achievable* level is among all symmetric encryption schemes, and which necessary and/or sufficient conditions on encryption schemes characterize when this level can be achieved.

For *key confidentiality* (KC), we show that the *optimal achievable* level is $I(K; C) = \text{negl}(|C|)$, i.e., the ciphertext reveals only negligible information about the key. Moreover, this holds if and only if, informally, decrypting under any key induces sampling of messages according to p_m . HC and HE following this principle achieve this level, whereas p_m -agnostic schemes do not in general. For *message confidentiality* (MC) against message-recovery attacks, we show that the *optimal bound* on the adversary’s success probability is $p_{\max} + \text{negl}(|C|)$, where $p_{\max}$ is the baseline success probability of guessing the most likely message or key under (p_m, p_k) . We construct a scheme OE tailored to (p_m, p_k) that attains $p_{\max} + O(2^{-l})$, and prove that p_k -agnostic schemes (including HC and HE) cannot in general achieve this bound. Our results on necessary and/or sufficient conditions characterize fundamental principles governing the use of randomness in probabilistic encryption.

The main technical challenge is to handle a discretization-induced negligible error that must be propagated throughout the derivation of our bounds and necessary and/or sufficient conditions. To facilitate the analysis, we introduce a continuous-ciphertext framework that separates structural constraints from discretization error.

1. Introduction

Low-entropy keys arise pervasively in practice, from user-chosen passwords [1] to biometrically derived secrets [2, 3], and they enable powerful offline guessing once ciphertexts leak. Empirical studies estimate password min-entropy at only 7 bits [4, 5], and recent attacks show that even large biometric dictionaries can succeed with only tens of thousands guesses [6]. In long-term confidentiality settings, this problem only worsens: advances in computation

can turn “today’s high-entropy keys” into “tomorrow’s low-entropy keys”, enabling harvest-now-decrypt-later attacks.

Conventional symmetric encryption does not address this threat in an information-theoretic sense. When decrypting under an incorrect key typically yields an explicit failure symbol or obvious garbage, an offline attacker can enumerate candidate keys and identify the unique key that produces a plausible message. Exhaustive-guessing resistance therefore requires *ambiguity*: a ciphertext should admit many plausible decryptions, making correct and wrong-key outputs statistically hard to distinguish.

We study this question in a distributional model where messages M and keys K are drawn independently from given distributions (p_m, p_k) over finite spaces $\mathcal{M}$ and $\mathcal{K}$. Consider the extreme case $|\mathcal{K}| = 2$: any scheme that hopes to provide information-theoretic security must ensure that decrypting under the wrong key yields a *plausible* message. This makes it impossible for a universal, distribution-agnostic construction to provide satisfactory guarantees for *all* message distributions.

Prior work on homophonic ciphers (HC) [7, 8, 9] and honey encryption (HE) [10, 11] provides an important clue: by randomizing encryption in a way tailored to the message distribution, *wrong-key decryption can be made to resemble sampling from p_m* . At a high level, these schemes can be viewed under a unified construction. *Encryption* first encodes a non-uniform message into a nearly uniform seed using a distribution-dependent randomized encoder, and then applies a conventional cipher to encrypt the seed. *Decryption* first recovers a seed under the given key, and then decodes it by using the seed as the randomness for sampling from p_m . The main difference between HC and HE is that HC typically relies on variable-length randomized encodings to achieve perfect uniformity of the seed, whereas HE works with fixed-length ciphertexts for near-uniformity.

Existing analyses of HC focus primarily on key confidentiality (KC) via Shannon-style key equivocation $H(K | C)$ [7, 8, 9], while HE studies focus on message confidentiality (MC) notions such as message-recovery security (MRS) [10] and target-distribution semantic security / non-malleability (TDSS/TDNM) [11]. However, HC analyses are carried out in an infinite-sequence setting, which can obscure leakage through the (variable) ciphertext length when one encrypts a single message, and they do not quantify the impact of using finite-length (hence imperfect) uniformization on KC. On the HE side, existing bounds for MRS/TDSS/TDNM are derived for particular constructions under *specific* message and key distributions and are not

tight for arbitrary (p_m, p_k) . Moreover, HE notions largely emphasize MC, while KC—crucial when keys are sensitive or reused across systems, as with passwords and biometrics—has received comparatively less attention. A detailed analysis is deferred to Section 2.

In this study, we investigate both KC and MC in a general framework and ask, for arbitrary message and key distributions (p_m, p_k) , what is the *optimal achievable* information-theoretic security over all (possibly probabilistic) symmetric encryption schemes—beyond the HC/HE paradigm—and which necessary and/or sufficient conditions characterize when this level can be achieved?

Symmetric encryption. We use the following simple formalism for (probabilistic) symmetric encryption. A scheme is a tuple $(\mathcal{M}, \mathcal{K}, \mathcal{C}, \mathcal{E}, \mathcal{D})$, where $\mathcal{C}$ is the finite ciphertext space; $\mathcal{E} = \{E_k \mid k \in \mathcal{K}\}$, where each E_k is a (possibly randomized) encryption algorithm that takes $m \in \mathcal{M}$ and outputs $c \in \mathcal{C}$; and $\mathcal{D} = \{D_k \mid k \in \mathcal{K}\}$, where each D_k is a deterministic decryption algorithm that takes $c \in \mathcal{C}$ and outputs either a message in $\mathcal{M}$ or a failure symbol $\perp \notin \mathcal{M}$ (we write $\mathcal{M}' = \mathcal{M} \cup \{\perp\}$). Correctness requires that for every $k \in \mathcal{K}$ and $m \in \mathcal{M}$,

$$\Pr[D_k(C) = m : C \leftarrow E_k(m)] = 1,$$

where the probability is over the internal randomness of E_k .

To make the use of randomness explicit, for each message–key pair (m, k) , define

$$\mathcal{C}(m, k) = \{c \in \mathcal{C} \mid D_k(c) = m\}, \quad n(m, k) = |\mathcal{C}(m, k)|.$$

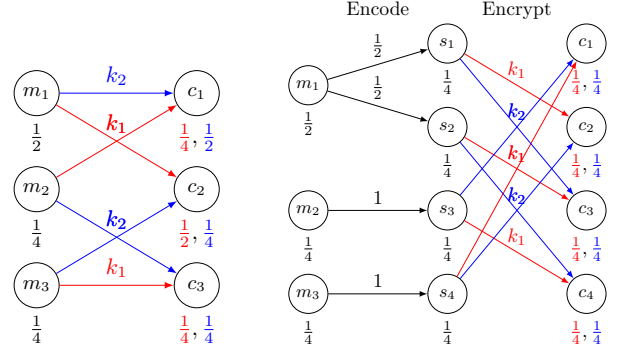
A probabilistic encryption algorithm E_k induces a conditional distribution $p_{c|m,k}(\cdot \mid m, k)$ over $\mathcal{C}(m, k)$. In general, $p_{c|m,k}(\cdot \mid m, k)$ may be pathological or degenerate; for instance, its probability mass may concentrate on a single ciphertext. We focus mainly on *uniform* probabilistic schemes, which are uniform over each ciphertext set:

$$p_{c|m,k}(c \mid m, k) = \frac{1}{n(m, k)} \quad \text{for all } c \in \mathcal{C}(m, k).$$

This restriction does not weaken the optimal security under the notions considered in this paper, as we later show that the optimum over all symmetric encryption schemes can be attained by uniform schemes.

With this formalization, we can articulate the common design principle behind HC and HE. Both instantiate *uniform* probabilistic encryption and use randomness to control the ciphertext-set sizes $n(m, k)$ as a function of the message distribution p_m , giving more ciphertexts to more likely messages (roughly, $n(m, k) \propto p_m(m)$). As illustrated in Fig. 1, this stands in sharp contrast to conventional encryption, where $n(m, k)$ is essentially fixed (e.g., $n(m, k) = 1$ for deterministic schemes, or the nonce/IV space size when appending a nonce/IV). By tailoring $n(m, k)$ to p_m , HC/HE make the key-conditioned ciphertext distributions $p_{c|k}(\cdot \mid k)$ nearly uniform over $\mathcal{C}$ for every key k , causing wrong-key decryption to approximate sampling from p_m .

In our definition, we require the ciphertext space $\mathcal{C}$ to be finite, typically the fixed-length space $\{0, 1\}^l$. Classical



(a) Deterministic cipher (b) Homophonic cipher (HC) / honey encryption (HE) with $n(m, k) \propto p_m(m)$.

Figure 1: Effect of randomization in encryption algorithm: for the message space $\{m_1, m_2, m_3\}$ with the distribution $(\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$, HC/HE make the ciphertext distributions $p_{c|k}(\cdot \mid k_1)$ and $p_{c|k}(\cdot \mid k_2)$ *uniform*, thereby causing wrong-key decryption to induce sampling from p_m using the uniform ciphertexts as the randomness.

HC is often formulated with an infinite ciphertext space $\mathcal{C} = \{0, 1\}^+ = \bigcup_{n \geq 1} \{0, 1\}^n$, and thus does not literally fit our definition. Nevertheless, it can be viewed as the limit of a sequence of fixed-length schemes as $l \rightarrow \infty$. Likewise, fixed-length HE may not achieve perfect uniformization of seeds/ciphertexts, but the approximation can be made arbitrarily accurate by increasing l . Accordingly, we take the ciphertext length $l = |\mathcal{C}|$ as the security parameter: we consider a sequence of encryption schemes indexed by l , and write negligible terms as functions of l .

Optimal attainable security and tight conditions. With this formalism in place, we analyze the optimal information-theoretic security achievable by *any* (possibly probabilistic) symmetric encryption scheme under given message and key distributions (p_m, p_k) . We consider both key confidentiality (KC) and message confidentiality (MC), and derive *tight* (necessary-and-sufficient) structural conditions under which these optima are achievable.

Key confidentiality (KC). We quantify key leakage via mutual information $I(K; C) = H(K) - H(K \mid C)$, which is the standard information-theoretic counterpart of Shannon-style key equivocation [12, 9]. We show that the optimal attainable level is $I(K; C) = \text{negl}(l)$, i.e., the ciphertext reveals only negligible information about the key; in particular, this implies strong guessing-based notions of KC. Moreover, we prove a tight characterization: $I(K; C) = \text{negl}(l)$ holds if and only if ciphertext distributions are statistically indistinguishable across keys. Equivalently,

$$C \stackrel{s}{=} E_k(M) \quad \text{for every } k \in \mathcal{K},$$

where $M \leftarrow_{p_m} \mathcal{M}$, $C \leftarrow_{p_c} \mathcal{C}$ (equivalently, $M, M' \leftarrow_{p_m} \mathcal{M}$, $K' \leftarrow_{p_k} \mathcal{K}$, $C \leftarrow_{E_{K'}}(M')$), and $X \stackrel{s}{=} Y$ denotes that the statistical distance between random variables X and Y is negligible in the security parameter l . For *uniform* schemes,

this ciphertext indistinguishability is further equivalent to the message indistinguishability, i.e.,

$$M \stackrel{s}{=} D_k(C) \quad \text{for every } k \in \mathcal{K},$$

which is the HC/HE design goal that decrypting under any key yields messages distributed (approximately) as p_m . This demonstrates that the structural principle behind HC/HE gives *tight* conditions for achieving optimal KC.

These two equivalences are proved using two distinct techniques. The first uses standard two-sided bounds between statistical distance and Kullback–Leibler divergence, via Pinsker’s inequality and a suitable converse bound. The second leverages uniformity to lift a statistical-distance bound from the message space to the ciphertext space, upgrading a bound of the form $\sum_i |\sum_j x_{i,j}|$ to an upper bound on $\sum_{i,j} |x_{i,j}|$, going beyond standard information-theoretic arguments.

Message confidentiality (MC). We focus on message-recovery security (MRS), measuring the adversary’s success probability in recovering M from C . We show that the optimal bound over all schemes is $p_{\max} + \text{negl}(l)$, where $p_m^{\max} = \max_m p_m(m)$, $p_k^{\max} = \max_k p_k(k)$, and $p_{\max} = \max\{p_m^{\max}, p_k^{\max}\}$. It captures the success probability of a trivial strategy: given a ciphertext c , outputs a guess $m_{\text{triv}}(c)$, defined as the most likely message under p_m if $p_m^{\max} \geq p_k^{\max}$, otherwise as the decryption under the most likely key under p_k . We show that the $p_{\max} + \text{negl}(l)$ bound is achieved if and only if $m_{\text{triv}}(c)$ (nearly) maximizes the posterior guessing probability for every ciphertext c . Based on this observation, we construct a new uniform scheme OE that achieves $p_{\max} + O(2^{-l})$ for general (p_m, p_k) by carefully arranging the ciphertext sets $\mathcal{C}(m, k)$. At a high level, OE (i) allocates $n(m, k) \propto p_m(m)p_k(k) + O(2^{-l})$, and (ii) avoids *decryption collisions* by ensuring $\mathcal{C}(m, k) \cap \mathcal{C}(m, k') = \emptyset$ for $k \neq k'$. As a result, for every ciphertext c and message m , there is at most one key k such that $D_k(c) = m$. If such k exists, the guessing probability

$$p_{m|c}(m|c) = p_{m,k|c}(m, k|c) \propto \frac{p_m(m)p_k(k)}{n(m, k)} \propto 1 + O(2^{-l}).$$

This implies that any attempt to guess a message other than $m_{\text{triv}}(c)$ gains only a negligible additional advantage.

For the stronger notion TDSS (capturing partial-information leakage), we show that achieving $\text{Adv}_{E, p_m, p_k}^{\text{tdss}} = \text{negl}(l)$ is typically impossible; determining the optimal attainable TDSS remains open.

A continuous-ciphertext framework. To cleanly propagate discretization-induced negligible errors throughout our bounds and tight conditions, we introduce a continuous-ciphertext abstraction over $[0, 1)$ that separates structural constraints (ciphertext allocation and collision avoidance) from discretization error. We establish two fundamental transformations: (i) any continuous scheme can be discretized to $\{0, 1\}^l$ with KC/MC degradation at most $O(2^{-l})$, and (ii) any sequence of discrete schemes admits subsequences that converge to a continuous limit with convergent security metrics. This enables us to reason in the continuous

domain without precision artifacts, while retaining tight discrete guarantees up to negligible error.

Within this framework, we derive several necessary conditions for optimal security, implying three impossibility results: (1) optimal KC is impossible for p_m -agnostic schemes; (2) optimal MRS is impossible for p_k -agnostic schemes (including HC/HE schemes); and (3) achieving optimal KC and optimal MRS simultaneously is impossible. For (3), we give a variant of HC/HE that achieves optimal KC and nearly optimal MRS, with success probability bounded by $2p_{\max} + O(2^{-l})$.

Practicality of distribution-tailored encryption. Tailoring to an exact pair (p_m, p_k) may appear impractical when these distributions are complex or only partially known. Nevertheless, prior HE applications—including honey password vaults [13, 14, 15, 16] and long-term genetic protection [17]—show that tailored encryption can be deployed with approximate statistical models of the underlying data. More recently, work on honey password vaults [16] shows that modern generative models (e.g., Transformer-based models [18]) can learn realistic low-entropy password distributions, and that even imperfect models can yield practical and meaningful security gains. More broadly, as deep generative models continue to improve, we expect increasingly accurate learning of complex distributions across domains (text, images, and structured data), and deployment settings where the data owner specifies message/key types while the encryption mechanism selects an appropriate learned model.

Our general constructions (including OE) and HC/HE-style schemes are specified for arbitrary (p_m, p_k) , and a naive implementation can be computationally expensive when p_m or p_k is complex. We therefore provide a complexity-reduction approach that bins messages or keys with similar probabilities, trading a small security loss for substantial speedups. Finally, in many practical settings p_m or p_k is represented by a structured probabilistic model such as Markov models, and one can often exploit this structure to further improve efficiency.

Supplementary proof artifact. Due to space limitations, most proofs and detailed calculations are deferred to an anonymous supplementary proof artifact, available at <https://anonymous.4open.science/r/sp-proof-artifact-2027-35F7/>.

When referring to this artifact in the main text, we use the form “Supplementary Section XXX”.

2. Related works

Information-theoretic notions and entropy requirements. Shannon introduced key and message equivocation, $H(K|C)$ and $H(M|C)$, as fundamental measures for information-theoretic security of encryption schemes [12]. In general, $H(K|C) \geq H(M|C)$, and equality holds for schemes without decryption collisions, i.e., for every pair (m, c) there is at most one key k such that $D_k(c) = m$. He defined *perfect secrecy* by $H(M|C) = H(M)$, equivalently $I(M; C) = 0$, capturing the ideal guarantee that C leaks no information about M . However, perfect secrecy requires key entropy comparable to message entropy, namely

$H(K) \geq H(M)$. More generally, even relaxing perfect secrecy to $I(M; C) \leq \epsilon$ still requires $H(K) \geq H(M) - \epsilon$ [19]. Entropic security [20, 21] further relaxes perfect secrecy by bounding the adversary’s advantage in predicting any function of the message from the ciphertext by the corresponding trivial (prior-only) baseline plus ϵ . For an n -bit message distribution with min-entropy t , achieving entropic security for *arbitrary* such distributions still requires essentially $H(K) \geq n - t$ [21]. These entropy requirements make perfect secrecy and entropic security typically unattainable under low-entropy keys.

Equivocation bounds for deterministic ciphers. A large body of work derives bounds on key/message equivocation for deterministic encryption with uniform keys, including Shannon’s pure and random ciphers and classical substitution ciphers [12, 22, 23, 24, 25]. These works typically consider a single key K used to encrypt a sequence of messages M^n and quantify how $H(K | C^n)$ or $H(M^n | C^n)$ decays as $n \rightarrow \infty$, yielding precise impossibility results for maintaining large equivocation in these deterministic schemes. More broadly, such schemes are distribution-agnostic (i.e., not tailored to a particular message distribution), and as argued in the Introduction, such designs generally cannot provide satisfactory information-theoretic security under low-entropy keys.

Distribution-tailored probabilistic encryption. In contrast, homophonic ciphers (HC) [8, 7, 9] and honey encryption (HE) [10, 11] exploit *probabilistic encryption* and tailor the randomization to the message distribution so as to create many plausible decryptions. At a high level, both share an *encode-then-encrypt* structure: a distribution-dependent randomized encoder maps m to a (nearly) uniform seed, a conventional cipher encrypts the seed, and a corresponding decoder maps the decrypted seed back to a message so that decrypting under wrong keys resembles sampling from the target message distribution.

HC. Classical HC analyses often allow variable-length and infinite ciphertext spaces to achieve perfect uniformization, and then argue strong key-equivocation guarantees such as $H(K | C^n) = H(K)$ for long/infinite message sequences [7, 9]. However, in a single-message setting (as considered in this paper), ciphertext length is directly observable; these works do not formally analyze $H(K | C)$ in this regime and do not quantify the key leakage induced by variable ciphertext length. In particular, they cannot in general guarantee perfect key secrecy, i.e., $I(K; C) = 0$ (equivalently $H(K | C) = H(K)$). When the idealized infinite-length construction is approximated by finite-length ciphertexts, they also do not quantify the resulting deviation from perfect uniformization (see Appendix A).

HE. HE was proposed with fixed-length ciphertexts explicitly for low-entropy keys [10]. Its cryptographic analysis focuses on three message-confidentiality notions: (1) message-recovery security (MRS) [10], measuring the adversary’s success probability of recovering the *entire* message; (2) target-distribution semantic security (TDSS) [11], measuring leakage of *partial information* about the message (via predicates); and (3) target-distribution non-malleability

(TDNM) [11], measuring resistance to *meaningful ciphertext manipulation*. Corresponding bounds for this fixed-length construction are proved in the random-oracle / ideal-cipher models [10, 11], under specific distributional assumptions, and are not tight for general message and key distributions (see Appendix B).

Limitations of prior HC/HE work. Overall, prior HC and HE studies leave open: (1) an information-theoretic characterization of the *optimal attainable* guarantees (and tight conditions) across *all* probabilistic symmetric encryption schemes under arbitrary (p_m, p_k) ; and (2) a formal analysis of key confidentiality, which is important when keys are sensitive or reused across systems (as with passwords and biometrics), but has received comparatively less attention.

3. Notation and Definitions

Notation. Let $\mathcal{M}$ denote the message space with distribution p_m and $\mathcal{K}$ the key space with distribution p_k . For a distribution p_x over a set $\mathcal{X}$, we write $X \leftarrow_{p_x} \mathcal{X}$ to sample X from $\mathcal{X}$ according to p_x , and write $X \leftarrow_{\S} \mathcal{X}$ when p_x is uniform. For $\mathcal{X}' \subseteq \mathcal{X}$, we denote $p(\mathcal{X}') = \sum_{x \in \mathcal{X}'} p(x)$. Throughout, script letters (e.g., $\mathcal{M}$) denote spaces, italic capitals (e.g., $M \leftarrow_{p_m} \mathcal{M}$) denote random variables, and italic lowercase letters (e.g., m) denote concrete values. We write $p_x^{\min} = \min_x p_x(x)$ and $p_x^{\max} = \max_x p_x(x)$. We write $p_{x,y}(x, y)$ for $\Pr[X = x, Y = y]$ and $p_{x|y}(x | y)$ for $\Pr[X = x | Y = y]$, and analogously for multiple variables. For convenience, we assume that p_m and p_k are strictly positive over $\mathcal{M}$ and $\mathcal{K}$.

Entropy and statistical distances. For a random variable X with distribution p , its Shannon entropy is $H(X) = -\sum_x p(x) \log p(x)$. For jointly distributed X, Y , the mutual information is

$$I(X; Y) = \sum_{x,y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)} = H(X) - H(X | Y).$$

We use base-2 log and natural ln. For random variables X_0, X_1 over the same space with distributions p_0, p_1 , the statistical distance is

$$\text{SD}(X_0, X_1) = \frac{1}{2} \sum_x |p_0(x) - p_1(x)|,$$

and the Kullback–Leibler divergence is

$$\text{KL}(X_0 \| X_1) = \sum_x p_0(x) \log \frac{p_0(x)}{p_1(x)}.$$

Negligibility and statistical indistinguishability. A function $\varepsilon(l)$ is *negligible* in l , denoted $\varepsilon(l) = \text{negl}(l)$, if for every positive polynomial $\text{poly}(\cdot)$ there exists l_0 such that for all $l > l_0$, $|\varepsilon(l)| < 1/\text{poly}(l)$. Two sequences of random variables $\{X_0^{(l)}\}_l$ and $\{X_1^{(l)}\}_l$ are *statistically indistinguishable* if $\text{SD}(X_0^{(l)}, X_1^{(l)}) = \text{negl}(l)$, denoted $X_0^{(l)} \stackrel{s}{\equiv} X_1^{(l)}$; X and Y are *perfectly indistinguishable*, denoted $X \equiv Y$, if the two random variables have identical distributions.

Encryption schemes indexed by l . We treat the ciphertext length l as the security parameter and view an encryption

scheme as a family $(\mathcal{M}, \mathcal{K}, \mathcal{C}^{(l)}, \mathcal{E}^{(l)}, \mathcal{D}^{(l)})$ with $\mathcal{C}^{(l)} = \{0, 1\}^l = \mathbb{Z}_{2^l}$, $\mathcal{E}^{(l)} = \{E_k^{(l)}\}_{k \in \mathcal{K}}$ and $\mathcal{D}^{(l)} = \{D_k^{(l)}\}_{k \in \mathcal{K}}$. When the context is clear, we omit the superscript (l) .

Ciphertext sets and uniformity. We use the ciphertext sets $\mathcal{C}(m, k)$ and their sizes $n(m, k)$ as defined in the Introduction. A probabilistic encryption algorithm E_k induces a conditional distribution $p_{c|m,k}(\cdot|m, k)$ on $\mathcal{C}(m, k)$. We call a scheme *uniform* if it is uniform on each ciphertext set, i.e., $p_{c|m,k}(c|m, k) = 1/n(m, k)$ for all $c \in \mathcal{C}(m, k)$.

Structural properties and classes of schemes. We further distinguish (probabilistic) schemes by how the ciphertext-set sizes $n(m, k)$ vary:

1. **Message independence (MID).** For each key k , the value $n(m, k)$ is constant over $m \in \mathcal{M}$.
2. **Key independence (KID).** For each message m , the value $n(m, k)$ is constant over $k \in \mathcal{K}$.

We write **MD** (message-dependent) and **KD** (key-dependent) for the negations of MID and KID, respectively.

Deterministic schemes have $n(m, k) = 1$ for all (m, k) , and thus satisfy both MID and KID. Standard probabilistic schemes that append a fixed-length nonce/IV also satisfy both MID and KID, since $n(m, k)$ equals the nonce/IV space size. In contrast, HC and HE schemes are MD but not KD, as they tailor $n(m, k)$ to p_m but not p_k . To the best of our knowledge, no prior work has studied KD schemes.

We use the following shorthand for spaces of schemes: Ω_E for all symmetric encryption schemes; Ω_{DE} and Ω_{PE} for deterministic and probabilistic schemes, respectively; and Ω_{EE} and Ω_{HE} for encode-then-encrypt constructions and HE schemes (defined below). For probabilistic encryption, we also consider spaces Ω_{PE}^P defined by attributes $P \in \{\mathbf{U}, \mathbf{MID}, \mathbf{MD}, \mathbf{KID}, \mathbf{KD}\}$ (or combinations thereof), where **U** denotes uniformity of encryption algorithms. Supplementary Section S.10 summarizes the optimal attainable security in different spaces of encryption schemes and compares these optima with the global optimum over the whole space Ω_E .

Encode-then-encrypt (EE) constructions. We formalize the *EE* construction as a combination of an encoder (encode, decode) and a standard cipher (encrypt, decrypt) with $E_k = \text{encrypt}_k \circ \text{encode}$ and $D_k = \text{decode} \circ \text{decrypt}_k$. Here encode is possibly randomized and maps a message $m \in \mathcal{M}$ to a seed $s \in \mathcal{S}$; correctness requires $\text{decode}(\text{encode}(m)) = m$ with probability 1. Moreover, we focus on *uniform* encoders: for each m , the random variable $\text{encode}(m)$ is supported on $\text{decode}^{-1}(m) = \{s \mid \text{decode}(s) = m\}$ and uniform over this set, i.e., $\Pr[\text{encode}(m) = s] = 1/|\text{decode}^{-1}(m)|$ for all $s \in \text{decode}^{-1}(m)$. The cipher *encrypt* inputs the seed s and outputs a ciphertext c ; *encrypt* may be deterministic, or randomized using message- and key-independent coins (e.g., a fixed-length nonce/IV), which induces a constant ciphertext-set size per seed.

If *encrypt* is randomized, we can absorb its randomness into the seed space by enlarging $\mathcal{S}$ (e.g., appending the nonce/IV to the seed) and adjusting encode and decode accordingly; this yields an equivalent EE representation with

Real	Honey
$M_1 \leftarrow_{p_m} \mathcal{M}$	$S_0 \leftarrow_{\$} \mathcal{S}$
$S_1 \leftarrow \text{encode}(M_1)$	$M_0 \leftarrow \text{decode}(S_0)$

Figure 2: Indistinguishability of real and honey message-seed pairs (**MS-IND**) for encoders: $(M_1, S_1) \stackrel{s}{=} (M_0, S_0)$

a deterministic cipher *encrypt'* and a modified encoder *encode'*. Accordingly, in this paper we treat the cipher in EE constructions as deterministic without loss of generality.

In fact, an encryption scheme can be presented as an EE construction if and only if it is message-dependent, uniform, and key-independent, i.e., $\Omega_{EE} = \Omega_E^{\mathbf{U}, \mathbf{MD}, \mathbf{KID}}$ (See Appendix C).

Homophonic ciphers and honey encryption. Homophonic ciphers (HC) and honey encryption (HE) can be viewed as EE constructions whose encoder aims to (approximately) *uniformize* a non-uniform message distribution by mapping messages to (nearly) uniform seeds. As discussed before, HC with infinite ciphertext space can be viewed as a limit of l -length HE schemes as $l \rightarrow \infty$. In the rest of the paper, we do not distinguish HC and HE.

To formalize this encoder requirement, we recall message-seed indistinguishability (**MS-IND**) [10]: the *real* pair (M_1, S_1) and the *honey* pair (M_0, S_0) in Fig. 2 are statistically indistinguishable. When their statistical distance is zero, we call this *perfect* MS-IND.

Following [10], we denote by $p_d(m)$ the induced honey distribution $\Pr[M_0 = m : S_0 \leftarrow_{\$} \mathcal{S}, M_0 \leftarrow \text{decode}(S_0)]$ for $m \in \mathcal{M}'$. Since $\mathcal{S}$ is finite, p_d has granularity $1/|\mathcal{S}|$ (e.g., $k/2^l$ when $|\mathcal{S}| = 2^l$), so perfect MS-IND need not hold for arbitrary p_m at a fixed l . Nevertheless, the HE encoders achieve $p_d(m) = p_m(m) + O(2^{-l})$ for all m , so that $(M_1, S_1) \stackrel{s}{=} (M_0, S_0)$. The correctness of the encoders requires $l \geq \log 1/p_m^{\min} + 1$, ensuring that the least likely message has corresponding seeds.

4. Key Confidentiality

Key confidentiality (KC) is crucial for user-based keys such as passwords and biometrics: since such keys are often reused across systems, any information about a reused key leaked in one setting may compromise security in another.

We quantify key leakage by the mutual information $I(K; C)$ between the key K and ciphertext C . Our main result is that the optimal attainable level of KC is $I(K; C) = \text{negl}(|C|)$. Moreover, we derive *tight* characterizations of when this optimum is achieved: for general schemes, $I(K; C) = \text{negl}(|C|)$ holds if and only if ciphertext distributions are statistically indistinguishable across keys; for *uniform* schemes, it is further equivalent to a message-indistinguishability condition that matches the HE design goal (informally, decrypting under any key yields messages distributed almost as real ones). Finally, we show that HE schemes achieve $I(K; C) = O(2^{-l})$.

4.1. Security Definition

We begin with an idealized notion of key confidentiality.

Definition 1. A symmetric encryption scheme $(\mathcal{M}, \mathcal{K}, \mathcal{C}, \mathcal{E}, \mathcal{D})$ achieves perfect KC if, for $M \leftarrow_{p_m} \mathcal{M}$, $K \leftarrow_{p_k} \mathcal{K}$, and $C \leftarrow E_K(M)$, it holds that $I(K; C) = 0$.

Perfect KC is an information-theoretic ideal in the spirit of Shannon’s framework. With finite ciphertext length l , perfect KC is generally unattainable for arbitrary (p_m, p_k) , because discretization and distribution-approximation effects introduce unavoidable error. We collect the corresponding conditions and impossibility statements in Supplementary Section S.1. Accordingly, we focus on the asymptotic relaxation of *statistical KC* defined next.

For optimal achievable KC, we define statistical KC for an encryption scheme consisting of a family $(\mathcal{M}, \mathcal{K}, \mathcal{C}^{(l)}, \mathcal{E}^{(l)}, \mathcal{D}^{(l)})$ with finite l and $\mathcal{C}^{(l)} = \{0, 1\}^l = \mathbb{Z}_{2^l}$. Unless otherwise stated, KC refers to statistical KC in this paper.

Definition 2. An encryption scheme achieves statistical KC if $I(K; C^{(l)}) = \text{negl}(l)$, where $M \leftarrow_{p_m} \mathcal{M}$, $K \leftarrow_{p_k} \mathcal{K}$, and $C^{(l)} \leftarrow E_K^{(l)}(M)$.

We note that requiring $I(K; C) = \text{negl}(|C|)$ is strong and in particular implies guessing-based notions of KC (see Supplementary Section S.5).

4.2. Necessary and/or Sufficient Conditions for Statistical KC

We begin with a high-level summary of our characterizations of statistical KC, and relate them to three indistinguishability notions: ciphertext indistinguishability across keys (C-IND in Fig. 3), message indistinguishability under wrong-key decryption (M-IND in Fig. 4), and message–seed indistinguishability for EE encoders (MS-IND in Fig. 2). The formal theorems appear below.

For general schemes, statistical KC holds if and only if C-IND holds. Moreover, C-IND implies M-IND: for every key k , decrypting a real ciphertext under k yields a message distribution statistically close to that of the real message (and hence close to p_m), matching the decoy-message intuition behind HE-style designs. The converse (M-IND $\Rightarrow$ C-IND) fails in general, but it does hold for *uniform* schemes. In that case, M-IND is equivalent to statistical KC and further implies a structural near-uniformity property: for each key, the key-conditioned ciphertext distribution is nearly uniform within each equivalence class of the decryption-induced partition of the ciphertext space.

For EE constructions, MS-IND of the encoder is sufficient for statistical KC but not necessary. In addition, MS-IND implies that the marginal ciphertext distribution p_c is nearly uniform over the entire ciphertext space.

In summary:

1. General schemes: Statistical KC $\Leftrightarrow$ C-IND (Fig. 3) $\nRightarrow$ M-IND (Fig. 4).
2. Uniform schemes: Statistical KC $\Leftrightarrow$ M-IND $\Rightarrow$ local marginal uniformity of ciphertexts.

3. EE constructions: Statistical KC $\nRightarrow$ MS-IND (of encoders) $\Rightarrow$ global marginal uniformity of ciphertexts.

Theorem 1. A symmetric encryption scheme achieves statistical KC if and only if it satisfies C-IND in Fig. 3, i.e., for every key k we have $C_1^{(l)} \stackrel{s}{=} C_{0,k}^{(l)}$. More specifically,

$$\begin{aligned} \text{SD}(C_{0,k}^{(l)}, C_1^{(l)}) &\leq \sqrt{\frac{\ln 2}{2p_k^{\min}} I(K; C^{(l)})} = O\left(\sqrt{I(K; C^{(l)})}\right), \quad \forall k \in \mathcal{K}; \\ I(K; C^{(l)}) &\leq \frac{\log(1/p_k^{\min})}{1 - p_k^{\min}} \max_k \text{SD}(C_{0,k}^{(l)}, C_1^{(l)}) = O\left(\max_k \text{SD}(C_{0,k}^{(l)}, C_1^{(l)})\right). \end{aligned}$$

The proof given in Appendix D combines Pinsker’s inequality [26] with a suitable reverse Pinsker’s inequality (Theorem 7 in [27]), relating statistical distance and KL divergence. Intuitively, this shows that for each k , $p_{c|k}(\cdot | k)$ of $E_k^{(l)}(M)$ is close to p_c . Thus, for every $k \in \mathcal{K}$, $D_k^{(l)}$ maps p_c back to nearly the same message distribution p_m , yielding Proposition 1. This reflects the central design principle of HE schemes: generating indistinguishable honey messages for each key.

Definition 3. An encryption scheme is M-IND if for every key k , the indistinguishability in Fig. 4 holds, i.e., $M_1 \stackrel{s}{=} M_{0,k}^{(l)}$.

Proposition 1. If a symmetric encryption scheme achieves statistical KC, then it is M-IND. Specifically, for each $k \in \mathcal{K}$,

$$\text{SD}(M_{0,k}^{(l)}, M_1) \leq \text{SD}(C_1^{(l)}, C_{0,k}^{(l)}) \leq \sqrt{\frac{\ln 2}{2p_k^{\min}} I(K; C^{(l)})} = O\left(\sqrt{I(K; C^{(l)})}\right).$$

The converse of Proposition 1 does not hold in general (see Supplementary Section S.4). Fortunately, if the encryption is uniform, the converse does hold.

Theorem 2. A uniform encryption scheme achieves statistical KC, if and only if it is M-IND. Specifically,

$$\max_k \text{SD}(C_{0,k}, C_1) \leq \left(2 + \frac{1}{p_k^{\min}}\right)^{|\mathcal{M}| \cdot |\mathcal{K}|} \max_k \text{SD}(M_{0,k}, M_1).$$

The proof given in Appendix D is subtle because M-IND only bounds the discrepancy of *aggregate* probability mass on each ciphertext set $\mathcal{C}^{(l)}(m, k)$, namely

$$\sum_m \left| \sum_{c \in \mathcal{C}^{(l)}(m, k)} (p_c(c) - p_{c|k}(c | k)) \right| = \text{negl}(l),$$

whereas statistical KC requires controlling the *sum of per-ciphertext deviations*,

$$\sum_m \sum_{c \in \mathcal{C}^{(l)}(m, k)} |p_c(c) - p_{c|k}(c | k)| = \text{negl}(l).$$

Equivalently, we need to upgrade a bound of the form $|\sum_i x_i|$ to one of the form $\sum_i |x_i|$; this is precisely where the uniformity property is exploited.

Our argument starts by choosing a pair (c_1, k_1) maximizing $p_{c|k}(c | k)$, and letting $m_1 = D_{k_1}^{(l)}(c_1)$. Uniformity implies that $p_{c|k}(\cdot | k_1)$ is flat on the corresponding set $\mathcal{C}^{(l)}(m_1, k_1)$. The maximizing choice further ensures that for every $k' \neq k_1$ and every $c \in \mathcal{C}^{(l)}(m_1, k_1)$ we have

Ciphertext $C_1^{(l)}$ under random key K	Ciphertext $C_{0,k}^{(l)}$ under given key k
$M \leftarrow_{p_m} \mathcal{M}$	$M \leftarrow_{p_m} \mathcal{M}$
$K \leftarrow_{p_k} \mathcal{K}$	$C_{0,k}^{(l)} \leftarrow E_k^{(l)}(M)$
$C_1^{(l)} \leftarrow E_K^{(l)}(M)$	

Figure 3: Indistinguishability of ciphertexts (**C-IND**): $C_1^{(l)} \stackrel{s}{\equiv} C_{0,k}^{(l)}$ for each k

$p_{c|k}(c|k_1) \geq p_{c|k}(c|k')$ and $p_{c|k}(c|k_1) \geq p_c(c)$. Thus, on this set the absolute values have a fixed sign, allowing us to move the absolute value inside the summation and bound the total variation on $\mathcal{C}^{(l)}(m_1, k_1)$ by the corresponding M-IND deviation. By the maximality of k_1 , the same bound extends to all other k' . Iterating the argument over the remaining parts of $\mathcal{C}^{(l)}$ at most $|\mathcal{M}| \cdot |\mathcal{K}|$ times covers the entire ciphertext space.

As $p_{c|k}(\cdot|k)$ is uniform within each subspace $\mathcal{C}^{(l)}(m, k)$, the distribution p_c is also nearly uniform on the same subspace. Defining the partition P_k by $P_k^{(l)} = \{\mathcal{C}^{(l)}(m, k)\}_m$ for each k , we obtain Proposition 2. Note that in such partitions, the number of subspaces is bounded by $|\mathcal{M}|$, which remains finite even as $l \rightarrow \infty$.

Proposition 2. *If a uniform encryption scheme is M-IND, then $\forall k, k' \in \mathcal{K}$, $C_{0,k}^{(l)} \stackrel{s}{\equiv} C_1^{(l)} \stackrel{s}{\equiv} U_{k'}^{(l)}$, where $C_{0,k}^{(l)} \leftarrow_{p_{c|k}(\cdot|k)} \mathcal{C}^{(l)}$, $C_1^{(l)} \leftarrow_{p_c} \mathcal{C}^{(l)}$, and $U_{k'}^{(l)}$ is uniform over each $S \in P_{k'}^{(l)}$.*

We attempt to construct a partition coarser than each P_k . A direct union of overlapping subspaces from $P_{k_1}^{(l)}$ and $P_{k_2}^{(l)}$ fails, as the probability of their intersection may be negligible. Moreover, $P_k^{(l)}$ can vary drastically with l , making coarse partitions difficult to define. Instead, by allocating more ciphertexts to higher-probability elements, the distribution within each subspace can be stretched toward uniformity over the entire ciphertext space—a property inherently achieved by HE schemes.

4.3. Security Analysis

The insufficiency of traditional encryption. For uniformly random messages, a deterministic cipher can satisfy Theorem 1. However, for non-uniform p_m , deterministic ciphers fail: E_k produces ciphertext distributions that vary with k . Fig. 1b shows an example with $(p_m(m_i))_i = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$, yielding $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ for k_1 and $(\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ for k_2 .

In general, let m^* be the most likely message and $c_k^* = E_k(m^*)$. If a deterministic cipher achieves KC, Theorem 1 requires $p_c(c_k^*) = p_{c|k}(c_k^*|k) = p_m(m^*)$ for all k , and further $p_m(D_{k_2}(c_{k_1}^*)) = p_m(m^*)$ for all k_1, k_2 . These equalities may extend over large subsets of $\mathcal{M}$, but they rarely hold except when p_m is uniform.

Some prior studies quantify key leakage in certain deterministic encryption schemes [22, 25, 24] under multiple-message models. In contrast, we derive a general bound for

Real message M_1	Honey message $M_{0,k}^{(l)}$ under given key k
$M_1 \leftarrow_{p_m} \mathcal{M}$	$C_0^{(l)} \leftarrow_{p_c} \mathcal{C}^{(l)}$
	$M_{0,k}^{(l)} \leftarrow D_k^{(l)}(C_0^{(l)})$

Figure 4: Indistinguishability of real and honey messages (**M-IND**) for encryption: $M_1 \stackrel{s}{\equiv} M_{0,k}^{(l)}$ for each key k

the ideal cipher in our single-message model, for arbitrary p_m and p_k ; see Theorem S.5 in Supplementary Section S.3. However, our bound is typically loose for specific p_m, p_k .

To illustrate the magnitude of key leakage in practice, we consider a password-vault example, a typical HE application [16], and estimate $H(K|C)$ by empirical sampling in the ideal-cipher model. Both the master password (key) and the stored password (message) are modeled as 4-digit numbers following a Zipf distribution with $p(\text{pin}_i) \propto i^{-r}$ and $r = 1.0$. This estimate gives $H(K|C) \approx 3.3$ bits. When two 4-digit passwords are stored, the message distribution becomes more skewed and the leakage rises to nearly 6 bits. Given that the key entropy $H(K)$ is only 9.5 bits, this leakage is substantial.

Security of HE schemes and EE constructions. Unlike traditional encryption, EE constructions incorporate message dependency through a randomized encoder. The encoder assigns $|S| \cdot p_d(m)$ seeds to each message m , which translates into $|C| \cdot p_d(m)$ ciphertexts under a deterministic cipher.

If the encoder achieves perfect MS-IND (i.e., $p_m = p_d$), then the ciphertext distribution is uniform under every key. By Theorem 1, such a construction achieves perfect KC. As shown in Fig. 1a with a three-message distribution $(\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$: the most probable message m_1 (probability $\frac{1}{2}$) is assigned two seeds, while the less probable messages m_2 and m_3 (probability $\frac{1}{4}$ each) are assigned one seed each. Thus, the ciphertext distribution remains uniform across all keys.

If the encoder is MS-IND (as in HE schemes), then the seeds are nearly uniform, which in turn ensures that the ciphertext distribution under each key is nearly uniform. Consequently, the construction achieves statistical KC.

Theorem 3. *An EE construction with an encoder and a cipher achieves statistical KC if the encoder is MS-IND. Specifically, with $l \geq \log 1/p_m^{\min} + 1$,*

$$I(K; C^{(l)}) \leq \frac{\text{SD}(M_0^{(l)}, M_1)(2p_m^{\min} - \text{SD}(M_0^{(l)}, M_1))}{\ln 2 \cdot p_m^{\min}(p_m^{\min} - \text{SD}(M_0^{(l)}, M_1))} = O(\text{SD}(M_0^{(l)}, M_1)).$$

where $M_1 \leftarrow_{p_m} \mathcal{M}$ and $M_0^{(l)} \leftarrow_{p_d} \mathcal{M}$.

The converse of Theorem 3 does not hold (see Supplementary Section S.4).

Proposition 3. *For an EE construction, if its encoder is MS-IND, then $C_{0,k}^{(l)} \stackrel{s}{\equiv} C_1^{(l)} \stackrel{s}{\equiv} U_{k'}^{(l)}$, where $C_{0,k}^{(l)} \leftarrow_{p_{c|k}(\cdot|k)} \mathcal{C}^{(l)}$,*

$C_1^{(l)} \leftarrow_{p_c} \mathcal{C}^{(l)}$, and $U_{k'}^{(l)} \leftarrow_{\mathcal{S}} \mathcal{C}^{(l)} = \{0,1\}^l$. Specifically, with $l \geq \log 1/p_m^{\min} + 1$, $M_0^{(l)} \leftarrow_{p_d} \mathcal{M}$ and $M_1 \leftarrow_{p_m} \mathcal{M}$,

$$\text{SD}(C_{0,k}^{(l)}, U^{(l)}) \leq \frac{\text{SD}(M_0^{(l)}, M_1)}{2(p_m^{\min} - \text{SD}(M_0^{(l)}, M_1))}.$$

Theorem 3 provides a bound for the EE construction by the statistical distance. However, with carefully designed encoders, such as IS-DTE [10] and IS-PMTE [28], it is possible to directly constrain the disparity $|p_d(m) - p_m(m)|$ for each $m \in \mathcal{M}$, thereby achieving a more precise bound.

Proposition 4. *For an EE construction, if for each m , $|p_d(m) - p_m(m)| \leq \epsilon$, where $0 < \epsilon < p_m^{\min}$, then*

$$I(K; C^{(l)}) \leq \frac{\epsilon(2p_m^{\min} - \epsilon)}{p_m^{\min}(p_m^{\min} - \epsilon) \ln 2} = O(\epsilon).$$

By this proposition, we can directly provide bounds of $O(2^{-l})$ for specific encoders such as IS-DTE [10] and IS-PMTE [28]; see Supplementary Section S.2.

Security implications of the cipher in HE schemes. In the above analysis, the only requirement on the cipher used in the HE scheme is determinism. Hence, the choice of deterministic cipher has no impact on KC, and the conclusions also hold in the random oracle and ideal cipher models.

Key confidentiality in multiple ciphertexts.

Theorem 4. *For n ciphertexts $C_1, C_2, \dots, C_n$ encrypted with the same key K (possibly under different encryption schemes), if the messages are independent and the encryption randomness is independent, then*

$$I(K; C_1, C_2, \dots, C_n) \leq \sum_{i=1}^n I(K; C_i).$$

If the messages are dependent, this bound may fail. For instance, when the messages are identical, the ciphertext interval of the message can be inferred, thereby leaking information about the key.

5. Message Confidentiality

We study message confidentiality primarily via message-recovery security (MRS). We show that the best achievable information-theoretic guarantee is to bound any adversary's message-recovery success probability by $p_{\max} + \text{negl}(l)$, where $p_{\max}$ is the success probability of a trivial strategy (guessing the most likely message or key).

We then construct a scheme OE that attains $p_{\max} + O(2^{-l})$ for general (p_m, p_k) . To improve practicality, we propose generic complexity-reduction techniques for real-world distributions. Finally, we show that HE schemes and more general EE constructions cannot in general achieve the $p_{\max} + \text{negl}(l)$ bound for arbitrary distributions, revealing that achieving optimal MRS security requires principles for using randomness in probabilistic encryption that are fundamentally different from those underlying HC/HE.

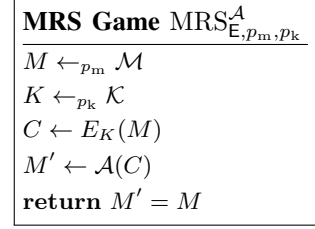


Figure 5: Game for message recovery security (MRS)

5.1. Security Definition

Message-recovery security (MRS) captures attacks that attempt to recover the entire message from the ciphertext. For a symmetric-encryption scheme E , MRS is defined by the game in Fig. 5. An adversary $\mathcal{A}$ wins if it outputs the correct message M given C . The MRS advantage of $\mathcal{A}$ is

$$\text{Adv}_{E, p_m, p_k}^{\text{mrs}}(\mathcal{A}) = \Pr[\text{MRS}_{E, p_m, p_k}^A \Rightarrow \text{true}],$$

and the maximum MRS advantage of E is

$$\text{Adv}_{E, p_m, p_k}^{\text{mrs}} = \max_{\mathcal{A}} \text{Adv}_{E, p_m, p_k}^{\text{mrs}}(\mathcal{A}).$$

Given a ciphertext c , the optimal strategy is to output a message maximizing the posterior $p_{m|c}(m|c)$. Therefore,

$$\text{Adv}_{E, p_m, p_k}^{\text{mrs}} = \sum_c p_c(c) \max_m p_{m|c}(m|c) = \sum_c \max_m p_{m,c}(m, c).$$

In particular, a trivial adversary already achieves a baseline $p_{\max}$ by always outputting $m_{\text{triv}}(c)$, defined as the most likely message under p_m if $p_m^{\max} \geq p_k^{\max}$, otherwise as the decryption under the most likely key under p_k .

Proposition 5. *For an arbitrary encryption scheme E ,*

$$\text{Adv}_{E, p_m, p_k}^{\text{mrs}} \geq p_{\max} = \max_k \{ \max_m p_k(k), \max_m p_m(m) \}.$$

Beyond MRS, we also consider target-distribution semantic security (TDSS); see Supplementary Section S.8.

5.2. Our Scheme OE with Optimal Security

Core design. Our core idea is to conceal alternative messages behind $m_{\text{triv}}(c)$ for each ciphertext c , such that $m_{\text{triv}}(c)$ is nearly the most likely message given c . Building on this idea, we propose a probabilistic encryption scheme OE that is uniform and tailored to the distributions p_m and p_k . The scheme assigns the ciphertext sets $\mathcal{C}^{(l)}(m, k)$ for each m and k to satisfy two key properties:

1. $\forall m, \forall k, |\mathcal{C}^{(l)}(m, k)| \propto p_m(m)p_k(k) + O(2^{-l})$.
2. $\forall m, \forall k \neq k', \mathcal{C}^{(l)}(m, k) \cap \mathcal{C}^{(l)}(m, k') = \emptyset$.

The second property ensures that ciphertext sets $\mathcal{C}^{(l)}(m, k)$ and $\mathcal{C}^{(l)}(m, k')$ are offset, thereby preventing any *decryption collisions*, i.e., cases where the same ciphertext under different keys maps to the same message. As a result, for any ciphertext c and message m , there exists at most one key k such that $D_k(c) = m$.

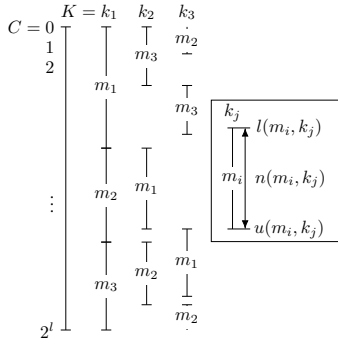


Figure 6: Ciphertext intervals in OE

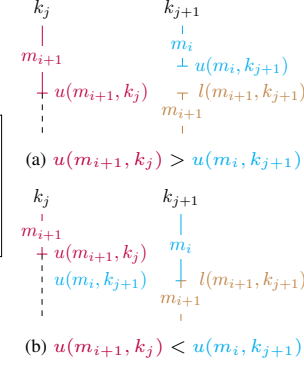


Figure 7: Illustration of Rule 5

The first property further guarantees that, if such a key exists, the guessing probability satisfies

$$p_{m|c}(m|c) = p_{m,k|c}(m, k|c) \propto \frac{p_m(m)p_k(k)}{|\mathcal{C}^{(l)}(m, k)|} \propto 1 + O(2^{-l}).$$

Thus, any attempt to guess a message other than $m_{\text{triv}}(c)$ yields only a negligible additional advantage of $O(2^{-l})$.

Notations. For clarity, we arrange messages and keys in descending order of probability. Let $N_m = |\mathcal{M}|$ and $N_k = |\mathcal{K}|$. Then we write $\mathcal{M} = \{m_1, m_2, \dots, m_{N_m}\}$ with $p_m(m_i) \geq p_m(m_{i+1})$, $\forall i$, and $\mathcal{K} = \{k_1, k_2, \dots, k_{N_k}\}$ with $p_k(k_i) \geq p_k(k_{i+1})$, $\forall i$. Thus, $p_m^{\max} = p_m(m_1)$, $p_k^{\max} = p_k(k_1)$. For a ciphertext c , the trivial guess $m_{\text{triv}}(c)$ is m_1 if $p_m(m_1) \geq p_k(k_1)$, and $D_{k_1}(c)$ otherwise.

We treat $\mathcal{C}^{(l)} = \{0, 1\}^l$ with size $N_c = 2^l$ as the additive group $\mathbb{Z}_{2^l}$, and denote intervals by $[a, b) = \{x \bmod N_c | x \in \mathbb{Z} \cap [a, b)\}$. For example, $\{a, a+1, \dots, N_c-1, 0, 1, \dots, b\}$ corresponds to $[a, b + N_c)$ in $\mathcal{C}^{(l)}$.

For each key k , the scheme assigns to each message m an interval $\mathcal{C}^{(l)}(m, k) = [l(m, k), u(m, k))$, and the ciphertext is drawn uniformly from this interval. Hence, ciphertexts satisfy $p_{c|m,k}(c | m, k) = 1/n(m, k)$ where $n(m, k) = |\mathcal{C}^{(l)}(m, k)|$.

Construction. The interval $\mathcal{C}^{(l)}(m, k) = [l(m, k), u(m, k))$ is defined recursively as follows:

1. $l(m_1, k_1) = 0$.
2. $u(m_i, k_j) = l(m_i, k_j) + n(m_i, k_j)$.
3. $l(m_{i+1}, k_1) = u(m_i, k_1)$, $\forall i$.
4. $l(m_1, k_{j+1}) = u(m_1, k_j)$, $\forall j$.
5. $l(m_{i+1}, k_{j+1}) = \max\{u(m_{i+1}, k_j), u(m_i, k_{j+1})\}$, $\forall i, j$.

Fig. 6 illustrates the recursive assignment of intervals. The first two rules are straightforward. Rules 3 and 5 ensure that $D_{k_{j+1}}(c)$ cannot correspond to two consecutive messages m_i and m_{i+1} , i.e., $\mathcal{C}^{(l)}(m_i, k_j) \cap \mathcal{C}^{(l)}(m_{i+1}, k_j) = \emptyset$. Rules 4 and 5 ensure that $\mathcal{K}(m_{i+1}, c) = \{k | D_k(c) = m_{i+1}\}$ does not contain both k_j and k_{j+1} , i.e., $\mathcal{C}^{(l)}(m_{i+1}, k_j) \cap \mathcal{C}^{(l)}(m_{i+1}, k_{j+1}) = \emptyset$. Fig. 7 shows that in both cases, the intersections vanish.

These rules guarantee only local non-overlap of adjacent intervals. Since $\mathcal{C}^{(l)}$ is circular, overlaps across the boundary 2^l must be prevented by the choice of $n(m, k)$.

As $m_{\text{triv}}(c)$ depends on whether $p_k^{\max} \geq p_m^{\max}$, we employ two different strategies to construct OE in the two respective cases.

In Key-dominant Case (Case K) with $p_k^{\max} \geq p_m^{\max}$:

1. For k_1 , set $n(m, k_1) = n(m)$, where $n(m)$ satisfies: (i) $\sum_m n(m) = N_c$, (ii) $|n(m) - N_c \cdot p_m(m)| < 1$ for each m , and (iii) $n(m_1) \leq N_c \cdot p_k(k_1)$. These are achieved by discretizing p_m . Condition (iii) holds since $p_k(k_1) \geq p_m(m_1)$.
2. For $j \neq 1$, set $n(m, k_j) = \lfloor n(m) \cdot p_k(k_j) / p_k(k_1) \rfloor$. Here, $\lfloor x \rfloor$ denotes the largest integer $\leq x$.

In Message-dominant Case (Case M) with $p_k^{\max} \leq p_m^{\max}$:

1. $n(m_1, k) = n(k)$, where $n(k)$ satisfies three conditions: (i) $\sum_k n(k) = N_c$, (ii) $|n(k) - p_k(k) \cdot N_c| < 1$ for each k , and (iii) $n(k_1) \leq N_c \cdot p_m(m_1)$.
2. For $m \neq m_1$, $n(m, k) = \lfloor n(k) \cdot p_m(m) / p_m(m_1) \rfloor$.

The correctness requires $l \geq \log \frac{p_m^{\max}}{p_m^{\min} p_k^{\min}} + 2$, ensuring that for each m, k , we have $n(m, k) \geq 1$. With these assignments of $\mathcal{C}^{(l)}(m, k)$ and $n(m, k)$, we ensure all intervals are globally non-overlapping, as in the following two lemmas.

Lemma 1. In OE, $\mathcal{C}^{(l)}(m, k) \cap \mathcal{C}^{(l)}(m', k) = \emptyset$, $\forall m \neq m', \forall k$.

Lemma 2. In OE, $\mathcal{C}^{(l)}(m, k) \cap \mathcal{C}^{(l)}(m, k') = \emptyset$, $\forall m, \forall k \neq k'$.

With the above $\mathcal{C}^{(l)}(m, k)$, we construct a symmetric encryption scheme OE with $E_k(m) \leftarrow_{\$} \mathcal{C}^{(l)}(m, k)$. Lemma 1 ensures that OE is well defined, and Lemma 2 guarantees the absence of decryption collisions.

Theorem 5. For our scheme OE,

$$\text{Adv}_{\text{OE}, p_m, p_k}^{\text{mrs}} < p_{\max} \left(1 + \frac{t}{2^l - t} \right) = p_{\max} + O(2^{-l}),$$

where t is a constant $\frac{2p_m^{\max}}{p_m^{\min} p_k^{\min}}$.

The proof of Theorem 5 is provided in Appendix D. Informally, the design of Case M mirrors that of Case K, with the roles of m and k swapped. Under this assignment, the security analysis is analogous. For brevity, we provide the proof only for Case K.

5.3. Our Schemes OE_U for Uniform Distributions

Our scheme OE is intricate for general message and key distributions. Fortunately, when either the message or the key distribution is uniform, OE reduces to a simpler variant, OE_U, whose details are given in Supplementary Section S.6.

For uniform message distributions, OE_U can be viewed as a simple shift cipher, yet it provides stronger security than the ideal HE scheme in the ideal-cipher model. Consider the classic HE scenario of encrypting credit-card CVVs (uniformly distributed over 3 digits) with PINs (non-uniform 6-digit numbers). If the PIN distribution follows Zipf's law $p(\text{pin}_i) = ci^{-r}$ with $r = \frac{1}{2}$ and constant c [29, 30, 31], then OE_U achieves an optimal advantage of 0.001, compared with roughly 0.00151 for the HE scheme in the ideal-cipher model. The latter value comes from sampling-based estimation, as no suitable provable bound is currently available.

5.4. Practical Variant OE' of OE

The OE schemes in Section 5.2 are theoretically optimal but impractical, as ciphertext assignment requires $O(|\mathcal{M}| \cdot |\mathcal{K}|)$ time and space. For many real-world distributions, however, we can reduce this cost to a practical level with only a slight loss in security.

This reduction applies to distributions with long-tailed characteristics, where a few elements have high probability while most others are much less likely. Such patterns commonly occur in low-entropy keys like passwords [29, 30]. Our approach is to approximate locally by uniform distributions, enabling the use of OE_U to cut down complexity.

Formally, we partition the key and message spaces into bins by probability values, ensuring that the ratio between any two probabilities in the same bin is at most e^ϵ . Within each bin, we apply a piecewise uniform approximation, giving equal probability to all elements. This produces approximated distributions p'_k and p'_m for the original p_k and p_m , with the guarantee that $p'/p \leq e^\epsilon$ for both keys and messages. We then construct OE' using p'_k and p'_m .

Theorem 6. *For the scheme OE' constructed via piecewise uniform approximation with parameter e^ϵ , we have*

$$\text{Adv}_{\text{OE}', p_m, p_k}^{\text{mrs}} \leq e^{3\epsilon} p_{\max} + O(2^{-l}),$$

with computational complexity $O(n_{\text{KBin}} \cdot n_{\text{MBin}})$, where n_{KBin} and n_{MBin} are the numbers of key and message bins, respectively.

As an example, consider passwords following a Zipf distribution $p(pw_i) = c i^{-r}$ with $r = 0.8$ and normalization constant c , defined over a space of size 10^{40} (20-character ASCII printable strings). Using 10,000 bins with $e^\epsilon = 1.01$ suffices to cover the whole space. This reduces complexity from 10^{40} to just 10^4 , while retaining near-optimal security.

5.5. Insufficiency of HE Schemes and EE Constructions for Optimal Security

In this section, we show that HE schemes and general EE constructions cannot achieve optimal security for certain key and message distributions.

Insufficiency of HE schemes with MS-IND encoders. Prior work [10, 11, 13, 17, 14, 28, 15] focused only on MS-IND encoders, implicitly treating them as optimal. We show this is not the case.

In HE schemes, each message m_i is assigned $p_m(m_i) \cdot |\mathcal{C}|$ ciphertexts. When a message has probability greater than $\frac{1}{2}$, some ciphertexts inevitably decrypt to the same message under multiple keys, yielding an advantage larger than $p_{\max}$.

Consider $\mathcal{M} = \{m_0, m_1\}$ with distribution (0.9, 0.1) and $\mathcal{K} = \{k_0, k_1\}$ with distribution (0.9, 0.1). The MS-IND encoder assigns $0.9 \cdot |\mathcal{C}|$ seeds to m_0 , resulting in $0.9 \cdot |\mathcal{C}|$ ciphertexts under both k_0 and k_1 . Thus, at least $0.8 \cdot |\mathcal{C}|$ ciphertexts decrypt to m_0 under both keys. This yields $\text{Adv}_{\text{HE}, p_m, p_k}^{\text{mrs}} \geq p_k(k_0) + p_k(k_1) \cdot 0.8 = 0.98 > p_{\max} = 0.9$. In contrast, a trivial encoder encode_t generating uniform

outputs achieves better security. Let $\mathcal{C} = \{0, 1\}$, define $\text{encode}_t(m_i) = i$, and set $E_{k_j}(m_i) = i \oplus j$. Then $\max_m p_{m,c}(m, 0) = 0.81$ and $\max_m p_{m,c}(m, 1) = 0.09$, giving $\text{Adv}_{\text{EE}, p_m, p_k}^{\text{mrs}} = 0.9 = p_{\max}$.

This result, while counterintuitive, highlights that insecurity stems from decryption collisions. In MS-IND encoders, high-probability messages are assigned more ciphertexts, creating collisions and increasing advantage. In contrast, EE_t avoids collisions and achieves optimal advantage. This suggests new directions for encoder design: *which encoder is optimal for EE constructions?*

Insufficiency of EE constructions with general encoders. Even without the MS-IND restriction, EE constructions cannot achieve optimal security. The analysis is more delicate for general encoders, especially as $l \rightarrow \infty$.

Proposition 6. *For $\mathcal{M} = \{a, b\}$ with $p_m = (\frac{1}{2}, \frac{1}{2})$ and any $\mathcal{K}$ such that $p_k^{\max} \leq \frac{1}{2}$ and for any subsets $\mathcal{K}_1 \neq \mathcal{K}_2$ with $p_k(\mathcal{K}_1) \neq p_k(\mathcal{K}_2)$, it holds that*

$$\inf_{\text{EE} \in \Omega_{\text{EE}}} \text{Adv}_{\text{EE}, p_m, p_k}^{\text{mrs}} > p_{\max} = \frac{1}{2}.$$

From the bound in the proof, if $p_k = (\frac{1}{4}, \frac{1}{3}, \frac{5}{12})$, then $\text{Adv}_{\text{EE}, p_m, p_k}^{\text{mrs}} \geq \frac{1}{2} + \frac{1}{48}$ for any EE construction EE, even as $l \rightarrow \infty$. EE constructions are key-independent probabilistic encryption schemes, unlike our key-dependent OE. This independence prevents them from reaching optimal security for general distributions.

In Proposition 6, the advantage is computed for an EE construction with a fixed deterministic cipher. In the random oracle and ideal cipher models [10, 11], the cipher is modeled as a random oracle or ideal cipher, and the advantage is taken in expectation over the choice of the cipher. Since $\inf_x \mathbb{E}_y f(x, y) \geq \inf_{x, y} f(x, y)$, the insufficiency of EE constructions still holds in these models.

5.6. Other Security Properties of Our Scheme OE

Key confidentiality. OE does not achieve KC for arbitrary message and key distributions, since the ciphertext distribution varies with the key (see Fig. 6).

Proposition 7. *OE achieves both optimal KC and optimal MRS if and only if the key distribution is uniform and $p_m^{\max} \leq \frac{1}{|\mathcal{K}|}$.*

For general encryption schemes, KC and MRS cannot be simultaneously optimal under arbitrary distributions. The formal statement is given in Theorem 15.

Semantic security. In the information-theoretic setting, semantic security is unattainable for low-entropy keys. A weaker notion, TDSS, has been studied for HE schemes, but the known bounds degenerate to trivial values when $p_k(k_1) \approx 0.01$, as in the case of passwords (see Appendix B). Whether meaningful TDSS guarantees can be achieved under such low-entropy conditions remains an open problem. For our scheme OE, the analysis of TDSS is also left for future work.

Computational security. The design of OE focuses solely on information-theoretic security against unbounded adversaries, and does not incorporate mechanisms for increasing the computational cost of offline guessing attacks. In particular, the assignment of $\mathcal{C}^{(l)}(m, k)$ is deterministic, allowing adversaries to exploit it at low computational cost.

A natural idea is to incorporate computationally intensive hash functions H (e.g., iterative or memory-hard hashes [32, 33]). Following [10], hashing the key together with a random salt s would force the adversary to evaluate H for every key candidate, thereby multiplying its cost. However, this approach is unsuitable for OE, as it inevitably introduces decryption collisions and increases the MRS advantage. As an alternative, applying H only to a random salt s and using $H(s)$ to uniformly shift all intervals $\mathcal{C}^{(l)}(m, k)$ preserves the information-theoretic guarantees, but increases the adversary's cost by only a fixed constant.

How to amplify the computational hardness of OE remains an open problem.

6. Encryption Schemes with Continuous Ciphertext Spaces

To simplify the analysis and design of HE schemes, we extend encryption schemes to a continuous ciphertext space $[0, 1)$. On the one hand, continuous schemes can be discretized onto $\{0, 1\}^l$ with only $O(2^{-l})$ precision loss, enabling the construction of practical HE schemes from secure continuous ones. On the other hand, any sequence of discrete schemes over $\{0, 1\}^l$ converges to one or more continuous schemes over $[0, 1)$, together with the convergence of their security metrics. This allows security analysis to be carried out on the asymptotic continuous scheme rather than on sequences of discrete ones, greatly simplifying the analysis.

In addition, we identify necessary and sufficient conditions under which continuous schemes achieve optimal key or message confidentiality.

6.1. Definition and Convergence

Definition. A continuous encryption scheme over $\mathcal{C} = [0, 1)$ is defined by E_k and $p_{c|m,k}$, where $E_k(m)$ samples a ciphertext c from $\mathcal{C}(m, k)$ according to $p_{c|m,k}$. In principle, $p_{c|m,k}$ may be any distribution. For our purposes, however, we restrict attention to continuous distributions that are piecewise uniform over intervals in a partition of $[0, 1)$. We also exclude schemes with $p_c = 0$ on some intervals, as such intervals can be naturally removed.

Convergence of schemes discretized from continuous ones. A continuous scheme E_c over $[0, 1)$ with a given ciphertext distribution can be discretized onto the ciphertext space $\{0, 1\}^l$, yielding a discrete scheme $E^{(l)}$. A ciphertext interval $[a, b)$ on which E_c is locally uniform corresponds to the discrete interval $[a', b')$ in $E^{(l)}$ with $a', b' \in \mathbb{Z}_{2^l}$.

Conversely, a discrete scheme $E^{(l)}$ can be lifted to a continuous one $E_c^{(l)}$ by spreading the probability mass at each ciphertext c uniformly over the interval $[\frac{c}{2^l}, \frac{c+1}{2^l})$ instead of concentrating it at the single point c .

Under this transformation, the statistical distance between the ciphertext distributions $p_{c|m,k}$ of $E_c^{(l)}$ and E_c is bounded by $O(2^{-l})$. Thus, $p_{c|m,k}$ of $E_c^{(l)}$ converges pointwise to that of E_c as $l \rightarrow \infty$. Accordingly, we say that $E^{(l)}$ (or equivalently, $E_c^{(l)}$) converges to E_c , and the convergence also extends to the KC and MRS security metrics.

Theorem 7. *For any continuous scheme E_c and its discretization $E^{(l)}$,*

$$I_{E^{(l)}}(K; C^{(l)}) = I_{E_c}(K; C) + O(2^{-l}) \text{ and } \text{Adv}_{E^{(l)}, p_m, p_k}^{\text{MRS}} = \text{Adv}_{E_c, p_m, p_k}^{\text{MRS}} + O(2^{-l}).$$

Therefore, to design secure HE schemes, it suffices to construct a continuous scheme E_c satisfying either $I(K; C) = 0$ or $\text{Adv}_{E_c, p_m, p_k}^{\text{MRS}} = p_{\max}$. Such a scheme can then be discretized into a secure discrete scheme $E^{(l)}$ with $I(K; C^{(l)}) = O(2^{-l})$ or $\text{Adv}_{E^{(l)}, p_m, p_k}^{\text{MRS}} = p_{\max} + O(2^{-l})$. Our OE follows exactly this approach.

Convergence of general discrete schemes. For a general encryption scheme E consisting of a sequence of discrete schemes $E^{(l)}$, convergence may fail, since $p_{c|m,k}$ can vary substantially with l . To address this, we introduce a rearrangement of $p_{c|m,k}$ that guarantees convergence.

This rearrangement is guided by the equivalence relation R_a , which partitions the ciphertext space by decryption outputs: two ciphertexts are equivalent if they decrypt to the same message under all keys. Formally, for each l , define $R_a^{(l)}$ on $\mathcal{C}^{(l)}$ such that $c_1 \sim c_2$ iff $D_k^{(l)}(c_1) = D_k^{(l)}(c_2)$ for all $k \in \mathcal{K}$. Each equivalence class $S \in \mathcal{C}^{(l)}/R_a^{(l)}$ can be expressed as $\bigcap_i \mathcal{C}^{(l)}(m_{j_i}, k_i)$, where $m_{j_i} = D_{k_i}^{(l)}(c)$ for any $c \in S$. We denote such a class by S_m , where $\mathbf{m} = (m_{j_i})_i$.

In the rearrangement, we first redistribute $p_{c|m,k}$ uniformly within each equivalence class S . This preserves both key and message confidentiality, as shown in Theorem S.8 (Supplementary Section S.9). Next, we aggregate the ciphertexts in each S_m into a contiguous interval S'_m over $\{0, 1\}^l$, and order all intervals lexicographically by $\mathbf{m}$ (see Supplementary Section S.9). This aggregation and sorting preserves security. Finally, each discrete interval is mapped to a continuous interval on $[0, 1)$ via the continuization process described earlier.

In the rearranged scheme $E_r^{(l)}$, $p_{c|m,k}$ is piecewise uniform over intervals in $[0, 1)$. Since interval endpoints lie in $[0, 1]$, they admit convergent subsequences as $l \rightarrow \infty$, and so does the probability mass within each interval. Hence, there exists a subsequence $(l_i)_i$ such that $E^{(l_i)}$ converges to a continuous scheme E_a . We refer to such limits as the *asymptotic schemes* of E , which may not be unique.

Theorem 8. *If a discrete scheme E satisfies $I_{E^{(l)}}(K; C^{(l)}) = o(1)$ or $\text{Adv}_{E^{(l)}, p_m, p_k}^{\text{MRS}} = p_{\max} + o(1)$, then every asymptotic scheme E_a satisfies $I_{E_a}(K; C) = 0$ or $\text{Adv}_{E_a, p_m, p_k}^{\text{MRS}} = p_{\max}$.*

Thus, the security of an encryption scheme E can be analyzed via its asymptotic scheme E_a , modulo the convergence rate. While convergence guarantees only $o(1)$ (rather than a negligible $\text{negl}(l)$ bound), this approach still greatly simplifies both the analysis and the design of secure schemes.

6.2. Sufficiency and Necessity of Continuous Schemes

As a secure continuous scheme naturally induces a secure discrete scheme, any sufficient condition for continuous schemes also applies to discrete schemes. Conversely, a secure discrete scheme converges to one or more secure continuous schemes. Thus, any necessary condition for discrete schemes also carries over to continuous schemes.

Theorem 9. A continuous scheme E_c achieves $I(K; C) = 0$, if and only if $\forall k \in \mathcal{K}, \forall c \in \mathcal{C}, p_{c|k}(c|k) = p_c(c)$.

Theorem 10. A uniform continuous scheme E_c achieves $I(K; C) = 0$, if and only if $\forall m \in \mathcal{M}, k \in \mathcal{K}, \Pr[D_k(C) = m | C \leftarrow_{p_c} C] = p_m(m)$.

Theorem 11. A continuous encryption scheme E_c achieves $\text{Adv}_{E_c, p_m, p_k}^{\text{mrs}} = p_{\max}$, if and only if: Let $m_1 \in \arg \max_m p_m(m)$ and $k_1 \in \arg \max_k p_k(k)$.

1. In Case K: 1) $\forall c, D_{k_1}(c)$ maximizes $p_{m,c}(\cdot, c)$ and 2) $\mathcal{C}(m, k_1) \cap \mathcal{C}(m, k') = \emptyset, \forall m, \forall k' \neq k_1$.
2. In Case M: $\forall c, m_1$ maximizes $p_{m,c}(\cdot, c)$.

Corollary 1. For a continuous encryption scheme E_c with $\text{Adv}_{E_c, p_m, p_k}^{\text{mrs}} = p_{\max}$, the following conditions hold: Let $m_1 \in \arg \max_m p_m(m)$ and $k_1 \in \arg \max_k p_k(k)$.

1. Case K with $p_k^{\max} \geq p_m^{\max}$: 1) $\{\mathcal{C}(m, k_1)\}_m$ is a partition of $\mathcal{C}$, 2) $\mathcal{C}(m, k_1) \cap \mathcal{C}(m, k') = \emptyset, \forall m, \forall k' \neq k_1$.
2. Case M with $p_m^{\max} \geq p_k^{\max}$: $\cup_k \mathcal{C}(m_1, k) = \mathcal{C}$.

These conditions are natural. The conditions in Case K mean: 1) There does not exist a ciphertext c such that $D_{k_1}(c) = \perp$; 2) There does not exist a ciphertext c such that $D_{k_1}(c) = D_{k'}(c) \in \mathcal{M}$ for some $k' \neq k_1$. If such ciphertexts exist, the advantage would exceed $p_k(k_1)$. The condition in Case M means that there does not exist a ciphertext c such that $D_k(c) \neq m_1$ for any k . If such a ciphertext exists, the advantage would exceed $p_m(m_1)$.

6.3. Analysis and Designs of Continuous Schemes

Building on the above theorems on continuous encryption, we obtain more concise insights and streamlined designs for encryption schemes.

Necessary tailoring to the key distribution for optimal MC. Using the necessary conditions, we demonstrate that tailoring the scheme to the key distribution is essential to achieving MC in general cases. In other words, no p_k -agnostic continuous encryption scheme can achieve optimal MC for all key distributions.

Theorem 12. For a message distribution p_m , there exists a p_k -agnostic continuous encryption scheme E_c such that $\text{Adv}_{E_c, p_m, p_k}^{\text{mrs}} = p_{\max}$ for any key distribution p_k (under given $|\mathcal{K}|$), only in the following three cases:

1. $|\mathcal{M}| = |\mathcal{K}| = 2$.
2. $|\mathcal{M}| \geq 3, |\mathcal{K}| = 2$, and $p_m(m_1) \geq 0.5$.
3. $p_m^{\max} \leq \frac{1}{|\mathcal{K}|}$.

In other cases, such a p_k -agnostic scheme does not exist.

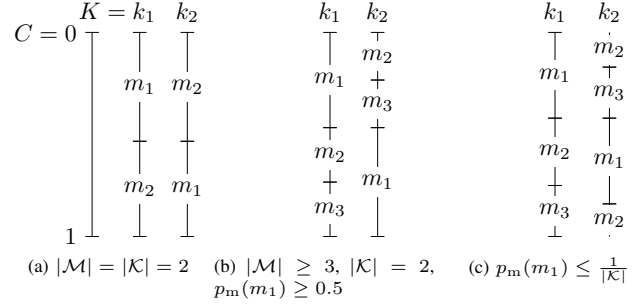


Figure 8: p_k -agnostic scheme for any key distribution p_k

Fig. 8 illustrates the constructions for the three cases. In Case 1 ($|\mathcal{M}| = |\mathcal{K}| = 2$), the two messages partition the ciphertext space into two equal parts, and the two keys swap the assignments of these parts. In Case 2 ($|\mathcal{K}| = 2$ and $p_m(m_1) \geq 0.5$), the most likely message m_1 occupies half of the ciphertext space, while the remaining messages jointly partition the other half. In Case 3 ($p_m^{\max} \leq \frac{1}{|\mathcal{K}|}$), we assign each message m_i an interval of length $p_m(m_i)$ under a reference key k_1 , and for each key k_j we shift all intervals by an offset of $(j-1)p_m^{\max}$. Formally, let $F_m(i) = \sum_{i' \leq i} p_m(m_{i'})$ with $F_m(0) = 0$. Then

$$\mathcal{C}(m_i, k_j) = [F_m(i-1) + (j-1)p_m^{\max}, F_m(i) + (j-1)p_m^{\max}).$$

We denote this HE-style shift construction by HES. Since $p_m^{\max} \leq \frac{1}{|\mathcal{K}|}$, these shifts avoid decryption collisions and yield optimal MRS.

Analogously to Case 3, universal schemes exist under a simple restriction on the key distribution.

Proposition 8. For a message distribution p_m , there exists a continuous scheme E_c such that $\text{Adv}_{E_c, p_m, p_k}^{\text{mrs}} = p_{\max}$ for any key distribution p_k with $p_k^{\max} \leq \frac{1}{|\mathcal{M}|}$.

Analysis of EE constructions. By the necessary conditions, we show that EE constructions cannot achieve optimal MRS, with a simpler analysis than Proposition 6.

Theorem 13. There exist distributions p_m and p_k , such that no continuous EE construction EE_c with a general encoder can achieve $\text{Adv}_{EE_c, p_m, p_k}^{\text{mrs}} = p_{\max}$.

By leveraging continuous encryption, we explicitly compute the MRS advantage $\text{Adv}_{EE, p_m, p_k}^{\text{mrs}}$ for general EE constructions under a simple message distribution, thereby illustrating their insecurity. We leave the details in Appendix E.

Design and security of our OE. Our OE can be naturally designed as a continuous scheme OE_c on the ciphertext space $[0, 1)$. For $\mathcal{C}(m, k)$, its length

$$n(m, k) = \frac{p_k(k)p_m(m)}{p_{\max}}.$$

Theorem 14. For arbitrary p_m and p_k , our continuous scheme OE_c achieves

$$\text{Adv}_{OE_c, p_m, p_k}^{\text{mrs}} = p_{\max}.$$

Proof. In this design, Lemmas 1 and 2 hold, meaning that OE is well-defined, and has no decryption collisions. Therefore, for each m, c , at most one k exists such that $D_k(c) = m$. We denote such k as $k_{\text{OE}}(m, c)$. Then, if $k_{\text{OE}}(m, c)$ exists, we have

$$p_{m,c}(m, c) = p_{k,c}(k_{\text{OE}}(m, c), c) = \frac{p_k(k_{\text{OE}}(m, c))p_m(m)}{n(m, k_{\text{OE}}(m, c))} = p_{\max},$$

which is a constant. Meanwhile, our OE satisfies the conditions in Corollary 1. Thus, in Case K, we have $p_{m,c}(m_{\text{triv}}(c), c) = p_{k,c}(k_1, c) = p_{\max}$ for each c ; in Case M, we have $p_{m,c}(m_{\text{triv}}(c), c) = p_{k,c}(k_{\text{OE}}(m_1, c), c) = p_{\max}$. Therefore, $m_{\text{triv}}(c)$ always maximizes $p_{m,c}(\cdot, c)$ for each c , meaning that our OE achieves the optimal MRS. $\square$

One might worry that the design of OE, which assigns many ciphertexts to $\perp$, appears to conflict with the usual honey-encryption principle that every ciphertext should look like a plausible message. This is a deliberate choice: it maximizes $p_{m,c} = p_{\max}$ on the support and minimizes the size of the support in order to eliminate decryption collisions. In more favorable parameter regimes—for instance, when $p_m^{\max} \leq \frac{1}{|\mathcal{K}|}$ —decryption collisions can be avoided without shrinking the support, as in the HES scheme, which achieves optimal MRS security without changing the support size.

Impossibility of simultaneously optimal KC and MRS.

For general message and key distributions, it is impossible to achieve both optimal KC and optimal MRS simultaneously.

Theorem 15. *There exist distributions p_m and p_k such that no continuous scheme E_c can satisfy both $I(K; C) = 0$ and $\text{Adv}_{E_c, p_m, p_k}^{\text{MRS}} = p_{\max}$.*

As a compromise between KC and MRS, HES offers a favorable trade-off while having lower computational complexity than OE. As an HE scheme, it achieves optimal KC. Its MRS advantage is bounded by $2p_{\max}$ for general distributions, and if $p_m^{\max} \leq \frac{1}{|\mathcal{K}|}$, it even achieves optimal MRS. This situation arises in many practical settings, such as when the message is a random secret (e.g., an RSA private key) and the key is a password.

Theorem 16. *For any message and key distributions (p_m, p_k) , HES satisfies*

$$I(K; C) = 0 \text{ and } \text{Adv}_{\text{HES}, p_m, p_k}^{\text{MRS}} \leq p_k^{\max} + \frac{1 - p_k^{\max}}{\lfloor 1/p_m^{\max} \rfloor} \mathbf{1}_{1/p_m^{\max} < |\mathcal{K}|} \leq 2p_{\max}.$$

6.4. Discussion

Randomness for information-theoretic security. Our results not only reaffirm the fundamental role of randomness in information-theoretic security but also clarify *how* it should be used: the ciphertext cardinality $|\mathcal{C}(m, k)|$ should vary across message–key pairs, with the allocation guided by the given message and key distributions; encryption then randomly samples from the assigned set $\mathcal{C}(m, k)$. For optimal KC, $|\mathcal{C}(m, k)|$ should scale proportionally with $p_m(m)$. For optimal MRS, the allocation must also account for $p_k(k)$. In the absence of collisions, choosing $|\mathcal{C}(m, k)| \propto p_m(m) p_k(k)$ achieves the optimal MRS bound.

Decryption collisions and message confidentiality. Collisions inherently limit the MRS security of HE schemes and EE constructions. For instance, with uniform message and key distributions and $|\mathcal{K}| = |\mathcal{M}|$, the HE scheme in the random oracle model has MRS advantage about $\frac{1}{|\mathcal{K}|} + \frac{3 \ln |\mathcal{K}|}{|\mathcal{K}| \ln \ln |\mathcal{K}|}$ [10], since the random oracle inherently induces collisions. By contrast, in this case, the one-time pad (OTP) achieves Shannon’s perfect security with MRS advantage exactly $p_{\max}$. Our OE can be seen as a variant of the OTP that uses addition instead of XOR, thereby avoiding collisions. Similarly, the HES scheme based on a shift cipher controls collisions, yielding an MRS bound of $2p_{\max}$.

The advantage of our OE and HES over HE schemes.

Existing analyses of HE provide meaningful bounds only under specific distributional regimes (see Appendix B), e.g., when $|\mathcal{M}| \gg |\mathcal{K}|$ and $p_m^{\max} \ll p_k^{\max}$. For general (p_m, p_k) , these bounds can become loose or uninformative—for instance, when encrypting a 3-digit CVV under a 6-digit PIN. In contrast, our OE and HES yield near-optimal provable security bounds for arbitrary (p_m, p_k) .

7. Conclusion and Future Work

We initiated a systematic study of the information-theoretic limits of *all* (possibly probabilistic) symmetric-encryption schemes under low-entropy keys, in a distributional model where messages and keys are drawn from arbitrary given distributions (p_m, p_k) . We identified the optimal attainable levels of key and message confidentiality, measured by $I(K; C)$ and MRS advantage, respectively, and tightly characterized when they are achievable. Technically, we introduced a continuous-ciphertext abstraction over $[0, 1)$ that separates structural constraints from discretization artifacts. Our main message is that in this regime, the optimal attainable levels and the appropriate design principles are governed by *how a scheme uses randomness*.

Several directions remain open. One open problem is to determine the optimal attainable guarantees for stronger notions, most notably TDSS and TDNM.

Another practical challenge is computational efficiency: HC/HE and our OE are designed for arbitrary p_m and p_k , which may be computationally prohibitive/impractical under complexity constraints. While our binning-based complexity reduction is a first step, there is substantial room for improvement by exploiting structured probabilistic models (e.g., Markov models) that arise in many practical settings.

A further challenge is robustness to distributional mismatch: HC/HE typically assume access to the target message distribution p_m , and our approach additionally requires the key distribution p_k , but in practice one often works with biased or misspecified models. Although recent studies on real datasets suggest that imperfect models can still yield impressive security gains, existing provable bounds are largely derived for the matched-model setting and are too coarse to precisely capture security under distributional mismatch; developing tight, general guarantees in the presence of mismatch remains an open question.

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Appendix A. Homophonic Ciphers

Homophonic substitution is a classical cryptographic technique, documented as early as 1401 and employed by the Duchy of Mantua to thwart frequency analysis [8, 34]. In its traditional setting, each plaintext symbol is mapped to one of several ciphertext symbols, with the mappings specifically chosen to uniformize the resulting ciphertext frequencies.

In the modern information-theoretic formulation of homophonic substitution [9], a message is first *encoded* into a perfectly uniform bit string before a conventional cipher is applied. To achieve *perfect* uniformization, the encoder utilizes variable-length codewords, which inherently yields ciphertexts with message-dependent lengths. For instance, given the toy example in Fig. 1 with a message space $\{m_1, m_2, m_3\}$ distributed as $(\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$, a variable-length homophonic encoder assigns codewords as follows: $m_1 \mapsto 0$, $m_2 \mapsto 10$, and $m_3 \mapsto 11$.

Furthermore, for general message distributions that cannot be expressed as finite dyadic fractions—such as the uniform distribution $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ —perfect uniformization requires codewords of potentially unbounded length. As illustrated by the infinite prefix tree in Fig. 9, the encoder must expand the mapping infinitely:

- $m_1 \mapsto (11)^k 00$ (e.g., 00, 1100, 111100, ...)
- $m_2 \mapsto (11)^k 01$ (e.g., 01, 1101, 111101, ...)
- $m_3 \mapsto (11)^k 10$ (e.g., 10, 1110, 111110, ...)

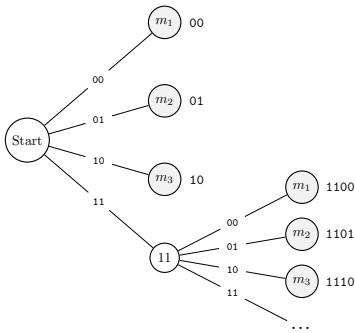


Figure 9: The infinite prefix tree illustrating how a variable-length homophonic encoder maps codewords to messages m_1, m_2, m_3 under a uniform distribution $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. The tree continues to expand infinitely along the 11 branches.

Classical analyses often work in a long/infinite-message-sequence model, where subsequent ciphertext blocks can be viewed as padding to mask such length leakage, and argue that strong Shannon-style key equivocation is achieved in that idealized setting. In our single-message setting, however, variable ciphertext length constitutes an explicit side channel and prevents perfect key confidentiality in general.

Appendix B. Existing Provable Bounds of HE Schemes

For MRS, Juels and Ristenpart [10] analyzed the HE scheme in the random oracle model only under several specific message and key distributions:

1. Uniform message and key distribution with $|\mathcal{K}| = |\mathcal{M}|$:

$$\text{Adv}_{\text{HE}, p_m, p_k}^{\text{mrs}} \leq \frac{1}{|\mathcal{K}|} + \frac{3 \ln |\mathcal{K}|}{|\mathcal{K}| \ln \ln |\mathcal{K}|} + O(2^{-l}).$$

2. Uniform message distribution with $|\mathcal{K}| \ll |\mathcal{M}|$:

$$\text{Adv}_{\text{HE}, p_m, p_k}^{\text{mrs}} \leq p_k^{\max} (1 + \delta) + O(2^{-l}),$$

where $\alpha = \lceil 1/p_k^{\max} \rceil$, $\delta = \frac{\alpha^2}{2|\mathcal{M}|} + \frac{e\alpha^4}{27|\mathcal{M}|^2} \cdot \left(1 - \frac{e\alpha^2}{|\mathcal{M}|^2}\right)^{-1}$, and e is the base of the natural logarithm. Here, δ is small when $|\mathcal{M}| \gg \alpha^2$.

3. General message and key distributions with $|\mathcal{K}| \ll |\mathcal{M}|$: If $p_m^{\max} \leq 3 - \sqrt{5}$, then

$$\text{Adv}_{\text{HE}, p_m, p_k}^{\text{mrs}} \leq p_k^{\max} (1 + \delta) + O(2^{-l}),$$

where $\delta = \frac{\alpha^2}{2b} + \frac{e\alpha^4}{27b^2} \cdot \left(1 - \frac{e\alpha^2}{b^2}\right)^{-1}$, $\alpha = \lceil 3/p_k^{\max} \rceil$, and $b = \lfloor 2/p_m^{\max} \rfloor$.

The MRS bounds are useful in many practical cases, but they do not achieve the optimal $p_{\max} + \text{negl}(l)$ even in the simplest setting with uniform messages and keys of equal length. In this setting, the one-time pad naturally satisfies Shannon's perfect secrecy, which in turn implies optimal MRS. For typical low-entropy keys such as passwords with $p_k^{\max} = 0.01$, the MRS bound is 0.0529 when $p_m^{\max} = 10^{-4}$ and 0.0125 when $p_m^{\max} = 10^{-5}$.

For TDSS, Jaeger et al. [11] has analyzed the HE scheme in the random oracle model for general message and key distributions. For any s satisfying $\alpha^{s-1} \leq \frac{1}{2} - O(2^{-l})$,

$$\text{Adv}_{\text{HE}, p_m, p_k}^{\text{tdss}} \leq p_k^{\max} + \alpha^{s-1} + 2e^{-\alpha^{2s-1}/3} + O(2^{-l}),$$

where $\alpha = \lceil 1/p_k^{\max} \rceil$.

The TDSS bound vanishes as $p_k^{\max} \rightarrow 0$, but it remains unsuitable for low-entropy keys such as passwords. For instance, when $p_k^{\max} = 0.01$ (a typical value for passwords), the bound is 0.5579. Yet by definition, $\frac{1}{2}$ already serves as a trivial bound for $\text{Adv}_{\text{E}, p_m, p_k}^{\text{tdss}}$.

Appendix C. Space of EE constructions.

EE constructions are inherently message-dependent, uniform, and key-independent. Conversely, the following proposition shows that any encryption scheme with these three properties can be represented as an EE construction. This fully characterizes the space of EE constructions.

Proposition 9. *If a probabilistic encryption scheme $(\mathcal{M}, \mathcal{K}, \mathcal{C}, \mathcal{E}, \mathcal{D})$ is uniform, message-dependent, and key-independent, then it can be represented as an EE construction $(\mathcal{M}, \mathcal{K}, \mathcal{C}, \mathcal{E}', \mathcal{D}')$ such that for all $m \in \mathcal{M}$, $k \in \mathcal{K}$, and $c \in \mathcal{C}$,*

$$\Pr[E_k(m) = c] = \Pr[E'_k(m) = c].$$

Proof. Since for each m , $|\mathcal{C}(m, k)|$ is independent of k , we can establish $p_d(m) = \frac{|\mathcal{C}(m, k)|}{2^l}$ with an appropriate encoder (encode, decode). Further, we can design a cipher (encrypt, decrypt) on $\{0, 1\}^l$ such that encrypt_k transforms the seeds in $\text{encode}(m)$ to the ciphertext in $\mathcal{C}(m, k)$. In this way, we construct an EE construction that, under the key k , encrypts the message m to ciphertexts in $\mathcal{C}(m, k)$ with the same probability $\frac{1}{|\mathcal{C}(m, k)|}$. Therefore, for each $m \in \mathcal{M}$, $k \in \mathcal{K}$: if $c \in \mathcal{C}(m, k)$, then $\Pr[E_k(m) = c] = \frac{1}{|\mathcal{C}(m, k)|} = \Pr[E'_k(m) = c]$; if $c \notin \mathcal{C}(m, k)$, then $\Pr[E_k(m) = c] = 0 = \Pr[E'_k(m) = c]$. In other words, the EE construction satisfies the requirement in the theorem. $\square$

Together with the insufficiency results for EE constructions, this proposition shows that achieving optimal security necessarily requires schemes *tailored* to the key distribution. We establish this formally in Section 6 with Theorem 12.

Appendix D.

Proofs of Theorems 1, 2, and 5

Proof of Theorem 1.

$$\begin{aligned} I(K; C^{(l)}) &= \sum_{k \in \mathcal{K}, c \in \mathcal{C}^{(l)}} p_k(k) p_{c|k}(c) \log \frac{p_k(k) p_{c|k}(c|k)}{p_k(k) p_c(c)} \\ &= \sum_{k \in \mathcal{K}} p_k(k) \sum_{c \in \mathcal{C}^{(l)}} p_{c|k}(c) \log \frac{p_{c|k}(c|k)}{p_c(c)} \\ &= \sum_{k \in \mathcal{K}} p_k(k) \text{KL}(C_{0,k}^{(l)} \| C_1^{(l)}) \\ &= \mathbb{E}_{K \leftarrow p_K \mathcal{K}} [\text{KL}(C_{0,K}^{(l)} \| C_1^{(l)})]. \end{aligned}$$

With the equation, we can easily prove the necessity (i.e., the negligibility of $I(K; C^{(l)})$ implies that of $\text{SD}(C_{0,k}^{(l)}, C_1^{(l)})$ for each $k \in \mathcal{K}$):

Since $I(K; C^{(l)}) = \mathbb{E}_{K \leftarrow p_K \mathcal{K}} [\text{KL}(C_{0,K}^{(l)} \| C_1^{(l)})]$, for each $k \in \mathcal{K}$, $\text{KL}(C_{0,k}^{(l)} \| C_1^{(l)}) \leq \frac{I(K; C^{(l)})}{p_k^{\min}}$. Further, by Pinsker's inequality [26],

$$\begin{aligned} \text{SD}(C_{0,k}^{(l)}, C_1^{(l)}) &\leq \sqrt{\frac{\ln 2}{2} \text{KL}(C_{0,k}^{(l)} \| C_1^{(l)})} \\ &\leq \sqrt{\frac{\ln 2}{2 p_k^{\min}}} \cdot I(K; C^{(l)}). \end{aligned}$$

Thus, the necessity holds.

By a reverse Pinsker's inequality (Theorem 7 in [27]), we naturally get the sufficiency (i.e., the negligibility of $\text{SD}(C_{0,k}^{(l)}, C_1^{(l)})$ for each $k \in \mathcal{K}$ implies that of $I(K; C^{(l)})$):

For each $k \in \mathcal{K}$, $p_c(c) = \sum_{k \in \mathcal{K}} p_k(k) p_{c|k}(c|k)$, and

$$p_{c|k}(c|k) \leq p_c(c) \cdot \frac{1}{p_k^{\min}},$$

where $\frac{1}{p_k^{\min}}$ is a constant. Applying this reverse Pinsker's inequality (X_0 and X_1 correspond to $C_{0,k}^{(l)}$ and $C_1^{(l)}$, meanwhile p_0 and p_1 correspond to $p_{c|k}$ and p_c),

$$\text{KL}(C_{0,k}^{(l)} \| C_1^{(l)}) \leq \frac{\log \frac{1}{p_k^{\min}}}{1 - p_k^{\min}} \text{SD}(C_{0,k}^{(l)}, C_1^{(l)})$$

Further,

$$I(K; C^{(l)}) \leq \frac{\log \frac{1}{p_k^{\min}}}{1 - p_k^{\min}} \max_k \text{SD}(C_{0,k}^{(l)}, C_1^{(l)})$$

Thus, the sufficiency holds. $\square$

Proof of Theorem 2. We prove sufficiency; necessity follows from Proposition 1.

By Proposition 1, it suffices to bound $\text{SD}(C_{0,k}, C_1) = \sum_{c \in \mathcal{C}} |p_c(c) - p_{c|k}(c|k)|$ for all $k \in \mathcal{K}$.

We partition the ciphertext space $\mathcal{C}$ into parts, i.e., $\mathcal{C} = \cup_i \mathcal{C}_i$. We exclude ciphertexts c with $p_c(c) = 0$ from $\mathcal{C}$, as they do not contribute to the statistical distance. For each $\mathcal{C}_i$, we aim to bound

$$\rho_{i,k} = \sum_{c \in \mathcal{C}_i} |p_c(c) - p_{c|k}(c|k)|.$$

Note that $D_k(C^{(l)})$ may fail and output $\perp$, thus the following analysis is on the extended message space $\mathcal{M}' = \mathcal{M} \cup \{\perp\}$ with $p_m(\perp) = 0$.

Let c_1, k_1 be the pair that maximizes $p_{c|k}(c|k)$ over all $k \in \mathcal{K}, c \in \mathcal{C}$, $m_1 = D_{k_1}(c_1)$, and $\mathcal{C}_1 = \mathcal{C}(m_1, k_1)$.

Iteratively, for each $i \geq 2$, let c_i, k_i be the pair that maximizes $p_{c|k}(c|k)$ over all $k \in \mathcal{K}$ and $c \in \mathcal{C} - \bigcup_{j=1}^{i-1} \mathcal{C}_j$, and let $m_i = D_{k_i}(c_i)$. Define $\mathcal{C}_i = \mathcal{C}(m_i, k_i) - \bigcup_{j=1}^{i-1} \mathcal{C}_j$.

Repeat this construction until $\bigcup_{i=1}^t \mathcal{C}_i = \mathcal{C}$, where t is the number of iterations. Since the family $\{\mathcal{C}(m, k)\}_{m \in \mathcal{M}, k \in \mathcal{K}}$ covers $\mathcal{C}$ (we exclude ciphertexts c with $p_c(c) = 0$), we can select at most $|\mathcal{M}| \cdot |\mathcal{K}|$ distinct pairs (m_i, k_i) . Hence $t \leq |\mathcal{M}| \cdot |\mathcal{K}|$.

Next, we bound $\rho_{i,k}$ separately. For each $i = 1, 2, \dots, t$ and for all $c \in \mathcal{C}_i$, $k \in \mathcal{K}$,

$$p_{c|k}(c|k) \leq p_{c|k}(c|k_i) = p_{c|k}(c|k_i).$$

Therefore $p_{c|k}(c|k) - p_{c|k}(c|k_i) \leq 0$. For any other $k' \in \mathcal{K} - \{k_i\}$, we have:

$$\begin{aligned} \rho_{i,k_i} &= \sum_{c \in \mathcal{C}_i} |p_c(c) - p_{c|k}(c|k_i)| \\ &= \sum_{c \in \mathcal{C}_i} \left| \sum_k p_k(k) p_{c|k}(c|k) - p_{c|k}(c|k_i) \right| \\ &= \sum_{c \in \mathcal{C}_i} \left| \sum_k p_k(k) (p_{c|k}(c|k) - p_{c|k}(c|k_i)) \right| \end{aligned}$$

$$\begin{aligned}
&= \sum_k p_k(k) \sum_{c \in \mathcal{C}_i} |p_{c|k}(c|k) - p_{c|k}(c|k_i)| \\
&\geq p_k(k') \sum_{c \in \mathcal{C}_i} |p_{c|k}(c|k') - p_{c|k}(c|k_i)|.
\end{aligned}$$

Thus:

$$\begin{aligned}
\rho_{i,k'} &= \sum_{c \in \mathcal{C}_i} |p_c(c) - p_{c|k}(c|k')| \\
&\leq \sum_{c \in \mathcal{C}_i} |p_c(c) - p_{c|k}(c|k_i)| \\
&\quad + \sum_{c \in \mathcal{C}_i} |p_{c|k}(c|k') - p_{c|k}(c|k_i)| \\
&\leq \left(1 + \frac{1}{p_k(k')}\right) \sum_{c \in \mathcal{C}_i} |p_c(c) - p_{c|k}(c|k_i)| \\
&= \left(1 + \frac{1}{p_k(k')}\right) \rho_{i,k_i}.
\end{aligned}$$

Thus $\rho_{i,k'}$ is bounded by ρ_{i,k_i} , so it only suffices to bound ρ_{i,k_i} . Again, by maximality of $p_{c|k}(c|k_i)$,

$$p_c(c) = \sum_k p_k(k) p_{c|k}(c|k) \leq p_{c|k}(c|k_i).$$

so $p_c(c) - p_{c|k}(c|k_i)$ also remains non-positive. Let $d = \max_k \text{SD}(M_{0,k}, M_1)$. For ρ_{1,k_1} , we obtain:

$$\begin{aligned}
\rho_{1,k_1} &= \sum_{c \in \mathcal{C}(m_1, k_1)} |p_c(c) - p_{c|k}(c|k_1)| \\
&= \left| \sum_{c \in \mathcal{C}(m_1, k_1)} (p_c(c) - p_{c|k}(c|k_1)) \right| \\
&= \left| \Pr[D_{k_1}(C_1) = m_1] - p_m(m_1) \right| \leq d.
\end{aligned}$$

For ρ_{i,k_i} with $2 \leq i \leq t$,

$$\begin{aligned}
\rho_{i,k_i} &= \sum_{c \in \mathcal{C}_i} |p_c(c) - p_{c|k}(c|k_i)| \\
&= \left| \sum_{c \in \mathcal{C}_i} (p_c(c) - p_{c|k}(c|k_i)) \right| \\
&= \left| \sum_{c \in \mathcal{C}(m_i, k_i)} (p_c(c) - p_{c|k}(c|k_i)) \right. \\
&\quad \left. - \sum_{j=1}^{i-1} \sum_{c \in \mathcal{C}(m_i, k_i) \cap \mathcal{C}_j} (p_c(c) - p_{c|k}(c|k_i)) \right| \\
&\leq \left| \sum_{c \in \mathcal{C}(m_i, k_i)} (p_c(c) - p_{c|k}(c|k_i)) \right| \\
&\quad + \sum_{j=1}^{i-1} \left| \sum_{c \in \mathcal{C}(m_i, k_i) \cap \mathcal{C}_j} (p_c(c) - p_{c|k}(c|k_i)) \right| \\
&\leq \left| \Pr[D_{k_i}(C_1) = m_i] - p_m(m_i) \right|
\end{aligned}$$

$$\begin{aligned}
&+ \sum_{j=1}^{i-1} \sum_{c \in \mathcal{C}_j} |p_c(c) - p_{c|k}(c|k_i)| \\
&\leq d + \sum_{j=1}^{i-1} \rho_{j,k_i} \leq d + \left(1 + \frac{1}{p_k(k_i)}\right) \sum_{j=1}^{i-1} \rho_{j,k_j}.
\end{aligned}$$

Define $\mu_i = \sum_{j=1}^i \rho_{j,k_j}$, we have $\mu_1 = \rho_{1,k_1} \leq d$ and $\mu_i - \mu_{i-1} \leq d + \left(1 + \frac{1}{p_k(k_i)}\right) \mu_{i-1}$. Then,

$$\mu_i \leq d + \left(2 + \frac{1}{p_k(k_i)}\right) \mu_{i-1} \leq d + \left(2 + \frac{1}{p_k^{\min}}\right) \mu_{i-1}.$$

Unrolling the recurrence, we obtain

$$\mu_i \leq \frac{p_k^{\min}}{1 + p_k^{\min}} \left(\left(2 + \frac{1}{p_k^{\min}}\right)^i - 1 \right) d.$$

Finally, for all $k \in \mathcal{K}$, we can bound $\text{SD}(C_{0,k}, C_1)$:

$$\begin{aligned}
\text{SD}(C_{0,k}, C_1) &= \sum_{c \in \mathcal{C}} |p_c(c) - p_{c|k}(c|k)| \\
&= \sum_{i=1}^t \sum_{c \in \mathcal{C}_i} |p_c(c) - p_{c|k}(c|k)| \\
&= \sum_{i=1}^t \rho_{i,k} \leq \sum_{i=1}^t \left(1 + \frac{1}{p_k(k)}\right) \rho_{i,k_i} \\
&\leq \left(1 + \frac{1}{p_k^{\min}}\right) \mu_t \leq \left(\left(2 + \frac{1}{p_k^{\min}}\right)^t - 1 \right) d \\
&\leq \left(2 + \frac{1}{p_k^{\min}}\right)^{|\mathcal{M}| \cdot |\mathcal{K}|} \max_k \text{SD}(M_{0,k}, M_1),
\end{aligned}$$

which finishes the proof. $\square$

Proof of Lemma 1 for Case K. In our assignment, $[l(m_i, k_i), u(m_i, k_i))$ for $i = 1, 2, \dots, N_m$ are non-overlapping intervals on $\mathbb{Z}$. Should any overlap occur within $\mathcal{C}^{(l)}$, meaning $\mathcal{C}^{(l)}(m_i, k) \cap \mathcal{C}^{(l)}(m_{i'}, k_j) \neq \emptyset$ for some $i < i'$, it implies that $u(m_{i'}, k_j) > l(m_i, k_j) + N_c$. Consequently, this would suggest $u(m_{N_m}, k_j) > l(m_1, k_j) + N_c$. However, our method ensures that $u(m_{N_m}, k_j) \leq l(m_1, k_j) + N_c$, affirming the validity of the lemma.

In the following, we prove $u(m_{N_m}, k_j) \leq l(m_1, k_j) + N_c$ for each j , recursively.

For $j = 1$: the proposition obviously holds.

For $j > 1$: Assume the proposition holds for $j - 1$. Let $i_0 = \max\{i \mid l(m_i, k_j) = u(m_i, k_{j-1})\}$. Note that such i exists, since $i = 1$ meets the condition. Thus, $l(m_i, k_j) = u(m_{i-1}, k_j) = l(m_{i-1}, k_j) + n(m_{i-1}) \frac{p_k(k_j)}{p_k(k_1)}$, for $i = i_0 + 1, \dots, N_m$. Summarizing $l(m_i, k_j) - l(m_{i-1}, k_j)$ over i yields

$$l(m_{N_m}, k_j) - l(m_{i_0}, k_j) = \sum_{i=i_0+1}^{N_m} \left\lfloor n(m_{i-1}) \frac{p_k(k_j)}{p_k(k_1)} \right\rfloor.$$

Moreover,

$$\begin{aligned}
& u(m_{N_m}, k_j) - l(m_{i_0}, k_j) \\
&= \sum_{i=i_0}^{N_m} \lfloor n(m_i) \frac{p_k(k_j)}{p_k(k_1)} \rfloor \leq \sum_{i=i_0}^{N_m} \lfloor n(m_i) \frac{p_k(k_{j-1})}{p_k(k_1)} \rfloor \\
&\leq \lfloor n(m_1) \frac{p_k(k_{j-1})}{p_k(k_1)} \rfloor + \sum_{i=i_0+1}^{N_m} \lfloor n(m_i) \frac{p_k(k_{j-1})}{p_k(k_1)} \rfloor \\
&= u(m_1, k_{j-1}) - l(m_1, k_{j-1}) + \sum_{i=i_0+1}^{N_m} (u(m_i, k_{j-1}) - l(m_i, k_{j-1})).
\end{aligned}$$

Applying $l(m_{i+1}, k_j) = \max\{u(m_{i+1}, k_{j-1}), u(m_i, k_j)\} \geq u(m_i, k_j)$ yields

$$\begin{aligned}
& u(m_{N_m}, k_j) - l(m_{i_0}, k_j) \\
&\leq u(m_1, k_{j-1}) - l(m_1, k_{j-1}) + \sum_{i=i_0+1}^{N_m} (u(m_i, k_{j-1}) - u(m_{i-1}, k_{j-1})) \\
&= u(m_1, k_{j-1}) - l(m_1, k_{j-1}) + u(m_{N_m}, k_{j-1}) - u(m_{i_0}, k_{j-1}).
\end{aligned}$$

By the inductive hypothesis $u(m_{N_m}, k_{j-1}) \leq l(m_1, k_{j-1}) + N_c$,

$$u(m_{N_m}, k_j) - l(m_{i_0}, k_j) \leq u(m_1, k_{j-1}) + N_c - u(m_{i_0}, k_{j-1}).$$

Further, since $l(m_{i_0}, k_j) = u(m_{i_0}, k_{j-1})$,

$$u(m_{N_m}, k_j) \leq u(m_1, k_{j-1}) + N_c = l(m_1, k_j) + N_c.$$

Namely, the proposition holds for j .

The proposition implies that the intervals $[l(m_i, k_j), u(m_i, k_j)]$ for $i = 1, 2, \dots, N_m$ are non-overlapping on the space $[l(m_1, k_j), l(m_1, k_j) + N_c]$. Therefore, the corresponding intervals on $\mathcal{C}$ are also non-overlapping. $\square$

The proof of Lemma 1 for Case M and that of Lemma 2 are similar, and are left in Supplementary Section S.7.

Proof of Theorem 5. We prove the bound for Case K, where $p_{\max} = p_k^{\max} = p_k(k_1)$. The proof for Case M is symmetric by exchanging the roles of messages and keys.

As discussed before,

$$p_{m,c}(m, c) = \sum_{k: D_k(c)=m} \frac{p_m(m)p_k(k)}{n(m, k)}.$$

Thus,

$$\begin{aligned}
\text{Adv}_{\text{OE}, p_m, p_k}^{\text{mrs}} &= \sum_{c \in \mathcal{C}} \max_m p_{m,c}(m, c) \\
&= \sum_{c \in \mathcal{C}} \max_m \sum_{k: D_k(c)=m} \frac{p_m(m)p_k(k)}{n(m, k)}.
\end{aligned}$$

Note that

$$\begin{aligned}
n(m, k) &= \lfloor n(m) \frac{p_k(k)}{p_k(k_1)} \rfloor > n(m) \frac{p_k(k)}{p_k(k_1)} - 1 \\
&> (p_m(m)N_c - 1) \frac{p_k(k)}{p_k(k_1)} - 1 > p_m(m)N_c \frac{p_k(k)}{p_k(k_1)} - 2.
\end{aligned}$$

Let $t = \frac{2p_{\max}}{p_m^{\min}p_k^{\min}}$, then,

$$\begin{aligned}
\frac{p_m(m)p_k(k)}{n(m, k)} &< \frac{p_m(m)p_k(k)}{p_m(m)N_c \frac{p_k(k)}{p_k(k_1)} - 2} = \frac{p_k(k_1)}{N_c - \frac{2p_k(k_1)}{p_m(m)p_k(k)}} \\
&\leq \frac{p_k(k_1)}{N_c - t}.
\end{aligned}$$

Further, by Lemmas 1 and 2, $|\{k \mid D_k(c) = m\}| \leq 1$, and

$$\text{Adv}_{\text{OE}, p_m, p_k}^{\text{mrs}} < N_c \cdot \frac{p_k(k_1)}{N_c - t} = p_k(k_1) \left(1 + \frac{t}{2^l - t}\right).$$

Note $N_c = 2^l$, thus $\text{Adv}_{\text{OE}, p_m, p_k}^{\text{mrs}} = p_k(k_1) + O(\frac{1}{2^l})$. $\square$

Appendix E.

Explicit Advantage of EE Constructions and HE Schemes

The detailed calculations in this section are left in Supplementary Section S.9.

Example 1. For $p_m = (\frac{1}{2}, \frac{1}{2})$ and $p_k = (p_1, p_2, p_3)$ with $\frac{1}{2} > p_1 \geq p_2 \geq p_3$: $\inf_{\text{EE} \in \Omega_{\text{EE}}} \text{Adv}_{\text{EE}, p_m, p_k}^{\text{mrs}}$ are

1. $\min \left\{ \frac{1-2p_1}{2}, \frac{(2p_1-1)(1-3p_1)}{2(1-p_1)p_1}, \frac{1-3p_3}{4}, \frac{(1-2p_2)(p_2-p_3)}{2(1-p_2)p_2} \right\} + \frac{1}{2}$,
if $p_2 \geq \frac{1}{3}$;
2. $\min \left\{ \frac{1-2p_1}{2}, \frac{3p_1-1}{4} \right\} + \frac{1}{2}$, if $p_2 \leq \frac{1}{3}$.

Then, $\inf_{\text{EE} \in \Omega_{\text{EE}}} \text{Adv}_{\text{EE}, p_m, p_k}^{\text{mrs}} \geq \frac{1}{2}$, where the equality holds if and only if $p_1 = p_2 = p_3 = \frac{1}{3}$.

Example 2. For $p_m = (\frac{1}{2}, \frac{1}{2})$ and $p_k = (p_1, p_2, p_3)$ with $\frac{1}{2} > p_1 \geq p_2 \geq p_3$:

$$\inf_{\text{HE} \in \Omega_{\text{HE}}} \text{Adv}_{\text{HE}, p_m, p_k}^{\text{mrs}} = \frac{1-2p_1}{2} + \frac{1}{2} > \frac{1}{2}.$$

The above analysis studies the optimal security of EE constructions with different ciphers, aiming to minimize decryption collisions. Previous work analyzed their *average* security in the random oracle (RO) or ideal cipher (IC) models, where the advantage is computed as an expectation over random ciphers.

We can naturally associate any discrete EE construction $\text{EE}^{(l)}$ with a continuous one EE_c by setting $|S_m| = \prod_i p_d(m_{j_i})$. Under this mapping, EE_c characterizes the expected collisions, and the expected advantage (for both MC and KC) of $\text{EE}^{(l)}$ converges to that of EE_c . Thus, analyzing EE constructions in the RO or IC models is equivalent to analyzing the corresponding continuous scheme.

Proposition 10. For any EE construction $\text{EE}^{(l)}$ in the random oracle or ideal cipher models and the associated continuous scheme EE_c , as $l \rightarrow \infty$,

$$\mathbb{E}[I_{\text{EE}^{(l)}}(K; C^{(l)})] \rightarrow I_{\text{EE}_c}(K; C) \text{ and } \mathbb{E}[\text{Adv}_{\text{EE}^{(l)}, p_m, p_k}^{\text{mrs}}] \rightarrow \text{Adv}_{\text{EE}_c, p_m, p_k}^{\text{mrs}}.$$

Example 3. For $p_m = (\frac{1}{2}, \frac{1}{2})$ and $p_k = (p_1, p_2, p_3)$ with $\frac{1}{2} > p_1 \geq p_2 \geq p_3$, in the random-oracle or ideal-cipher model, the continuous HE scheme HE_c (with $p_d = p_m$) and EE constructions EE_c (with general p_d) satisfy the bounds:

$$\text{Adv}_{\text{HE}_c, p_m, p_k}^{\text{mrs}} = \frac{3}{4} \text{ and } \inf_{\text{EE}_c \in \Omega_{\text{EE}}} \text{Adv}_{\text{EE}_c, p_m, p_k}^{\text{mrs}} = \frac{1}{2} + p_1(1-p_1) \geq \frac{13}{18}.$$

The Supplementary Proof Artifact

S.1. Results on Perfect KC

We explore the concept of perfect KC, which requires that the ciphertext C reveals no information about the key K . This concept is analogous to the perfect security defined by Shannon for message confidentiality [12].

Theorem S.1. *A symmetric encryption scheme $(\mathcal{M}, \mathcal{K}, \mathcal{C}, \mathcal{E}, \mathcal{D})$ achieves perfect KC, if and only if for each $k \in \mathcal{K}$ and $c \in \mathcal{C}$, $p_{c|k}(c|k) = p_c(c)$, i.e., $p_{c|k}(c|k)$ is independent of k .*

Proof. Echoing Shannon's proof for perfect security [12], $I(K; C) = 0$ if and only if K and C are independent, i.e., for each $k \in \mathcal{K}$ and $c \in \mathcal{C}$, $p_{c|k}(c|k) = p_c(c)$. $\square$

This means that, for each $k \in \mathcal{K}$, E_k transforms the message distribution p_m into the same ciphertext distribution p_c . Accordingly, for each $k \in \mathcal{K}$, D_k transforms the ciphertext distribution p_c back into the message distribution p_m . Thus, the honey-message distribution under each key coincides with the real distribution. Thus, we have the following proposition.

Definition S.1. *A symmetric encryption scheme is perfect M-IND, if for each $k \in \mathcal{K}$, $M_1 \equiv M_{0,k}$, where $C_0 \leftarrow_{p_c} \mathcal{C}$, $M_{0,k} \leftarrow D_k(C_0)$ and $M_1 \leftarrow_{p_m} \mathcal{M}$.*

Proposition S.1. *If a symmetric encryption scheme achieves perfect KC, then it is an HE scheme with the property of perfect M-IND.*

However, unlike Theorem S.1, the converse of Proposition S.1 does not hold, because E_k can be randomized whereas D_k is deterministic. A randomized E_k may map p_m to different ciphertext distributions $p_{c|k}(\cdot|k)$ under different keys, even though the corresponding D_k still maps the overall ciphertext distribution p_c back to p_m . Fig. S.2 provides such a counterexample.

Fortunately, if E_k is uniform, meaning it assigns equal probability to all possible ciphertexts in $\mathcal{C}(m, k)$ for each message m and key k , the converse of Proposition S.1 holds.

Theorem S.2. *A uniform symmetric encryption scheme achieves perfect KC, if and only if it is perfect M-IND.*

More interestingly, a stronger result holds: $p_{c|k}(c|k)$ and $p_c(c)$ are not only equal but also uniformly distributed over certain subspaces of $\mathcal{C}$. These subspaces are defined by a partition $\mathcal{P}$ of $\mathcal{C}$ induced by an equivalence relation R , thus $\mathcal{P} = \mathcal{C}/R$. Here, R is the equivalence closure where $c_1, c_2 \in \mathcal{C}$ are related if and only if there exist $m \in \mathcal{M}$ and $k \in \mathcal{K}$ such that $c_1, c_2 \in \mathcal{C}(m, k)$.

Proposition S.2. *If a uniform symmetric encryption scheme is perfect M-IND, then for every equivalence class S of R , the distributions p_c and $p_{c|k}$ are identical and uniform over S , i.e., $p_c(c_1) = p_c(c_2) = p_{c|k}(c_1|k) = p_{c|k}(c_2|k)$ for all $c_1, c_2 \in S$ and $k \in \mathcal{K}$.*

Proof of Proposition S.2. Since

$$\sum_m \Pr[D_k(C) = m] = \sum_m p_m(m) = 1,$$

it follows that for each k , $D_k(C)$ never fails ($D_k(C) \neq \perp$).

The condition is equivalent to the following identity: for all $m \in \mathcal{M}, k \in \mathcal{K}$,

$$\begin{aligned} \sum_{c \in \mathcal{C}(m, k)} p_c(c) &= p_m(m) \\ &= \sum_{c \in \mathcal{C}(m, k)} p_m(m) p_{c|m, k}(c|m, k) \\ &= \sum_{c \in \mathcal{C}(m, k)} p_{c|k}(c|k). \end{aligned}$$

Here, $p_{c|k}(c|k)$ is constant over $\mathcal{C}(m, k)$, given by

$$p_{c|k}(c|k) = \frac{p_m(m)}{|\mathcal{C}(m, k)|}.$$

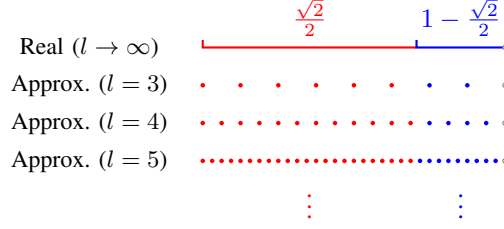


Figure S.1: Approximating the probability distribution $(\frac{\sqrt{2}}{2}, 1 - \frac{\sqrt{2}}{2})$ over $\{m_1, m_2\}$ by random l bits (i.e., l -bit ciphertexts/seeds).

Intuitively, $p_c(c)$ is the weighted average of $p_{c|k}(c|k)$ over all k , with weights $p_k(k)$. Typically, the maximum of $p_{c|k}(c|k)$ is greater than the average $p_c(c)$. However, the equation above implies that the maximum is equal to the average, forcing $p_{c|k}(c|k) = p_c(c)$. Based on this intuition, we now present a formal proof.

Formally, consider any equivalence class S of R , and let c_1, k_1 be the pair that maximizes $p_{c|k}(c|k)$ over all $k \in \mathcal{K}, c \in S$. Define $m_1 = D_{k_1}(c_1)$.

By the uniformity of encryption, for all $c \in \mathcal{C}(m_1, k_1)$,

$$p_{c|k}(c|k_1) = p_{c|k}(c_1|k_1) = \frac{p_m(m_1)}{|\mathcal{C}(m_1, k_1)|}.$$

Therefore, we derive the following:

$$\begin{aligned} \sum_{c \in \mathcal{C}(m_1, k_1)} p_{c|k}(c|k_1) &= \sum_{c \in \mathcal{C}(m_1, k_1)} p_c(c) \\ &= \sum_{c \in \mathcal{C}(m_1, k_1)} \sum_k p_k(k) p_{c|k}(c|k) \\ &\leq \sum_{c \in \mathcal{C}(m_1, k_1)} \sum_k p_k(k) p_{c|k}(c|k_1) \\ &= \sum_{c \in \mathcal{C}(m_1, k_1)} p_{c|k}(c|k_1). \end{aligned}$$

Since both sides are equal, the inequality must hold with equality, forcing

$$p_{c|k}(c|k) = p_{c|k}(c|k_1) = p_{c|k}(c_1|k_1)$$

for all $k \in \mathcal{K}$ and $c \in \mathcal{C}(m_1, k_1)$. Furthermore, for all $c \in \mathcal{C}(m_1, k_1)$,

$$p_c(c) = \sum_k p_k(k) p_{c|k}(c|k) = p_{c|k}(c_1|k_1).$$

This result shows that the maximum value $p_{c|k}(c_1|k_1)$ propagates to all $p_c(c)$ and $p_{c|k}(c|k)$ within $\mathcal{C}(m_1, k_1)$. By the definition of equivalence classes, this propagation extends to the entire S through the same argument. In other words, p_c and $p_{c|k}$ are identical and uniform over S . $\square$

Impossibility of perfect KC. For a general message distribution, such as the two-message distribution $(\frac{\sqrt{2}}{2}, 1 - \frac{\sqrt{2}}{2})$ illustrated in Fig. S.1, it is impossible to achieve perfect KC with a finite ciphertext length l and uniform encryption, except for trivial encryption schemes. Specifically, in these trivial schemes, E_k is the identity mapping id or any mapping independent of k , ensuring perfect KC. However, such trivial schemes fail to provide message confidentiality, as the ciphertext completely determines the message without requiring the key. Except for these trivial schemes, we establish by contradiction that no uniform encryption scheme can achieve perfect KC.

Assume that a uniform encryption scheme achieves perfect KC. Then, by Proposition S.2, $p_c = p_{c|k}(\cdot|k)$ must be uniform over each equivalence class $S \in \mathcal{C}/R$. Thus, for any $\mathcal{C}(m, k)$ and $\mathcal{C}(m', k')$ belonging to the same S ,

$$\frac{p_m(m)}{p_m(m')} = \frac{|\mathcal{C}(m, k)|}{|\mathcal{C}(m', k')|} \in \mathbb{Q}.$$

However, since $\frac{p_m(m_1)}{p_m(m_2)} = 1 + \sqrt{2} \notin \mathbb{Q}$, it follows that $\mathcal{C}(m_1, k)$ and $\mathcal{C}(m_2, k')$ cannot belong to the same S for any k, k' . Thus, $\mathcal{C}(m_1, k) \cap \mathcal{C}(m_2, k') = \emptyset$. Further, by $\mathcal{C}(m_1, k) \cup \mathcal{C}(m_2, k) = \mathcal{C}$, $\mathcal{C}(m_1, k)$ and $\mathcal{C}(m_2, k)$ remain unchanged across different keys, which are the only two equivalence classes of $\mathcal{C}$. This implies that the encryption scheme is trivial.

A non-uniform (and non-trivial) encryption scheme may achieve perfect KC, but it must assign *irrational* probabilities to ciphertexts, which still requires an infinite number of uniform random bits.

S.2. Bounds for HE schemes with Specific Well-designed Encoders

IS-DTE. Juels and Ristenpart [10] introduced a general approach to build encoders for arbitrary non-uniform distributions, resulting in what are known as IS-DTEs. For given p_m over $\mathcal{M}$, we denote its cumulative distribution function by F , where $F(i) = \sum_{t=1}^i p_m(m_t)$. The construction of IS-DTE leverages the fact that, for a uniform random variable U on $[0, 1)$, the inverse function $F^{-1}(U)$ follows the distribution p_m . By discretizing $[0, 1)$ to the set $\mathcal{S} = \{0, 1, \dots, 2^l - 1\}$ (i.e., $\{0, 1\}^l$), IS-DTE randomly assigns an integer seed in the range $[\text{round}(2^l F(i-1)), \text{round}(2^l F(i))]$ to the message m_i . Here, round represents the rounding function, with $F(0) = 0$, and $l \geq \lceil -\log \min_m p_m(m) \rceil$ ¹. With this design, $|p_d(m) - p_m(m)| \leq \frac{1}{2^l}$ for each $m \in \mathcal{M}$.

Corollary S.1. *Built on a cipher and an IS-DTE with the MS-IND property, the HE scheme achieves KC. Specifically,*

$$I(K; C^{(l)}) \leq \frac{2p_m^{\min} - \frac{1}{2^l}}{2^l \ln 2 p_m^{\min} (p_m^{\min} - \frac{1}{2^l})} = O\left(\frac{1}{2^l}\right).$$

For messages with complex distributions, such as natural language texts, directly applying IS-DTE often leads to prohibitive time and space complexity. To address this, Cheng et al. [28] proposed a method to construct an efficient encoder based on a probability model, termed the probability-model-transforming encoder (PMTE). Similar to IS-DTE, IS-PMTE is designed using inverse sampling. During encoding, IS-PMTE parses the message into a sequence of generating rules and encodes each rule individually. If the probability model accurately characterizes the message distribution, IS-PMTE achieves MS-IND. Without accounting for the differences between the probability model and the actual message distribution, $|p_d(m) - p_m(m)| \leq \frac{l_{\text{seq}} n_r}{2^{l/l_{\text{seq}}}}$ holds for each message m , where l_{seq} represents the maximum sequence length and n_r denotes the number of rules.

Corollary S.2. *Built on a cipher and an IS-PMTE with the MS-IND property, the HE scheme achieves KC. Specifically,*

$$I(K; C^{(l)}) \leq \frac{l_{\text{seq}} n_r \left(2p_m^{\min} - \frac{l_{\text{seq}} n_r}{2^{l/l_{\text{seq}}}}\right)}{2^{l/l_{\text{seq}}} \ln 2 p_m^{\min} \left(p_m^{\min} - \frac{l_{\text{seq}} n_r}{2^{l/l_{\text{seq}}}}\right)} = O\left(2^{-l/l_{\text{seq}}}\right).$$

S.3. Proofs in Section 4

Theorem S.3 (Pinsker's inequality [26]). *For two random variables X_0 and X_1 on the space $\mathcal{X}$ with the distributions p_0 and p_1 ,*

$$\text{SD}(X_0, X_1) \leq \sqrt{\frac{\ln 2}{2} \text{KL}(X_0 \| X_1)}.$$

Theorem S.4 (Theorem 7 in [27]). *For two random variables X_0 and X_1 on the space $\mathcal{X}$ with the distributions p_0 and p_1 such that for each $x \in \mathcal{X}$, $p_0(x) \leq c \cdot p_1(x)$, where $c \geq 1$ is a constant:*

$$\text{KL}(X_0 \| X_1) \leq \frac{c \log c}{c - 1} \text{SD}(X_0, X_1).$$

Theorem 1 (restated). *A symmetric encryption scheme achieves statistical KC if and only if it satisfies C-IND in Fig. 3, i.e., for every key k we have $C_1^{(l)} \stackrel{s}{\equiv} C_{0,k}^{(l)}$. More specifically,*

$$\begin{aligned} \text{SD}(C_{0,k}^{(l)}, C_1^{(l)}) &\leq \sqrt{\frac{\ln 2}{2p_k^{\min}} I(K; C^{(l)})} = O\left(\sqrt{I(K; C^{(l)})}\right), \quad \forall k \in \mathcal{K}; \\ I(K; C^{(l)}) &\leq \frac{\log(1/p_k^{\min})}{1 - p_k^{\min}} \max_k \text{SD}(C_{0,k}^{(l)}, C_1^{(l)}) = O\left(\max_k \text{SD}(C_{0,k}^{(l)}, C_1^{(l)})\right). \end{aligned}$$

1. $\lceil x \rceil$ represents the smallest integer greater than or equal to x .

Proof of Theorem 1. Note $I(K; C^{(l)}) = \mathbb{E}_{K \leftarrow p_K} \mathcal{K}[\text{KL}(C_{0,K}^{(l)} \| C_1^{(l)})]$, since

$$\begin{aligned} I(K; C^{(l)}) &= \sum_{k \in \mathcal{K}, c \in \mathcal{C}^{(l)}} p_k(k) p_{c|k}(c) \log \frac{p_k(k) p_{c|k}(c|k)}{p_k(k) p_c(c)} \\ &= \sum_{k \in \mathcal{K}} p_k(k) \sum_{c \in \mathcal{C}^{(l)}} p_{c|k}(c) \log \frac{p_{c|k}(c|k)}{p_c(c)} \\ &= \sum_{k \in \mathcal{K}} p_k(k) \text{KL}(C_{0,k}^{(l)} \| C_1^{(l)}) \\ &= \mathbb{E}_{K \leftarrow p_K} \mathcal{K}[\text{KL}(C_{0,K}^{(l)} \| C_1^{(l)})]. \end{aligned}$$

With the equation, we can easily prove the necessity (i.e., the negligibility of $I(K; C^{(l)})$ implies that of $\text{SD}(C_{0,k}^{(l)}, C_1^{(l)})$ for each $k \in \mathcal{K}$):

Since $I(K; C^{(l)}) = \mathbb{E}_{K \leftarrow p_K} \mathcal{K}[\text{KL}(C_{0,K}^{(l)} \| C_1^{(l)})]$, for each $k \in \mathcal{K}$, $\text{KL}(C_{0,k}^{(l)} \| C_1^{(l)}) \leq \frac{I(K; C^{(l)})}{p_k^{\min}}$. Further, by Pinsker's inequality,

$$\begin{aligned} \text{SD}(C_{0,k}^{(l)}, C_1^{(l)}) &\leq \sqrt{\frac{\ln 2}{2} \text{KL}(C_{0,k}^{(l)} \| C_1^{(l)})} \\ &\leq \sqrt{\frac{\ln 2}{2 p_k^{\min}} \cdot I(K; C^{(l)})}. \end{aligned}$$

Thus, the necessity holds.

By Theorem S.4, we naturally get the sufficiency (i.e., the negligibility of $\text{SD}(C_{0,k}^{(l)}, C_1^{(l)})$ for each $k \in \mathcal{K}$ implies that of $I(K; C^{(l)})$):

Note that for each $k \in \mathcal{K}$, $p_c(c) = \sum_{k \in \mathcal{K}} p_k(k) p_{c|k}(c|k)$, and thus

$$p_{c|k}(c|k) \leq p_c(c) \cdot \frac{1}{p_k^{\min}},$$

where $\frac{1}{p_k^{\min}}$ is a constant. Applying a reverse Pinsker's inequality, i.e., Theorem S.4 (X_0 and X_1 correspond to $C_{0,k}^{(l)}$ and $C_1^{(l)}$, meanwhile p_0 and p_1 correspond to $p_{c|k}$ and p_c),

$$\text{KL}(C_{0,k}^{(l)} \| C_1^{(l)}) \leq \frac{\log \frac{1}{p_k^{\min}}}{1 - p_k^{\min}} \text{SD}(C_{0,k}^{(l)}, C_1^{(l)})$$

Further,

$$I(K; C^{(l)}) \leq \frac{\log \frac{1}{p_k^{\min}}}{1 - p_k^{\min}} \max_k \text{SD}(C_{0,k}^{(l)}, C_1^{(l)})$$

Thus, the sufficiency holds. $\square$

Theorem 2 (restated). A uniform symmetric encryption scheme achieves statistical KC, if and only if it is M-IND. Specifically,

$$\max_k \text{SD}(C_{0,k}, C_1) \leq \left(2 + \frac{1}{p_k^{\min}}\right)^{|\mathcal{M}| \cdot |\mathcal{K}|} \max_k \text{SD}(M_{0,k}, M_1).$$

Proof of Theorem 2. We prove sufficiency; necessity follows from Proposition 1.

By Proposition 1, it suffices to bound $\text{SD}(C_{0,k}, C_1) = \sum_{c \in \mathcal{C}} |p_c(c) - p_{c|k}(c|k)|$ for all $k \in \mathcal{K}$.

We partition the ciphertext space $\mathcal{C}$ into parts, i.e., $\mathcal{C} = \cup_i \mathcal{C}_i$. We exclude ciphertexts c with $p_c(c) = 0$ from $\mathcal{C}$, as they do not contribute to the statistical distance. For each $\mathcal{C}_i$, we aim to bound

$$\rho_{i,k} = \sum_{c \in \mathcal{C}_i} |p_c(c) - p_{c|k}(c|k)|.$$

Note that $D_k(C^{(l)})$ may fail and output $\perp$, thus the following analysis is on the extended message space $\mathcal{M}' = \mathcal{M} \cup \{\perp\}$ with $p_m(\perp) = 0$.

Let c_1, k_1 be the pair that maximizes $p_{c|k}(c|k)$ over all $k \in \mathcal{K}, c \in \mathcal{C}$, and let $m_1 = D_{k_1}(c_1)$. Define $\mathcal{C}_1 = \mathcal{C}(m_1, k_1)$.

Iteratively, for each $i \geq 2$, let c_i, k_i be the pair that maximizes $p_{c|k}(c|k)$ over all $k \in \mathcal{K}$ and $c \in \mathcal{C} - \bigcup_{j=1}^{i-1} \mathcal{C}_j$, and let $m_i = D_{k_i}(c_i)$. Define $\mathcal{C}_i = \mathcal{C}(m_i, k_i) - \bigcup_{j=1}^{i-1} \mathcal{C}_j$.

Repeat this construction until $\bigcup_{i=1}^t \mathcal{C}_i = \mathcal{C}$, where t is the number of iterations. Since the family $\{\mathcal{C}(m, k)\}_{m \in \mathcal{M}, k \in \mathcal{K}}$ covers $\mathcal{C}$ (we exclude ciphertexts c with $p_c(c) = 0$), we can select at most $|\mathcal{M}| \cdot |\mathcal{K}|$ distinct pairs (m_i, k_i) . Hence $t \leq |\mathcal{M}| \cdot |\mathcal{K}|$.

Next, we bound $\rho_{i,k}$ separately. For each $i = 1, 2, \dots, t$ and for all $c \in \mathcal{C}_i, k \in \mathcal{K}$,

$$p_{c|k}(c|k) \leq p_{c|k}(c_i|k_i) = p_{c|k}(c|k_i).$$

Therefore $p_{c|k}(c|k) - p_{c|k}(c|k_i) \leq 0$. For any other $k' \in \mathcal{K} - \{k_i\}$, we have:

$$\begin{aligned} \rho_{i,k_i} &= \sum_{c \in \mathcal{C}_i} |p_c(c) - p_{c|k}(c|k_i)| = \sum_{c \in \mathcal{C}_i} \left| \sum_k p_k(k) p_{c|k}(c|k) - p_{c|k}(c|k_i) \right| \\ &= \sum_{c \in \mathcal{C}_i} \left| \sum_k p_k(k) (p_{c|k}(c|k) - p_{c|k}(c|k_i)) \right| \\ &= \sum_k p_k(k) \sum_{c \in \mathcal{C}_i} |p_{c|k}(c|k) - p_{c|k}(c|k_i)| \\ &\geq p_k(k') \sum_{c \in \mathcal{C}_i} |p_{c|k}(c|k') - p_{c|k}(c|k_i)|. \end{aligned}$$

Thus:

$$\begin{aligned} \rho_{i,k'} &= \sum_{c \in \mathcal{C}_i} |p_c(c) - p_{c|k}(c|k')| \\ &\leq \sum_{c \in \mathcal{C}_i} |p_c(c) - p_{c|k}(c|k_i)| \\ &\quad + \sum_{c \in \mathcal{C}_i} |p_{c|k}(c|k') - p_{c|k}(c|k_i)| \\ &\leq \left(1 + \frac{1}{p_k(k')}\right) \sum_{c \in \mathcal{C}_i} |p_c(c) - p_{c|k}(c|k_i)| \\ &= \left(1 + \frac{1}{p_k(k')}\right) \rho_{i,k_i}. \end{aligned}$$

Thus $\rho_{i,k'}$ is bounded by ρ_{i,k_i} , so it only suffices to bound ρ_{i,k_i} . Again, by maximality of $p_{c|k}(c|k_i)$,

$$p_c(c) = \sum_k p_k(k) p_{c|k}(c|k) \leq p_{c|k}(c|k_i).$$

so $p_c(c) - p_{c|k}(c|k_i)$ also remains non-positive. Let $d = \max_k \text{SD}(M_{0,k}, M_1)$. For ρ_{1,k_1} , we obtain:

$$\begin{aligned} \rho_{1,k_1} &= \sum_{c \in \mathcal{C}(m_1, k_1)} |p_c(c) - p_{c|k}(c|k_1)| \\ &= \left| \sum_{c \in \mathcal{C}(m_1, k_1)} (p_c(c) - p_{c|k}(c|k_1)) \right| \\ &= \left| \Pr[D_{k_1}(C_1) = m_1] - p_m(m_1) \right| \leq d. \end{aligned}$$

For ρ_{i,k_i} with $2 \leq i \leq t$,

$$\begin{aligned} \rho_{i,k_i} &= \sum_{c \in \mathcal{C}_i} |p_c(c) - p_{c|k}(c|k_i)| \\ &= \left| \sum_{c \in \mathcal{C}_i} (p_c(c) - p_{c|k}(c|k_i)) \right| \\ &= \left| \sum_{c \in \mathcal{C}(m_i, k_i)} (p_c(c) - p_{c|k}(c|k_i)) \right| \end{aligned}$$

$$\begin{aligned}
& \left| - \sum_{j=1}^{i-1} \sum_{c \in \mathcal{C}(m_i, k_i) \cap \mathcal{C}_j} (p_c(c) - p_{c|k}(c|k_i)) \right| \\
& \leq \left| \sum_{c \in \mathcal{C}(m_i, k_i)} (p_c(c) - p_{c|k}(c|k_i)) \right| \\
& \quad + \sum_{j=1}^{i-1} \left| \sum_{c \in \mathcal{C}(m_i, k_i) \cap \mathcal{C}_j} (p_c(c) - p_{c|k}(c|k_i)) \right| \\
& \leq \left| \Pr[D_{k_i}(C_1) = m_i] - p_m(m_i) \right| + \sum_{j=1}^{i-1} \sum_{c \in \mathcal{C}_j} |p_c(c) - p_{c|k}(c|k_i)| \\
& \leq d + \sum_{j=1}^{i-1} \rho_{j, k_i} \leq d + \left(1 + \frac{1}{p_k(k_i)}\right) \sum_{j=1}^{i-1} \rho_{j, k_j}.
\end{aligned}$$

Define $\mu_i = \sum_{j=1}^i \rho_{j, k_j}$, we have $\mu_1 = \rho_{1, k_1} \leq d$ and $\mu_i - \mu_{i-1} \leq d + \left(1 + \frac{1}{p_k(k_i)}\right) \mu_{i-1}$. Then,

$$\mu_i \leq d + \left(2 + \frac{1}{p_k(k_i)}\right) \mu_{i-1} \leq d + \left(2 + \frac{1}{p_k^{\min}}\right) \mu_{i-1}.$$

Unrolling the recurrence, we obtain

$$\mu_i \leq \frac{p_k^{\min}}{1 + p_k^{\min}} \left(\left(2 + \frac{1}{p_k^{\min}}\right)^i - 1 \right) d.$$

Finally, for all $k \in \mathcal{K}$, we can bound $\text{SD}(C_{0,k}, C_1)$:

$$\begin{aligned}
\text{SD}(C_{0,k}, C_1) &= \sum_{c \in \mathcal{C}} |p_c(c) - p_{c|k}(c|k)| \\
&= \sum_{i=1}^t \sum_{c \in \mathcal{C}_i} |p_c(c) - p_{c|k}(c|k)| \\
&= \sum_{i=1}^t \rho_{i, k} \leq \sum_{i=1}^t \left(1 + \frac{1}{p_k(k)}\right) \rho_{i, k_i} \\
&\leq \left(1 + \frac{1}{p_k^{\min}}\right) \mu_t \leq \left(\left(2 + \frac{1}{p_k^{\min}}\right)^t - 1 \right) d \\
&\leq \left(2 + \frac{1}{p_k^{\min}}\right)^{|\mathcal{M}| \cdot |\mathcal{K}|} \max_k \text{SD}(M_{0,k}, M_1),
\end{aligned}$$

which finishes the proof. $\square$

Proposition 2 (restated). *If a uniform symmetric encryption scheme is M-IND, then for each $k, k' \in \mathcal{K}$, $C_{0,k}^{(l)} \stackrel{s}{=} C_1^{(l)} \stackrel{s}{=} U_{k'}^{(l)}$, where $C_{0,k}^{(l)} \leftarrow_{p_{c|k}(\cdot|k)} \mathcal{C}^{(l)}$, $C_1^{(l)} \leftarrow_{p_c} \mathcal{C}^{(l)}$, and $U_{k'}^{(l)}$ is uniform over each $S \in P_{k'}^{(l)}$.*

Proof of Proposition 2. It is naturally implied by Theorem 2. $\square$

Theorem 3 (restated). *An EE construction with an encoder and a cipher achieves statistical KC if the encoder is MS-IND. Specifically, with $l \geq \log 1/p_m^{\min} + 1$,*

$$I(K; C^{(l)}) \leq \frac{\text{SD}(M_0^{(l)}, M_1)(2p_m^{\min} - \text{SD}(M_0^{(l)}, M_1))}{\ln 2 \cdot p_m^{\min}(p_m^{\min} - \text{SD}(M_0^{(l)}, M_1))} = O(\text{SD}(M_0^{(l)}, M_1)).$$

where $M_1 \leftarrow_{p_m} \mathcal{M}$ and $M_0^{(l)} \leftarrow_{p_d} \mathcal{M}$.

Proof of Theorem 3. With the approach in the proof of Theorem 1, we aim to bound $I(K; C^{(l)})$ by $\text{KL}(C_{0,k}^{(l)} || C_1^{(l)})$, as $I(K; C^{(l)}) = \mathbb{E}_{K \leftarrow_{p_k} \mathcal{K}} [\text{KL}(C_{0,K}^{(l)} || C_1^{(l)})]$, where $C_{0,k}^{(l)} \leftarrow_{p_{c|k}(\cdot|k)} \mathcal{C}^{(l)}$ and $C_1^{(l)} \leftarrow_{p_c} \mathcal{C}^{(l)}$.

Further, since $\text{KL}(C_{0,k}^{(l)} || C_1^{(l)}) = \sum_{c \in \mathcal{C}^{(l)}} p_{c|k}(c|k) \log \frac{p_{c|k}(c|k)}{p_c(c)}$, our strategy focuses on constraining $\text{KL}(C_{0,k}^{(l)} || C_1^{(l)})$ by $\frac{p_{c|k}(c|k)}{p_c(c)}$.

Denoting the encoding scheme as $(\text{encode}^{(l)}, \text{decode}^{(l)})$ and the cipher as $(\text{encrypt}^{(l)}, \text{decrypt}^{(l)})$, we have

$$\begin{aligned} p_{c|k}(c|k) &= \Pr[C^{(l)} = c | K = k] \\ &= \Pr[C^{(l)} = c, S^{(l)} = \text{decrypt}_k^{(l)}(C^{(l)}), M = \text{decode}^{(l)}(S^{(l)}) | K = k] \\ &= \Pr[M = D_k^{(l)}(c)] \\ &\quad \cdot \Pr[S^{(l)} = \text{decrypt}_k^{(l)}(C^{(l)}) | M = D_k^{(l)}(c)] \\ &= p_m(D_k^{(l)}(c)) \frac{1}{|\text{encode}^{(l)}(D_k^{(l)}(c))|} \\ &= \frac{1}{2^l} \cdot \frac{p_m(D_k^{(l)}(c))}{p_d(D_k^{(l)}(c))}. \end{aligned}$$

For each $m \in \mathcal{M}$, $|p_d(m) - p_m(m)| \leq \text{SD}(M_0^{(l)}, M_1)$. Let $d = \text{SD}(M_0^{(l)}, M_1)$, thus

$$\left| \frac{p_m(m)}{p_d(m)} - 1 \right| \leq \frac{d}{p_d(m)}.$$

Let $p_m^{\min} = \min_m p_m(m)$, then $p_d(m) \geq p_m(m) - d \geq p_m^{\min} - d$, further

$$\left| \frac{p_m(m)}{p_d(m)} - 1 \right| \leq \frac{d}{p_m^{\min} - d}.$$

Thus,

$$\begin{aligned} \left| p_{c|k}(c|k) - \frac{1}{2^l} \right| &= \left| \frac{1}{2^l} \cdot \frac{p_m(D_k^{(l)}(c))}{p_d(D_k^{(l)}(c))} - \frac{1}{2^l} \right| \\ &\leq \frac{1}{2^l} \cdot \frac{d}{p_m^{\min} - d}, \end{aligned}$$

and

$$\begin{aligned} \left| p_c(c) - \frac{1}{2^l} \right| &= \left| \sum_{k \in \mathcal{K}} p_k(k) p_{c|k}(c|k) - \frac{1}{2^l} \right| \\ &\leq \sum_{k \in \mathcal{K}} p_k(k) \left| p_{c|k}(c|k) - \frac{1}{2^l} \right| \\ &\leq \frac{1}{2^l} \cdot \frac{d}{p_m^{\min} - d}. \end{aligned}$$

Denoting $U^{(l)}$ as the uniformly random variable on $\{0, 1\}^l$, then

$$\begin{aligned} \text{SD}(C_{0,k}^{(l)}, U^{(l)}) &= \frac{1}{2} \sum_{c \in \mathcal{C}^{(l)}} \left| p_{c|k}(c|k) - \frac{1}{2^l} \right| \\ &\leq \frac{1}{2} \cdot 2^l \cdot \frac{1}{2^l} \cdot \frac{d}{p_m^{\min} - d} = \frac{d}{2(p_m^{\min} - d)}, \end{aligned}$$

and

$$\text{SD}(C_1^{(l)}, U^{(l)}) \leq \frac{d}{2(p_m^{\min} - d)}.$$

This means the ciphertext $C_{0,k}^{(l)}$ and $C_0^{(l)}$ are both indistinguishable from $U^{(l)}$, and

$$\begin{aligned} \text{SD}(C_{0,k}^{(l)}, C_1^{(l)}) &\leq \text{SD}(C_{0,k}^{(l)}, U^{(l)}) + \text{SD}(C_1^{(l)}, U^{(l)}) \\ &\leq \frac{d}{p_m^{\min} - d} \end{aligned}$$

is negligible. By Theorem 1, the EE construction achieves KC.

However, the upper bound provided by Theorem 1 grows with the increase of $\frac{1}{p_k^{\min}}$. We can give a tighter one for EE constructions which is independent of p_k .

Similar to $\frac{p_m(m)}{p_d(m)}$, we can get

$$\left| \frac{p_d(m)}{p_m(m)} - 1 \right| \leq \frac{d}{p_m(m)} \leq \frac{d}{p_m^{\min}}.$$

Denoting $\epsilon_1 = \frac{d}{p_m^{\min}}$ and $\epsilon_2 = \frac{d}{p_m^{\min} - d}$, then for each m ,

$$\frac{p_d(m)}{p_m(m)} \leq 1 + \epsilon_1, \text{ and } \frac{p_m(m)}{p_d(m)} \leq 1 + \epsilon_2.$$

Thus,

$$\frac{1}{2^l} \cdot \frac{1}{1 + \epsilon_1} \leq p_{c|k}(c|k) \leq \frac{1}{2^l} \cdot (1 + \epsilon_2),$$

and

$$\frac{1}{2^l} \cdot \frac{1}{1 + \epsilon_1} \leq p_c(c) = \sum_{k \in \mathcal{K}} p_k(k) p_{c|k}(c|k) \leq \frac{1}{2^l} \cdot (1 + \epsilon_2).$$

Further,

$$\frac{p_{c|k}(c|k)}{p_c(c)} \leq (1 + \epsilon_1)(1 + \epsilon_2),$$

and

$$\log \frac{p_{c|k}(c|k)}{p_c(c)} \leq \log(1 + \epsilon_1) + \log(1 + \epsilon_2) \leq \frac{\epsilon_1 + \epsilon_2}{\ln 2}.$$

Thus,

$$\begin{aligned} \text{KL}(C_{0,k}^{(l)} || C_1^{(l)}) &= \sum_{c \in \mathcal{C}^{(l)}} p_{c|k}(c|k) \log \frac{p_{c|k}(c|k)}{p_c(c)} \\ &\leq \frac{\epsilon_1 + \epsilon_2}{\ln 2}, \end{aligned}$$

and

$$\begin{aligned} I(K; C^{(l)}) &= \mathbb{E}_{K \leftarrow_{p_k} \mathcal{K}} [\text{KL}(C_{0,k}^{(l)} || C_1^{(l)})] \\ &\leq \frac{\epsilon_1 + \epsilon_2}{\ln 2} = \frac{d(2p_m^{\min} - d)}{p_m^{\min}(p_m^{\min} - d) \ln 2} = O(d) \end{aligned}$$

is negligible. $\square$

Proposition 3 (restated). For an EE construction, if its encoder is MS-IND, then $C_{0,k}^{(l)} \stackrel{s}{=} C_1^{(l)} \stackrel{s}{=} U_{k'}^{(l)}$, where $C_{0,k}^{(l)} \leftarrow_{p_{c|k}(\cdot|k)} \mathcal{C}^{(l)}$, $C_1^{(l)} \leftarrow_{p_c} \mathcal{C}^{(l)}$, and $U_{k'}^{(l)} \leftarrow_{\S} \mathcal{C}^{(l)} = \{0, 1\}^l$. Specifically, with $l \geq \log 1/p_m^{\min} + 1$, $M_0^{(l)} \leftarrow_{p_d} \mathcal{M}$ and $M_1 \leftarrow_{p_m} \mathcal{M}$,

$$\text{SD}(C_{0,k}^{(l)}, U^{(l)}) \leq \frac{\text{SD}(M_0^{(l)}, M_1)}{2(p_m^{\min} - \text{SD}(M_0^{(l)}, M_1))}.$$

Proof of Proposition 3. From the proof of Theorem 3, it becomes evident that the ciphertext distribution is almost uniform. $\square$

Proposition 4 (restated). For an EE construction, if for each m , $|p_d(m) - p_m(m)| \leq \epsilon$, where $0 < \epsilon < p_m^{\min}$, then

$$I(K; C^{(l)}) \leq \frac{\epsilon(2p_m^{\min} - \epsilon)}{p_m^{\min}(p_m^{\min} - \epsilon) \ln 2} = O(\epsilon).$$

Proof of Proposition 4. The proof is similar to that of Theorem 3 which only uses $\text{SD}(M_0, M_1)$ to bound $|p_d(m) - p_m(m)|$, for each $m \in \mathcal{M}$. $\square$

Theorem 4 (restated). For n ciphertexts $C_1, C_2, \dots, C_n$ encrypted with the same key K (possibly under different encryption schemes), if the messages are independent and the encryption randomness is independent, then

$$I(K; C_1, C_2, \dots, C_n) \leq \sum_{i=1}^n I(K; C_i).$$

Proof of Theorem 4.

$$\begin{aligned}
& I(K; C_1, C_2, \dots, C_n) \\
&= H(C_1, C_2, \dots, C_n) - H(C_1, C_2, \dots, C_n | K) \\
&= H(C_1, C_2, \dots, C_n) - \sum_i H(C_i | K) \\
&\leq \sum_i H(C_i) - \sum_i H(C_i | K) \\
&= \sum_i I(K; C_i).
\end{aligned}$$

□

Theorem S.5. *In the ideal-cipher model where $\mathcal{C} = \mathcal{M}$ and each E_k is an independent uniformly random permutation over $\mathcal{M}$, the expected key leakage satisfies*

$$\mathbb{E}[I(K; C)] \geq \frac{|\mathcal{M}| \sum_m p_m(m)^2 - 1}{2 \ln 2 \cdot |\mathcal{M}| \cdot p_m^{\max}} \left(1 - \sum_k p_k(k)^2 \right),$$

where the expectation is over the random choice of the ideal cipher.

Proof of Theorem S.5. In the ideal cipher model, the randomness arises from the fact that each encryption function E_k is an independently and uniformly chosen permutation over $\mathcal{M}$. For any fixed set of permutations $\{E_k\}_k$, the cipher becomes deterministic.

Using the equivalence relation R_a , defined in the proof of Theorem 2, we partition the ciphertext space $\mathcal{C}$ into $|\mathcal{M}|^{|\mathcal{K}|}$ equivalence classes. For each class $S_{\mathbf{m}}$ with $\mathbf{m} = (m_{j_i})_i$, we have $D_{k_i}(c) = m_{j_i}$ for each key k_i and each $c \in S_{\mathbf{m}}$.

Since the cipher is deterministic and $E_{k_i}(m_{j_i}) = c$, we have:

$$\begin{aligned}
p_{c|k}(c | k_i) &= p_m(m_{j_i}) p_{c|m, k}(c | m_{j_i}, k_i) \\
&= p_m(m_{j_i}).
\end{aligned}$$

Thus $I(K; C)$ can be rewritten as:

$$\begin{aligned}
I(K; C) &= \sum_c \sum_i p_k(k_i) p_{c|k}(c | k_i) \log \frac{p_{c|k}(c | k_i)}{p_c(c)} \\
&= \sum_{\mathbf{m}} \sum_{c \in S_{\mathbf{m}}} \sum_i p_k(k_i) p_{c|k}(c | k_i) \log \frac{p_{c|k}(c | k_i)}{\sum_{i'} p_k(k_{i'}) p_{c|k}(c | k_{i'})} \\
&= \sum_{\mathbf{m}} |S_{\mathbf{m}}| \sum_i p_k(k_i) p_m(m_{j_i}) \log \frac{p_m(m_{j_i})}{\sum_{i'} p_k(k_{i'}) p_m(m_{j_{i'}})}.
\end{aligned}$$

Since the randomness of the ideal cipher model only affects the value of $|S_{\mathbf{m}}|$, so

$$\begin{aligned}
\mathbb{E}[I(K; C)] &= \sum_{\mathbf{m}} \mathbb{E}[|S_{\mathbf{m}}|] \sum_i p_k(k_i) p_m(m_{j_i}) \\
&\quad \cdot \log \frac{p_m(m_{j_i})}{\sum_{i'} p_k(k_{i'}) p_m(m_{j_{i'}})}.
\end{aligned}$$

For the former part, noticed that $|S_{\mathbf{m}}|$ is binary, thus

$$\begin{aligned}
\mathbb{E}[|S_{\mathbf{m}}|] &= \Pr[|S_{\mathbf{m}}| = 1] \\
&= \Pr[E_{k_i}(m_{j_i}) = E_{k_1}(m^{(1)}), \forall i \geq 2] \\
&= \frac{1}{|\mathcal{M}|^{|\mathcal{K}|-1}}.
\end{aligned}$$

For the latter part, it can be transformed into

$$\sum_i p_k(k_i) p_m(m_{j_i}) \log \frac{p_m(m_{j_i})}{\sum_{i'} p_k(k_{i'}) p_m(m_{j_{i'}})}$$

$$\begin{aligned}
&= \sum_i p_k(k_i) p_m(m_{j_i}) \log p_m(m_{j_i}) \\
&\quad - \sum_i p_k(k_i) p_m(m_{j_i}) \log \left(\sum_{i'} p_k(k_{i'}) p_m(m_{j_{i'}}) \right) \\
&= \sum_i \lambda_i f(x_i) - f(\bar{x}),
\end{aligned}$$

where $\lambda_i = p_k(k_i)$, $x_i = p_m(m_{j_i})$, $\bar{x} = \sum_i \lambda_i x_i$, $f(x) = x \log x$. According to Taylor's expansion,

$$\begin{aligned}
\sum_i \lambda_i f(x_i) - f(\bar{x}) &= \sum_i \lambda_i \left(f(\bar{x}) + f'(\bar{x})(x_i - \bar{x}) + \frac{1}{2} f''(\xi_i)(x_i - \bar{x})^2 \right) - f(\bar{x}) \\
&= f(\bar{x}) + f'(\bar{x}) \sum_i \lambda_i (x_i - \bar{x}) + \frac{1}{2} \sum_i \lambda_i f''(\xi_i)(x_i - \bar{x})^2 - f(\bar{x}) \\
&= \frac{1}{2} \sum_i \lambda_i f''(\xi_i)(x_i - \bar{x})^2,
\end{aligned}$$

where ξ_i lies between x_i and $\bar{x}$. It's clear that both of them can be upper-bounded by $p_m^{\max}$. So $f''(\xi_i)$ has the following lower bound:

$$f''(\xi_i) = \frac{1}{\xi_i \ln 2} \geq \frac{1}{p_m^{\max} \ln 2}.$$

Based on the above results, we have

$$\mathbb{E}[I(K; C)] \geq \frac{\sum_m \sum_i p_k(k_i) (p_m(m_{j_i}) - \sum_{i'} p_k(k_{i'}) p_m(m_{j_{i'}}))^2}{2 \ln 2 |\mathcal{M}|^{|\mathcal{K}|-1} p_m^{\max}}.$$

For the numerator part,

$$\begin{aligned}
&\sum_m \sum_i p_k(k_i) \left(p_m(m_{j_i}) - \sum_{i'} p_k(k_{i'}) p_m(m_{j_{i'}}) \right)^2 \\
&= \sum_m \left(\sum_i p_k(k_i) p_m(m_{j_i})^2 - \left(\sum_i p_k(k_i) p_m(m_{j_i}) \right)^2 \right) \\
&= \sum_m \left(\sum_i (p_k(k_i) - p_k(k_i)^2) p_m(m_{j_i})^2 \right. \\
&\quad \left. - \sum_{i \neq i'} p_k(k_i) p_k(k_{i'}) p_m(m_{j_i}) p_m(m_{j_{i'}}) \right) \\
&= \sum_i (p_k(k_i) - p_k(k_i)^2) \sum_m p_m(m_{j_i})^2 \\
&\quad - \sum_{i \neq i'} p_k(k_i) p_k(k_{i'}) \sum_m p_m(m_{j_i}) p_m(m_{j_{i'}}) \\
&= \left(1 - \sum_i p_k(k_i)^2 \right) |\mathcal{M}|^{|\mathcal{K}|-1} \sum_m p_m(m)^2 \\
&\quad - \left(1 - \sum_i p_k(k_i)^2 \right) |\mathcal{M}|^{|\mathcal{K}|-2} \left(\sum_m p_m(m) \right)^2 \\
&= |\mathcal{M}|^{|\mathcal{K}|-2} \left(1 - \sum_i p_k(k_i)^2 \right) \left(|\mathcal{M}| \sum_m p_m(m)^2 - 1 \right).
\end{aligned}$$

Therefore,

$$\mathbb{E}[I(K; C)] \geq \frac{|\mathcal{M}| \sum_m p_m(m)^2 - 1}{2 \ln 2 |\mathcal{M}| p_m^{\max}} \left(1 - \sum_k p_k(k)^2 \right).$$

□

S.4. Counterexamples to the converses of Theorems 2 and 3

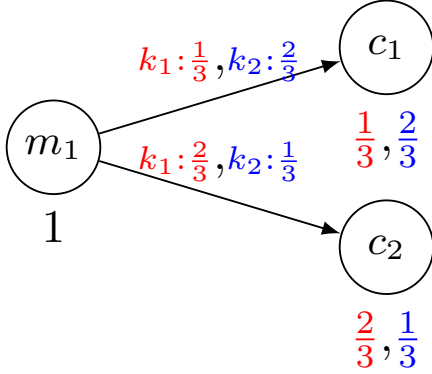


Figure S.2: Counterexample to the converse of Theorem 2

The converse of Proposition 1 does not hold in general. As illustrated in Fig. S.2, a simple counterexample arises when, for the same message, E_k assigns different probabilities to different ciphertexts, resulting in distinct ciphertext distributions.

The converse of Theorem 3 does not hold. For certain ciphers with small key sizes, the ciphertext distribution $p_{c|k}(\cdot|k)$ may remain consistent across different keys k , even when there are significant differences between p_d and p_m . Fig. S.3 presents a trivial counterexample, which naturally leads to more complex ones.

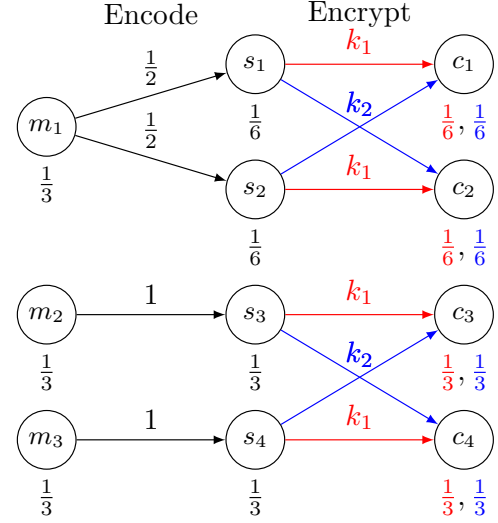


Figure S.3: Counterexample to the converse of Theorem 3: $p_m = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, and $p_d = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$

S.5. Implications of Mutual Information for Guessing-Based Security Notions

n -GS Game $\text{GS}_{p_{x,y}}^{\mathcal{A}_n}$
$X, Y \leftarrow p_{x,y} \mathcal{X} \times \mathcal{Y}$
$S \leftarrow \mathcal{A}_n(Y)$, where $ S = n$
return $X \in S$

Figure S.4: n -Guessing Security Game (GS): guessing X given Y with n attempts

Security notions defined via mutual information inherently bound guessing-based security. We formalize this connection for general random variables X and Y , rather than restricting to specific cryptographic instantiations such as keys and ciphertexts.

Consider the n -guessing security game $\text{GS}_{p_{x,y}}^{\mathcal{A}_n}$ in Fig. S.4. Given Y , the adversary $\mathcal{A}_n$ outputs a set S of n candidate values for X . The advantage of $\mathcal{A}_n$ is defined as

$$\text{Adv}_{p_{x,y}}^{n\text{-gs}}(\mathcal{A}_n) = \Pr[\text{GS}_{p_{x,y}}^{\mathcal{A}_n} \Rightarrow \text{True}],$$

and the maximum advantage is

$$\text{Adv}_{p_{x,y}}^{n\text{-gs}} = \max_{\mathcal{A}_n} \text{Adv}_{p_{x,y}}^{n\text{-gs}}(\mathcal{A}_n).$$

The optimal strategy of $\mathcal{A}_n$ is to output the n most likely values under the posterior distribution $p_{x|y}(\cdot|y)$. Let $S_o(y, n)$ denote this optimal set for a given y , and let $S_o(n)$ denote the set of n most likely values under the prior $p_x(\cdot)$.

Define

$$\text{Guess}_n(X|Y) = \Pr[X \in S_o(Y, n)], \quad \text{Guess}_n(X) = \Pr[X \in S_o(n)].$$

The following theorem quantifies the gain from observing Y .

Theorem S.6. For any random variables X, Y and any integer $n \geq 1$,

$$\mathbf{Adv}_{p_{x,y}}^{n\text{-gs}} = \text{Guess}_n(X|Y) \leq \text{Guess}_n(X) + \sqrt{\frac{\ln 2}{2} I(X; Y)}.$$

Proof. By Pinsker's inequality, for any y ,

$$\text{KL}(p_{x|y}(\cdot|y) \| p_x(\cdot)) \geq \frac{2}{\ln 2} \text{SD}(p_{x|y}(\cdot|y), p_x(\cdot))^2.$$

Hence,

$$\text{SD}(p_{x|y}(\cdot|y), p_x(\cdot)) \leq \sqrt{\frac{\ln 2}{2} \text{KL}(p_{x|y}(\cdot|y) \| p_x(\cdot))}.$$

Moreover, by the definition of statistical distance,

$$\sum_{x \in S_o(y,n)} (p_{x|y}(x|y) - p_x(x)) \leq \text{SD}(p_{x|y}(\cdot|y), p_x(\cdot)).$$

Combining the above, we obtain

$$\begin{aligned} & \text{Guess}_n(X|Y) \\ &= \sum_y p_y(y) \sum_{x \in S_o(y,n)} p_{x|y}(x|y) \\ &\leq \sum_y p_y(y) \sum_{x \in S_o(y,n)} p_x(x) + \sum_y p_y(y) \sqrt{\frac{\ln 2}{2} \text{KL}(p_{x|y}(\cdot|y) \| p_x(\cdot))} \\ &\leq \sum_{x \in S_o(n)} p_x(x) + \sqrt{\frac{\ln 2}{2} \sum_y p_y(y) \text{KL}(p_{x|y}(\cdot|y) \| p_x(\cdot))} \\ &= \text{Guess}_n(X) + \sqrt{\frac{\ln 2}{2} I(X; Y)}, \end{aligned}$$

where the second inequality follows from Jensen's inequality. $\square$

We next extend the result to partial-information recovery. Many guessing-based notions (e.g., TDSS) aim to recover a function of X rather than X itself.

Corollary S.3. For any random variables X, Y and any function $f : \mathcal{X} \rightarrow \mathcal{Z}$, let $\omega_f = \max_z \Pr[f(X) = z]$. Then

$$\text{Guess}_1(f(X)|Y) \leq \omega_f + \sqrt{\frac{\ln 2}{2} I(X; Y)}.$$

Proof. Applying Theorem S.6 with $n = 1$ and replacing X by $f(X)$ gives

$$\text{Guess}_1(f(X)|Y) \leq \omega_f + \sqrt{\frac{\ln 2}{2} I(f(X); Y)}.$$

Since $Y \rightarrow X \rightarrow f(X)$ forms a Markov chain, the data processing inequality implies $I(f(X); Y) \leq I(X; Y)$. $\square$

Theorem S.6 and Corollary S.3 together show that statistical key confidentiality, expressed via mutual information, automatically bounds all guessing-based key recovery attacks, including those targeting arbitrary partial information and allowing multiple guesses.

S.6. Our OE_U for Uniform Key or Message Distributions

For simplicity, we denote the cumulative distribution functions of keys and messages as F_k and F_m , respectively. Thus, $F_k(j) = \sum_{j'=1}^j p_k(k_{j'})$ and $F_m(i) = \sum_{i'=1}^i p_m(m_{i'})$. Formally, OE_U has the following ciphertext intervals on $[0, 1)$. The corresponding discretized intervals on $\{0, 1\}^l$ can be naturally obtained.

In Case K: for a uniform key distribution,

$$\mathcal{C}(m_i, k_j) = [F_m(i-1) + (j-1)F_m(1), F_m(i) + (j-1)F_m(1));$$

for a uniform message distribution,

$$\mathcal{C}(m_i, k_j) = \left[\frac{F_m(1)}{F_k(1)} F_k(j-1) + (i-1)F_m(1), \frac{F_m(1)}{F_k(1)} F_k(j) + (i-1)F_m(1) \right).$$

In Case M: for a uniform key distribution,

$$\mathcal{C}(m_i, k_j) = \left[\frac{F_k(1)}{F_m(1)} F_m(i-1) + (j-1)F_k(1), \frac{F_k(1)}{F_m(1)} F_m(i) + (j-1)F_k(1) \right);$$

for a uniform message distribution,

$$\mathcal{C}(m_i, k_j) = [F_k(j-1) + (i-1)F_k(1), F_k(j) + (i-1)F_k(1)].$$

S.7. Proofs in Section 5

Lemma 1 (restated). *In our design, $\mathcal{C}^{(l)}(m, k) \cap \mathcal{C}^{(l)}(m', k) = \emptyset$, for each $m \neq m'$ and k .*

Proof of Lemma 1 for Case K. In our assignment, $[l(m_i, k_i), u(m_i, k_i))$ for $i = 1, 2, \dots, N_m$ are non-overlapping intervals on $\mathbb{Z}$. Should any overlap occur within $\mathcal{C}^{(l)}$, meaning $\mathcal{C}^{(l)}(m_i, k) \cap \mathcal{C}^{(l)}(m_{i'}, k_j) \neq \emptyset$ for some $i < i'$, it implies that $u(m_{i'}, k_j) > l(m_i, k_j) + N_c$. Consequently, this would suggest $u(m_{N_m}, k_j) > l(m_1, k_j) + N_c$. However, our method ensures that $u(m_{N_m}, k_j) \leq l(m_1, k_j) + N_c$, affirming the validity of the lemma.

In the following, we prove $u(m_{N_m}, k_j) \leq l(m_1, k_j) + N_c$ for each j , recursively.

For $j = 1$: the proposition obviously holds.

For $j > 1$: Assume the proposition holds for $j-1$. Let $i_0 = \max\{i \mid l(m_i, k_j) = u(m_i, k_{j-1})\}$. Note that such i exists, since $i = 1$ meets the condition. Thus, $l(m_i, k_j) = u(m_{i-1}, k_j) = l(m_{i-1}, k_j) + n(m_{i-1}) \frac{p_k(k_j)}{p_k(k_1)}$, for $i = i_0 + 1, \dots, N_m$. Summarizing $l(m_i, k_j) - l(m_{i-1}, k_j)$ over i yields

$$l(m_{N_m}, k_j) - l(m_{i_0}, k_j) = \sum_{i=i_0+1}^{N_m} \lfloor n(m_{i-1}) \frac{p_k(k_j)}{p_k(k_1)} \rfloor.$$

Moreover,

$$\begin{aligned} & u(m_{N_m}, k_j) - l(m_{i_0}, k_j) \\ &= \sum_{i=i_0}^{N_m} \lfloor n(m_i) \frac{p_k(k_j)}{p_k(k_1)} \rfloor \leq \sum_{i=i_0}^{N_m} \lfloor n(m_i) \frac{p_k(k_{j-1})}{p_k(k_1)} \rfloor \\ &\leq \lfloor n(m_1) \frac{p_k(k_{j-1})}{p_k(k_1)} \rfloor + \sum_{i=i_0+1}^{N_m} \lfloor n(m_i) \frac{p_k(k_{j-1})}{p_k(k_1)} \rfloor \\ &= u(m_1, k_{j-1}) - l(m_1, k_{j-1}) + \sum_{i=i_0+1}^{N_m} (u(m_i, k_{j-1}) - l(m_i, k_{j-1})). \end{aligned}$$

Applying $l(m_{i+1}, k_j) = \max\{u(m_{i+1}, k_{j-1}), u(m_i, k_j)\} \geq u(m_i, k_j)$ yields

$$\begin{aligned} & u(m_{N_m}, k_j) - l(m_{i_0}, k_j) \\ &\leq u(m_1, k_{j-1}) - l(m_1, k_{j-1}) + \sum_{i=i_0+1}^{N_m} (u(m_i, k_{j-1}) - u(m_{i-1}, k_{j-1})) \\ &= u(m_1, k_{j-1}) - l(m_1, k_{j-1}) + u(m_{N_m}, k_{j-1}) - u(m_{i_0}, k_{j-1}). \end{aligned}$$

By the inductive hypothesis $u(m_{N_m}, k_{j-1}) \leq l(m_1, k_{j-1}) + N_c$,

$$u(m_{N_m}, k_j) - l(m_{i_0}, k_j) \leq u(m_1, k_{j-1}) + N_c - u(m_{i_0}, k_{j-1}).$$

Further, since $l(m_{i_0}, k_j) = u(m_{i_0}, k_{j-1})$,

$$u(m_{N_m}, k_j) \leq u(m_1, k_{j-1}) + N_c = l(m_1, k_j) + N_c.$$

Namely, the proposition holds for j .

The proposition implies that the intervals $[l(m_i, k_j), u(m_i, k_j))$ for $i = 1, 2, \dots, N_m$ are non-overlapping on the space $[l(m_1, k_j), l(m_1, k_j) + N_c)$. Therefore, the corresponding intervals on $\mathcal{C}$ are also non-overlapping. $\square$

Lemma 2 (restated). *In our design, $\mathcal{C}^{(l)}(m, k) \cap \mathcal{C}^{(l)}(m, k') = \emptyset$, for each m and $k \neq k'$.*

Proof of Lemma 2 for Case K. Similar to the proof of Lemma 1, we only need to prove that $u(m_i, k_{N_k}) \leq l(m_i, k_1) + N_c$ for each i .

In the following, we prove this recursively.

For $i = 1$: for $j = 1, 2, \dots, N_k - 1$,

$$\begin{aligned} u(m_1, k_{j+1}) &= l(m_1, k_{j+1}) + \lfloor n(m_1) \frac{p_k(k_{j+1})}{p_k(k_1)} \rfloor \\ &\leq u(m_1, k_j) + n(m_1) \frac{p_k(k_{j+1})}{p_k(k_1)}. \end{aligned}$$

Iterating over j gets

$$\begin{aligned} u(m_1, k_{N_k}) &\leq u(m_1, k_1) + \sum_{j=2}^{N_k} n(m_1) \frac{p_k(k_j)}{p_k(k_1)} = l(m_1, k_1) + \sum_{j=1}^{N_k} n(m_1) \frac{p_k(k_j)}{p_k(k_1)} \\ &= l(m_1, k_1) + \frac{n(m_1)}{p_k(k_1)} \leq l(m_1, k_1) + N_c. \end{aligned}$$

Namely, the proposition holds for $i = 1$.

For $i > 1$: Assume the proposition holds for $i - 1$, i.e., $u(m_{i-1}, k_{N_k}) \leq l(m_{i-1}, k_1) + N_c$. Let $j_0 = \max\{j \mid l(m_i, k_j) = u(m_i, k_{j-1})\}$. Note such j exists, because $j = 1$ satisfies the condition. Then,

$$\begin{aligned} l(m_i, k_j) &= u(m_i, k_{j-1}) \\ &= l(m_i, k_{j-1}) + n(m_i) \frac{p_k(k_{j-1})}{p_k(k_1)}, \quad j = j_0 + 1, \dots, N_k. \end{aligned}$$

Iterating over j yields

$$l(m_i, k_{N_k}) = l(m_i, k_{j_0}) + \sum_{j=j_0+1}^{N_k} \lfloor n(m_i) \frac{p_k(k_{j-1})}{p_k(k_1)} \rfloor.$$

Further,

$$\begin{aligned} &u(m_i, k_{N_k}) - l(m_i, k_{j_0}) \\ &= \sum_{j=j_0}^{N_k} \lfloor n(m_i) \frac{p_k(k_j)}{p_k(k_1)} \rfloor \leq \sum_{j=j_0}^{N_k} \lfloor n(m_{i-1}) \frac{p_k(k_j)}{p_k(k_1)} \rfloor \\ &\leq \lfloor n(m_{i-1}) \frac{p_k(k_1)}{p_k(k_1)} \rfloor + \sum_{j=j_0+1}^{N_k} \lfloor n(m_{i-1}) \frac{p_k(k_j)}{p_k(k_1)} \rfloor \\ &= u(m_{i-1}, k_1) - l(m_{i-1}, k_1) + \sum_{j=j_0+1}^{N_k} (u(m_{i-1}, k_j) - l(m_{i-1}, k_j)). \end{aligned}$$

By $l(m_i, k_{j+1}) = \max\{u(m_i, k_j), u(m_{i-1}, k_{j+1})\} \geq u(m_i, k_j)$,

$$\begin{aligned} &u(m_i, k_{N_k}) - l(m_i, k_{j_0}) \\ &\leq u(m_{i-1}, k_1) - l(m_{i-1}, k_1) + \sum_{j=j_0+1}^{N_k} (u(m_{i-1}, k_j) - u(m_{i-1}, k_{j-1})) \\ &= u(m_{i-1}, k_1) - l(m_{i-1}, k_1) + u(m_{i-1}, k_{N_k}) - u(m_{i-1}, k_{j_0}). \end{aligned}$$

With the inductive assumption $u(m_{i-1}, k_{N_k}) \leq l(m_{i-1}, k_1) + N_c$,

$$u(m_i, k_{N_k}) - l(m_i, k_{j_0}) \leq u(m_{i-1}, k_1) + N_c - u(m_{i-1}, k_{j_0})$$

Further, since $l(m_i, k_{j_0}) = u(m_i, k_{j_0-1})$, we have

$$u(m_i, k_{N_k}) \leq u(m_{i-1}, k_1) + N_c = l(m_i, k_1) + N_c.$$

Namely, the proposition holds for i .

The proposition naturally implies that the intervals $\mathcal{C}^{(l)}(m_i, k_j) = [l(m_i, k_j), u(m_i, k_j))$ for $j = 1, 2, \dots, N_k$ are non-overlapping on $\mathcal{C}^{(l)}$. In other words, the lemma holds. $\square$

Theorem 5 (restated). *For our scheme OE,*

$$\text{Adv}_{\text{OE}, p_m, p_k}^{\text{mrs}} < p_{\max} \left(1 + \frac{t}{2^l - t} \right) = p_{\max} + O(2^{-l}),$$

where t is a constant $\frac{2p_{\max}}{p_m^{\min} p_k^{\min}}$.

Proof of Theorem 5. We prove the bound for Case K, where $p_{\max} = p_k^{\max} = p_k(k_1)$. The proof for Case M is symmetric by exchanging the roles of messages and keys.

As discussed before,

$$p_{m,c}(m, c) = \sum_{k: D_k(c)=m} \frac{p_m(m)p_k(k)}{n(m, k)}.$$

Thus,

$$\begin{aligned} \text{Adv}_{\text{OE}, p_m, p_k}^{\text{mrs}} &= \sum_{c \in \mathcal{C}} \max_m p_{m,c}(m, c) \\ &= \sum_{c \in \mathcal{C}} \max_m \sum_{k: D_k(c)=m} \frac{p_m(m)p_k(k)}{n(m, k)}. \end{aligned}$$

Note that

$$\begin{aligned} n(m, k) &= \lfloor n(m) \frac{p_k(k)}{p_k(k_1)} \rfloor > n(m) \frac{p_k(k)}{p_k(k_1)} - 1 \\ &> (p_m(m)N_c - 1) \frac{p_k(k)}{p_k(k_1)} - 1 > p_m(m)N_c \frac{p_k(k)}{p_k(k_1)} - 2. \end{aligned}$$

Therefore,

$$\begin{aligned} \frac{p_m(m)p_k(k)}{n(m, k)} &< \frac{p_m(m)p_k(k)}{p_m(m)N_c \frac{p_k(k)}{p_k(k_1)} - 2} = \frac{p_k(k_1)}{N_c - \frac{2p_k(k_1)}{p_m(m)p_k(k)}} \\ &\leq \frac{p_k(k_1)}{N_c - \frac{2p_k(k_1)}{p_m(mN_m)p_k(k_{N_k})}}. \end{aligned}$$

Further, by Lemmas 1 and 2, $|\{k \mid D_k(c) = m\}| \leq 1$, and

$$\begin{aligned} \text{Adv}_{\text{OE}, p_m, p_k}^{\text{mrs}} &< N_c \cdot \frac{p_k(k_1)}{N_c - \frac{2p_k(k_1)}{p_m(mN_m)p_k(k_{N_k})}} \\ &= p_k(k_1) + \frac{\frac{2p_k(k_1)^2}{p_m(mN_m)p_k(k_{N_k})}}{N_c - \frac{2p_k(k_1)}{p_m(mN_m)p_k(k_{N_k})}}. \end{aligned}$$

Note $N_c = 2^l$, thus $\text{Adv}_{\text{OE}, p_m, p_k}^{\text{mrs}} = p_k(k_1) + O(\frac{1}{2^l})$. In Case M, the same argument with m and k interchanged gives $\text{Adv}_{\text{OE}, p_m, p_k}^{\text{mrs}} = p_m(m_1) + O(2^{-l})$. Therefore, in both cases,

$$\text{Adv}_{\text{OE}, p_m, p_k}^{\text{mrs}} < p_{\max} \left(1 + \frac{t}{2^l - t} \right) = p_{\max} + O(2^{-l}),$$

where $t = \frac{2p_{\max}}{p_m^{\min} p_k^{\min}}$ is a constant. $\square$

Theorem 6 (restated). *For the scheme OE' constructed via piecewise uniform approximation with parameter e^ϵ , we have*

$$\text{Adv}_{\text{OE}', p_m, p_k}^{\text{mrs}} \leq e^{3\epsilon} p_{\max} + O(2^{-l}),$$

with computational complexity $O(n_{\text{KBin}} \cdot n_{\text{MBin}})$, where n_{KBin} and n_{MBin} are the numbers of key and message bins, respectively.

Proof of Theorem 6. Using the continuous variants OE_c and OE'_c , we have

$$|\mathcal{C}(m, k)| = \frac{p_m(m)p_k(k)}{p_{\max}}, \quad |\mathcal{C}'(m, k)| = \frac{p'_m(m)p'_k(k)}{p'_{\max}}.$$

By construction, for all m and k , the piecewise uniform approximations satisfy

$$e^{-\epsilon} \leq \frac{p'_m(m)}{p_m(m)} \leq e^\epsilon, \quad e^{-\epsilon} \leq \frac{p'_k(k)}{p_k(k)} \leq e^\epsilon, \quad e^{-\epsilon} \leq \frac{p'_{\max}}{p_{\max}} \leq e^\epsilon.$$

Therefore, the ratio $\frac{|\mathcal{C}(m, k)|}{|\mathcal{C}'(m, k)|} \leq e^{3\epsilon}$.

Then, under p_m and p_k , it holds that

$$\begin{aligned} \text{Adv}_{\text{OE}'_c, p_m, p_k}^{\text{mrs}} &= \int_c \max_m p'_{m,c}(m, c) dc \\ &= \int_c \max_m \frac{p_m(m)p_k(k)}{|\mathcal{C}'(m, k)|} dc \\ &\leq \int_c \max_m \frac{p_m(m)p_k(k)}{|\mathcal{C}(m, k)|} e^{3\epsilon} dc \\ &= e^{3\epsilon} \cdot \text{Adv}_{\text{OE}_c, p_m, p_k}^{\text{mrs}} = e^{3\epsilon} p_{\max}. \end{aligned}$$

Hence,

$$\text{Adv}_{\text{OE}', p_m, p_k}^{\text{mrs}} \leq e^{3\epsilon} p_{\max} + O(2^{-l}).$$

Within each bin, ciphertext intervals have the same size for all messages and keys, and their endpoints can be computed as in OE_U , requiring only $O(1)$ time. Across the entire space, OE' only needs to store $l(m, k)$ and $n(m, k)$ for each representative (m, k) pair at the beginning of each bin. The ciphertext intervals for other (m, k) pairs can then be derived in $O(1)$ time using these values. Therefore, the overall complexity of OE' is $O(n_{\text{KBin}} \cdot n_{\text{MBin}})$. $\square$

Proposition 6 (restated). For $\mathcal{M} = \{a, b\}$ with $p_m = (\frac{1}{2}, \frac{1}{2})$ and any $\mathcal{K}$ such that $p_k^{\max} \leq \frac{1}{2}$ and for any subsets $\mathcal{K}_1 \neq \mathcal{K}_2$ with $p_k(\mathcal{K}_1) \neq p_k(\mathcal{K}_2)$, it holds that

$$\inf_{\text{EE} \in \Omega_{\text{EE}}} \text{Adv}_{\text{EE}, p_m, p_k}^{\text{mrs}} > p_{\max} = \frac{1}{2}.$$

Proof of Proposition 6. Let $n_m = n(m, k)$ as in EE constructions, $n(m, k)$ is independent of k . Without loss of generality, assume that $n_a \geq n_b$. Then

$$\begin{aligned} \text{Adv}_{\text{EE}, p_m, p_k}^{\text{mrs}} &= \sum_{c \in \mathcal{C}} \max_m \left\{ \sum_{k: D_k(c)=m} \frac{p_k(k)p_m(m)}{n(m, k)} \right\} \\ &= \sum_{c \in \mathcal{C}} \max \left\{ \sum_{k: D_k(c)=a} \frac{p_k(k)p_m(a)}{n(a, k)}, \sum_{k: D_k(c)=b} \frac{p_k(k)p_m(b)}{n(b, k)} \right\} \\ &= \sum_{c \in \mathcal{C}} \max \left\{ \frac{p_m(a)}{n_a} \sum_{k: D_k(c)=a} p_k(k), \frac{p_m(b)}{n_b} \sum_{k: D_k(c)=b} p_k(k) \right\}. \end{aligned}$$

Note that each term of the summation is related to its decryption result under each key. Define $f: \mathcal{C} \rightarrow \mathcal{M}^{N_k}$ such that $f(c) = (D_{k_i}(c))_{i=1}^{N_k}$. Using f , the ciphertext space is partitioned into distinct equivalence classes $\mathcal{C}_x$ that for each $x \in \mathcal{M}^{N_k}$, $\mathcal{C}_x = \{c \mid f(c) = x\}$. Further, by $g(x) = \sum_{i: x_i=a} p_k(k_i)$, we have

$$\begin{aligned} \text{Adv}_{\text{EE}, p_m, p_k}^{\text{mrs}} &= \sum_{c \in \mathcal{C}} \max \left\{ \frac{p_m(a)}{n_a} \sum_{k: D_k(c)=a} p_k(k) \frac{p_m(b)}{n_b} \sum_{k: D_k(c)=b} p_k(k) \right\} \\ &= \sum_{x \in \mathcal{M}^{N_k}} \sum_{c \in \mathcal{C}_x} \max \left\{ \frac{p_m(a)}{n_a} \sum_{k: x_i=a} p_k(k) \frac{p_m(b)}{n_b} \sum_{k: x_i=b} p_k(k) \right\} \\ &= \sum_{x \in \mathcal{M}^{N_k}} |\mathcal{C}_x| \cdot \max \left\{ \frac{p_m(a)}{n_a} g(x), \frac{p_m(b)}{n_b} (1 - g(x)) \right\} \end{aligned}$$

$$\begin{aligned}
&= \sum_{x \in \mathcal{M}^{N_k}} |\mathcal{C}_x| \cdot \left(\frac{p_m(a)}{n_a} g(x) + \frac{p_m(b)}{n_b} (1 - g(x)) + \left| \frac{p_m(a)}{n_a} g(x) - \frac{p_m(b)}{n_b} (1 - g(x)) \right| \right) \\
&= \frac{1}{2} + \frac{1}{2} \sum_{x \in \mathcal{M}^{N_k}} |\mathcal{C}_x| \cdot \left| \left(\frac{p_m(a)}{n_a} + \frac{p_m(b)}{n_b} \right) g(x) - \frac{p_m(b)}{n_b} \right| \\
&= \frac{1}{2} + \frac{1}{2} \sum_{x \in \mathcal{M}^{N_k}} |\mathcal{C}_x| \cdot |(\alpha + \beta) g(x) - \beta|,
\end{aligned}$$

here $\alpha = \frac{p_m(a)}{n_a} = \frac{1}{2n_a}$, $\beta = \frac{p_m(b)}{n_b} = \frac{1}{2n_b}$. Notice that $\alpha \leq \beta$ and

$$\alpha + \beta = \frac{1}{2n_a} + \frac{1}{2n_b} \geq \frac{2}{n_a + n_b} = \frac{2}{N_c}.$$

Denote $\mathcal{T} = \{g(x) \mid x \in \mathcal{M}^{N_k}\}$. Since $p_k(\mathcal{K}_1) \neq p_k(\mathcal{K}_2)$ for any subsets $\mathcal{K}_1 \neq \mathcal{K}_2$, $\mathcal{T}$ consists of $|\mathcal{M}^{N_k}|$ distinct elements. Assume that the elements in $\mathcal{T}$ are arranged in ascending order as $\{t_1, t_2, \dots, t_{N_t}\}$. Further, each $t_i \in \mathcal{T}$ corresponds to a unique element in $\mathcal{M}^{N_k}$, denoted by x_i . As a simple observation, $x_0 = (b, b, \dots, b)$ and $t_0 = 0$, while $x_{N_t} = (a, a, \dots, a)$ and $t_{N_t} = 1$. Denoting $\delta = \min_{t, t' \in \mathcal{T}} |t - t'|$, we provide a lower bound of $\text{Adv}_{\text{EE}, p_m, p_k}^{\text{mrs}}$ based on δ . We discuss the bound in two cases: 1) $n_a \leq t_{N_t-1} N_c$ and 2) $n_a > t_{N_t-1} N_c$.

Case 1: $n_a \leq t_{N_t-1} N_c$:

Define i by $t_i < n_a/N_c \leq t_{i+1}$, and let $\mathcal{S} = \mathcal{M}^{N_k} \setminus \{x_i, x_{i+1}\}$. Since $\frac{\beta}{\alpha+\beta} = n_a/N_c \in (t_i, t_{i+1}]$, we have $(\alpha + \beta)t_i < \beta$ and $(\alpha + \beta)t_{i+1} \geq \beta$.

If $|\mathcal{C}_{x_i}| + |\mathcal{C}_{x_{i+1}}| < 3N_c/4$: $\sum_{x \in \mathcal{S}} |\mathcal{C}_x| \geq N_c/4$. For $j > i+1$, $|(\alpha + \beta)g(x_j) - \beta| = (\alpha + \beta)t_j - \beta \geq (\alpha + \beta)(t_{i+1} + \delta) - \beta \geq (\alpha + \beta)\delta \geq 2\delta/N_c$. Similarly, for $j < i$, $|(\alpha + \beta)g(x_j) - \beta| = \beta - (\alpha + \beta)t_j \geq \beta - (\alpha + \beta)(t_i - \delta) \geq (\alpha + \beta)\delta \geq 2\delta/N_c$. Combining the inequalities together yields

$$\begin{aligned}
\text{Adv}_{\text{EE}, p_m, p_k}^{\text{mrs}} &\geq \frac{1}{2} + \frac{1}{2} \sum_{x \in \mathcal{M}^{N_k}} |\mathcal{C}_x| \cdot |(\alpha + \beta)g(x) - \beta| \\
&\geq \frac{1}{2} + \frac{1}{2} \sum_{x \in \mathcal{S}} |\mathcal{C}_x| \cdot |(\alpha + \beta)g(x) - \beta| \\
&\geq \frac{1}{2} + \frac{1}{2} \cdot \frac{2\delta}{N_c} \sum_{x \in \mathcal{T}} |\mathcal{C}_x| \\
&\geq \frac{1}{2} + \frac{\delta}{4}.
\end{aligned}$$

If $|\mathcal{C}_{x_i}| + |\mathcal{C}_{x_{i+1}}| \geq 3N_c/4$: Since $t_i < n_a/N_c \leq t_{N_t-1}$, we have $i < N_t - 1$. Besides, $t_{i+1} \geq n_a/N_c \geq \frac{1}{2} > t_1$ implies $i > 1$. Note that for $x \in \mathcal{M}^{N_k} \setminus \{x_0, x_{N_t}\}$, $|\mathcal{C}_x| \leq N_c/2$ (otherwise, there exist j, j' such that $x_j = a$ and $x_{j'} = b$, and further $n_a > |\mathcal{C}_x| > N_c/2$ and $n_b > |\mathcal{C}_x| > N_c/2$, leading to contradiction). Therefore, $|\mathcal{C}_{x_i}| \geq 3N_c/4 - |\mathcal{C}_{x_{i+1}}| \geq N_c/4$. Similarly, $|\mathcal{C}_{x_{i+1}}| \geq N_c/4$. Thus,

$$\begin{aligned}
\text{Adv}_{\text{EE}, p_m, p_k}^{\text{mrs}} &\geq \frac{1}{2} + \frac{1}{2} \sum_{x \in \mathcal{M}^{N_k}} |\mathcal{C}_x| \cdot |(\alpha + \beta)g(x) - \beta| \\
&\geq \frac{1}{2} + \frac{1}{2} |\mathcal{C}_{x_i}| \cdot |(\alpha + \beta)g(x_i) - \beta| \\
&\quad + \frac{1}{2} |\mathcal{C}_{x_{i+1}}| \cdot |(\alpha + \beta)g(x_{i+1}) - \beta| \\
&\geq \frac{1}{2} + \frac{N_c}{8} |(\alpha + \beta)g(x_i) - \beta| \\
&\quad + \frac{N_c}{8} |(\alpha + \beta)g(x_{i+1}) - \beta| \\
&\geq \frac{1}{2} + \frac{N_c}{8} |(\alpha + \beta)g(x_i) - (\alpha + \beta)g(x_{i+1})| \\
&\geq \frac{1}{2} + \frac{N_c}{8} \cdot \delta(\alpha + \beta) \\
&\geq \frac{1}{2} + \frac{\delta}{4}.
\end{aligned}$$

Case 2: $n_a > t_{N_t-1} N_c$:

Note $\alpha g(x) > \beta(1 - g(x))$ only holds if $x = x_{N_t}$. Therefore,

$$\begin{aligned}
\mathbf{Adv}_{\mathbf{EE}, p_m, p_k}^{\text{mrs}} &= \sum_{x \in \mathcal{M}^{N_k}} |\mathcal{C}_x| \cdot \max\{\alpha g(x), \beta(1 - g(x))\} \\
&= \alpha |\mathcal{C}_{x_0}| + \sum_{x \in \mathcal{M}^{N_k} \setminus \{x_0\}} |\mathcal{C}_x| \cdot \beta(1 - g(x)) \\
&= \alpha |\mathcal{C}_{x_0}| + \sum_{c \in \mathcal{C}} \sum_{k: D_k(c)=b} \beta \\
&= p_m(b) + \alpha |\mathcal{C}_{x_0}| \\
&= \frac{1}{2} + \frac{|\mathcal{C}_{x_0}|}{2n_a}.
\end{aligned}$$

On one hand, $\sum_{k \in \mathcal{K}} \sum_{c \in \mathcal{C}} 1_{D_k(c)=a} = \sum_{k \in \mathcal{K}} n_a = N_k n_a$. On the other hand,

$$\begin{aligned}
\sum_{c \in \mathcal{C}} \sum_{k \in \mathcal{K}} 1_{D_k(c)=a} &= \sum_{c \in \mathcal{C}_{x_0}} \sum_{k \in \mathcal{K}} 1_{D_k(c)=a} + \sum_{c \in \mathcal{C} \setminus \mathcal{C}_{x_0}} \sum_{k \in \mathcal{K}} 1_{D_k(c)=a} \\
&\leq N_k |\mathcal{C}_{x_0}| + (N_k - 1)(N_c - |\mathcal{C}_{x_0}|) \\
&= (N_k - 1)N_c + |\mathcal{C}_{x_0}| \\
&< (N_k - 1) \frac{n_a}{t_{N_t-1}} + |\mathcal{C}_{x_0}| \\
&= \frac{n_a(N_k - 1)}{1 - p_k^{\min}} + |\mathcal{C}_{x_0}|.
\end{aligned}$$

Thus, $\frac{|\mathcal{C}_{x_0}|}{n_a} > \frac{1 - N_k \cdot p_k^{\min}}{1 - p_k^{\min}} > 0$. Let $\lambda = \frac{1 - N_k \cdot p_k^{\min}}{1 - p_k^{\min}}$, we have

$$\mathbf{Adv}_{\mathbf{EE}, p_m, p_k}^{\text{mrs}} \geq \frac{1}{2} + \frac{\lambda}{2}.$$

With the bounds in the two cases,

$$\inf_{\mathbf{EE} \in \Omega_{\mathbf{EE}}} \mathbf{Adv}_{\mathbf{EE}, p_m, p_k}^{\text{mrs}} \geq \frac{1}{2} + \frac{1}{2} \min\left\{\frac{\delta}{2}, \lambda\right\} > \frac{1}{2}.$$

Namely, the theorem holds. $\square$

Proposition 9 (restated). *If a probabilistic encryption scheme $(\mathcal{M}, \mathcal{K}, \mathcal{C}, \mathcal{E}, \mathcal{D})$ is uniform, message-dependent, and key-independent, then it can be represented as an EE construction $(\mathcal{M}, \mathcal{K}, \mathcal{C}, \mathcal{E}', \mathcal{D}')$ such that for all $m \in \mathcal{M}$, $k \in \mathcal{K}$, and $c \in \mathcal{C}$,*

$$\Pr[E_k(m) = c] = \Pr[E'_k(m) = c].$$

Proof of Proposition 9. Since for each m , $|\mathcal{C}(m, k)|$ is independent of k , we can establish $p_d(m) = \frac{|\mathcal{C}(m, k)|}{2^l}$ with an appropriate encoder (encode, decode). Further, we can design a cipher (encrypt, decrypt) on $\{0, 1\}^l$ such that encrypt_k transforms the seeds in $\text{encode}(m)$ to the ciphertext in $\mathcal{C}(m, k)$. In this way, we construct an EE construction that, under the key k , encrypts the message m to ciphertexts in $\mathcal{C}(m, k)$ with the same probability $\frac{1}{|\mathcal{C}(m, k)|}$. Therefore, for each $m \in \mathcal{M}$, $k \in \mathcal{K}$: if $c \in \mathcal{C}(m, k)$, then $\Pr[E_k(m) = c] = \frac{1}{|\mathcal{C}(m, k)|} = \Pr[E'_k(m) = c]$; if $c \notin \mathcal{C}(m, k)$, then $\Pr[E_k(m) = c] = 0 = \Pr[E'_k(m) = c]$. In other words, the EE construction satisfies the requirement in the theorem. $\square$

Proposition 7 (restated). *OE achieves both optimal KC and optimal MRS if and only if the key distribution is uniform and $p_m^{\max} \leq \frac{1}{|\mathcal{K}|}$.*

Proof of Proposition 7. We demonstrate this by analyzing the following three cases:

1. When the key distribution is non-uniform, the ciphertext set size in OE satisfies

$$|\mathcal{C}^{(l)}(m, k)| \propto p_m(m) p_k(k) + O(2^{-l}),$$

which deviates from the requirement for optimal KC:

$$|\mathcal{C}^{(l)}(m, k)| \propto p_m(m) + \text{negl}(l).$$

TDSS Game $\text{TDSS}_{\mathbf{E}, p_m, p_k}^{\mathcal{A}, f}$
$M \leftarrow_{p_m} \mathcal{M}$
$K \leftarrow_{p_k} \mathcal{K}$
$C \leftarrow E_K(M)$
$b \leftarrow \mathcal{A}(C)$
return $b = f(M)$

Figure S.5: Game for TDSS security

2. If the key distribution is uniform but $p_m(m_1) > \frac{1}{|\mathcal{K}|}$, the ciphertext support varies across different keys, violating the condition for optimal KC. More specifically,

$$|\mathcal{C}^{(l)}(m, k)| \approx 2^l \cdot p_k(k) \cdot \frac{p_m(m)}{p_m(m_1)} = \frac{2^l p_m(m)}{p_m(m_1) |\mathcal{K}|},$$

$$\sum_m |\mathcal{C}^{(l)}(m, k)| \approx \frac{2^l}{p_m(m_1) |\mathcal{K}|} < 2^l.$$

This implies that in order to prevent collisions on m_1 under different keys, OE must assign a non-empty $\mathcal{C}^{(l)}(\perp, k)$, which differs across keys. Consequently, the ciphertext distributions vary with the key, violating KC.

3. When the key distribution is uniform and $p_m(m_1) \leq \frac{1}{|\mathcal{K}|}$, we have $\mathcal{C}^{(l)}(\perp, k) = \emptyset$, and the ciphertext distributions across keys are nearly uniform on $\mathcal{C}^{(l)}$. Therefore, OE satisfies both optimal KC and MRS in this case. □

S.8. Target-distribution Semantic Security

As a stronger notion of MC [11], target-distribution semantic-security (TDSS) models the leakage of partial information about the message from the ciphertext.

Formally, for a predicate $f : \mathcal{M} \rightarrow \{0, 1\}$, the TDSS game is shown in Fig. S.5. An adversary $\mathcal{A}$ is going to predict $f(M)$ given the ciphertext C of M . Clearly, a trivial adversary can simply output the most likely value of $f(M)$ and achieves the success probability of $\omega_f = \max_b \Pr[f(M) = b]$. Considering this, Jaeger et al. [11] defined the TDSS advantage as:

$$\text{Adv}_{\mathbf{E}, p_m, p_k}^{\text{tdss}} = \max_{\mathcal{A}, f} \left\{ \Pr[\text{TDSS}_{\mathbf{E}, p_m, p_k}^{\mathcal{A}, f} \Rightarrow \text{true}] - \omega_f \right\}.$$

Subtracting ω_f aims to ensure that the minimum TDSS advantage is zero. However, we show that zero TDSS advantage is generally unattainable for arbitrary message and key distributions.

Theorem S.7. *If $|\mathcal{M}| = |\mathcal{K}| = 2$ and $p_k(k_1) > p_m(m_1)$, then for any symmetric encryption scheme $\mathbf{E}$, $\text{Adv}_{\mathbf{E}, p_m, p_k}^{\text{tdss}} \geq p_k(k_1) - p_m(m_1) > 0$.*

Proof. Since $|\mathcal{M}| = 2$, it suffices to consider the trivial predicate $f(m_i) = i - 1$. In this case, the TDSS adversary must predict the exact message, making the TDSS game equivalent to the MRS game. Since the trivial MRS advantage is $p_k(k_1)$, we have

$$\text{Adv}_{\mathbf{E}, p_m, p_k}^{\text{tdss}} = \max_{\mathcal{A}} \Pr[\text{TDSS}_{\mathbf{E}, p_m, p_k}^{\mathcal{A}, f} \Rightarrow 1] - \omega_f \geq p_k(k_1) - p_m(m_1) > 0.$$

□

We leave the study of the optimal attainable TDSS advantage under general message and key distributions as future work.

S.9. Details of Continuous Honey Encryption Schemes and Proofs

Security of uniformized encryption schemes. Intuitively, uniformizing $p_{c|m, k}$ over each equivalence class S of R_a induces a new ciphertext C' from the original ciphertext C via a post-processing step: first identify the equivalence class S to which C belongs, and then sample C' uniformly from S . As a result, C' reveals only the equivalence class of C . Since C' is obtained by post-processing C , it cannot reveal more information about the underlying variables than C .

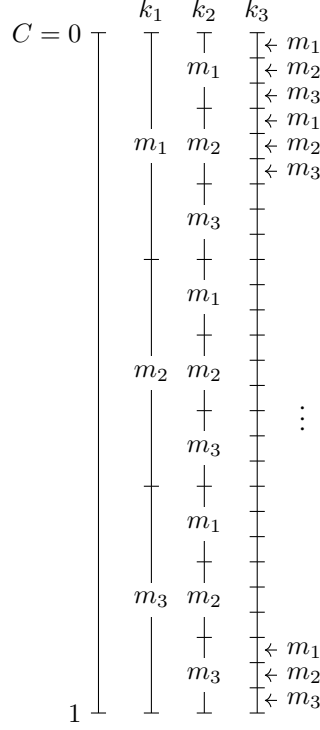


Figure S.6: Rearrangement of ciphertext intervals, where $m_i \in \mathcal{M}' = \mathcal{M} \cup \{\perp\}$

To formalize this intuition, we consider a general measure $M(X; Y)$ that quantifies the amount of information about a sensitive variable X revealed by an observation Y . Formally, M is a functional that maps the joint distribution $P_{X,Y}$ to a real value:

$$M : \mathcal{P}(\mathcal{X} \times \mathcal{Y}) \rightarrow \mathbb{R}.$$

Depending on the context, M may represent an information-theoretic quantity (e.g., mutual information or divergence) or an operational quantity (e.g., optimal guessing success probability).

Definition S.2 (Post-processing monotonicity). *A measure $M(X; Y)$ is said to be monotone under post-processing if for any Markov chain $X \rightarrow Y \rightarrow Z$, it holds that*

$$M(X; Z) \leq M(X; Y).$$

This property is a standard abstraction of the data processing inequality (DPI) in information theory [35], and is closely related to Blackwell's comparison of statistical experiments [36]. A wide class of measures satisfy this property, including f -divergences [37, 38], Shannon mutual information, Rényi divergence and Sibson mutual information [39, 40], as well as decision-theoretic quantities such as optimal guessing success probability. More recently, maximal leakage has also been shown to satisfy post-processing monotonicity [41]. In this paper, we focus on mutual information and guessing success probability as security measures, both of which satisfy post-processing monotonicity.

Proposition S.3. *Guess_n($X|Y$) and $I(X; Y)$ are monotone under post-processing.*

Theorem S.8. *Let E be any symmetric encryption scheme, and let E' be the scheme obtained by uniformizing $p_{c|m,k}$ on each equivalence class S of R_a . Then for any measure M that is monotone under post-processing, we have*

$$M(M, K; C') \leq M(M, K; C), \quad M(M; C') \leq M(M; C), \quad M(K; C') \leq M(K; C).$$

In particular,

$$I(K; C') \leq I(K; C), \quad \text{Adv}_{E', p_m, p_k}^{\text{mrs}} \leq \text{Adv}_{E, p_m, p_k}^{\text{mrs}}, \quad \text{Adv}_{E', p_m, p_k}^{\text{tdss}} \leq \text{Adv}_{E, p_m, p_k}^{\text{tdss}}.$$

Rearrangement of $p_{c|m,k}$. After uniformizing $p_{c|m,k}$ on S_m , we aggregate the ciphertexts in S_m into a continuous interval on $\{0, 1\}^l$. We then rearrange these intervals $S_m = \cap_j \mathcal{C}^{(l)}(m_{i_j}, k_j)$ on $\mathcal{C}^{(l)}$ in lexicographic order of $(i_j)_j$, where $m = (m_{i_j})_j$. As shown in Fig. S.6, the intervals are ordered as follows:

$$S_{(m_1, m_1, \dots, m_1, m_1)}, S_{(m_1, m_1, \dots, m_1, m_2)}, S_{(m_1, m_1, \dots, m_1, m_3)}, \dots,$$

$$\begin{aligned}
& S_{(m_1, m_1, \dots, m_2, m_1)}, S_{(m_1, m_1, \dots, m_2, m_2)}, S_{(m_1, m_1, \dots, m_2, m_3)}, \dots, \\
& \dots, \\
& S_{(m_2, m_1, \dots, m_1, m_1)}, S_{(m_2, m_1, \dots, m_1, m_2)}, S_{(m_2, m_1, \dots, m_1, m_3)}, \dots, \\
& \dots
\end{aligned}$$

Convergence of interval and $p_{c|m,k}$. For a sequence of intervals $[a^{(l)}, b^{(l)}]$ on $[0, 1)$, both $a^{(l)}$ and $b^{(l)}$ must have convergent subsequences. Assume $a^{(l)}$ converges to a and $b^{(l)}$ converges to b . If $a \neq b$, then $p_{c|m,k}$ naturally converges. However, if $a = b$, then $p_{c|m,k}$ converges to an infinite probability density at a single point, which yields a degenerate asymptotic structure.

Fortunately, we can rearrange these intervals so that they converge to continuous intervals with finite probability density. Specifically, for intervals where $p_{c|m,k} \rightarrow \infty$, $p_{c|m,k}$ may exhibit different orders of infinity, such as $O(l)$ and $O(l^2)$. Due to the finite nature of the intervals, these orders are also finite. We label the order with the index i . For intervals $[a^{(l)}, b^{(l)}]$ with the i -th order, we move them into $[i, i+1)$ and extend their lengths by the same factor, ensuring that $p_{c|m,k}$ converges to a finite value.

Thus, the asymptotic scheme maintains finite probability density on a continuous ciphertext space $[0, n)$, which can then be naturally rescaled to $[0, 1)$, where n is the number of orders.

We assert that this rearrangement does not affect the necessary conditions of asymptotic schemes, as the intersection of two intervals with different orders has a probability that converges to zero.

Corollary 1 (restated). *For a continuous encryption scheme E_c with $\text{Adv}_{E_c, p_m, p_k}^{\text{mrs}} = p_{\max}$, the following conditions hold: Let $m_1 \in \arg \max_m p_m(m)$ and $k_1 \in \arg \max_k p_k(k)$.*

1. *In Case K with $p_k^{\max} \geq p_m^{\max}$: 1) $\{\mathcal{C}(m, k_1)\}_m$ is a partition of $\mathcal{C}$ and 2) $\mathcal{C}(m, k_1) \cap \mathcal{C}(m, k') = \emptyset$ for each m and each $k' \neq k_1$.*
2. *In Case M with $p_m^{\max} \geq p_k^{\max}$: $\cup_k \mathcal{C}(m_1, k) = \mathcal{C}$.*

Proof of Corollary 1. In Case K: if the condition does not hold, then there exists a ciphertext c such that 1) $D_{k_1}(c) = \perp$ or 2) $D_{k_1}(c) = D_{k'}(c)$ for some $k' \neq k_1$. For R_a , let S be the equivalence class including c , and $|S|$ be the interval length of S . If $D_{k_1}(c) = \perp$ holds, then the MRS advantage is not less than

$$p_k(k_1) + |S| \max_m p_{m,c}(m, c).$$

If $D_{k_1}(c) = D_{k'}(c) \in \mathcal{M}$ for some $k' \neq k_1$, then the MRS advantage is not less than

$$p_k(k_1) + |S| p_{m,k,c}(m_{\text{triv}}(c), k', c).$$

In Case M, if the condition does not hold, then there exists a ciphertext c such that $D_k(c) \neq m_1$ for each k . Therefore, the MRS advantage is not less than

$$p_m(m_1) + |S| \max_m p_{m,c}(m, c).$$

Thus, the theorem holds. □

Theorem 12 (restated). *For a message distribution p_m , there exists a p_k -agnostic continuous encryption scheme E_c such that $\text{Adv}_{E_c, p_m, p_k}^{\text{mrs}} = p_{\max}$ for any key distribution p_k , only in the following three cases:*

1. $|\mathcal{M}| = |\mathcal{K}| = 2$.
2. $|\mathcal{M}| \geq 3$, $|\mathcal{K}| = 2$, and $p_m(m_1) \geq 0.5$.
3. $p_m^{\max} \leq \frac{1}{|\mathcal{K}|}$.

In other cases, such a p_k -agnostic scheme does not exist.

Proof of Theorem 12. If $\mathcal{M} = \mathcal{K} = \{0, 1\}$, then $E_k(m) = m \oplus k$ with $\mathcal{C} = \{0, 1\}$ achieves the optimal MRS. This scheme can be naturally extended to $[0, 1)$, yielding E_2 (as shown in Fig. 8a).

If $|\mathcal{M}| \geq 3$, $|\mathcal{K}| = 2$, and $p_m(m_1) \geq 0.5$, we use the scheme E_2 for $|\mathcal{M}| = 2$ and assign m_i for $i \neq 1$ within the intervals of m_2 in E_2 according to the probabilities (as shown in Fig. 8b). In other words, $\mathcal{C}(m_1, k) = \cup_{i \neq 1} \mathcal{C}(m_i, k')$ for each $k \neq k'$.

If $|\mathcal{M}| \geq 3$, $|\mathcal{K}| = 2$, and $p_m(m_1) < 0.5$, then $p_m(m_1) < \frac{1}{|\mathcal{K}|}$, which can be reduced to the next case.

If $p_m(m_1) \leq \frac{1}{|\mathcal{K}|}$, we can use an EE construction with the following interval structure as the general scheme (as shown in Fig. 8c):

$$\mathcal{C}(m_i, k_j) = [F_m(i-1) + (j-1)F_m(1), F_m(i) + (j-1)F_m(1)).$$

Since $|\mathcal{K}|p_m(m_1) \leq 1$, the scheme is well-defined and avoids collisions.

We now analyze the remaining case, namely $|\mathcal{K}| \geq 3$ and $p_m(m_1) > \frac{1}{|\mathcal{K}|}$. In this case, general schemes do not exist. Otherwise, we derive contradictions by exploiting the arbitrariness of p_k and applying Corollary 1.

Let $p_k(k')$ approach 1 for some k' . This corresponds to Case K, where $p_k(k') > p_m(m_1)$. Therefore, the following holds: 1) $\{\mathcal{C}(m, k')\}_m$ is a partition of $\mathcal{C}$, and 2) $\mathcal{C}(m, k') \cap \mathcal{C}(m, k'') = \emptyset$ for each $k'' \neq k'$ and each m . Since k' is arbitrary, each c must correspond to different messages under different keys. We next analyze two cases: 1) $p_m(m_1) > \frac{1}{|\mathcal{K}|-1}$, and 2) $\frac{1}{|\mathcal{K}|} < p_m(m_1) \leq \frac{1}{|\mathcal{K}|-1}$.

Case 1: $p_m(m_1) > \frac{1}{|\mathcal{K}|-1}$:

Let $p_k(k)$ be uniform. Since $p_m(m_1) > \frac{1}{|\mathcal{K}|}$, this corresponds to Case M. Therefore, $\{\mathcal{C}(m_1, k)\}_k$ covers $\mathcal{C}$. Furthermore, by the above analysis that each c corresponds to different m under different k , $\{\mathcal{C}(m_1, k)\}_k$ forms a partition of $\mathcal{C}$. Then, let p_k be uniform over $\mathcal{K} \setminus \{k'\}$ for some k' and $p_k(k') = 0$. As $|\mathcal{K}| \geq 3$, this yields a general scheme on $\mathcal{K} \setminus \{k'\}$. Because $p_m(m_1) > \frac{1}{|\mathcal{K}|-1}$, it still corresponds to Case M. However, $\{\mathcal{C}(m_1, k)\}_{k \neq k'}$ no longer forms a partition of $\mathcal{C}$, which contradicts Corollary 1. Thus, general schemes do not exist.

Case 2: $\frac{1}{|\mathcal{K}|} < p_m(m_1) \leq \frac{1}{|\mathcal{K}|-1}$: Let $p_k(k_i) = \frac{1}{2} = p_k(k_j)$ for some $k_i \neq k_j$. Since $p(m_1) \leq \frac{1}{|\mathcal{K}|-1} \leq \frac{1}{3-1} = \frac{1}{2}$, this falls into Case K. By Theorem 11, $p_{m,c}(D_{k_i}(c), c) = p_{m,c}(D_{k_j}(c), c)$ for each c .

Given that each c corresponds to a different m under different k , it holds that $p_{m,c}(D_k(c), c) = p_k(k)p_m(D_k(c))p_{c|m,k}(c|D_k(c), k)$. Thus, due to the arbitrariness of k_i and k_j , the product $p_m(D_k(c)) \cdot p_{c|m,k}(c|D_k(c), k)$ must be constant for all k and all c in any equivalence class S of R_a . It is easy to stretch equivalence classes such that this product remains constant for all k and c .

Moreover,

$$\sum_c p_m(D_k(c))p_{c|m,k}(c|D_k(c), k) = \sum_m p_m(m) \sum_{c \in \mathcal{C}(m, k)} p_{c|m,k}(c|m, k) = 1,$$

implying $p_m(D_k(c))p_{c|m,k}(c|D_k(c), k) = 1$.

Now, let $p_k(k) = \frac{1}{|\mathcal{K}|}$ for each k . Since $p_m(m_1) > \frac{1}{|\mathcal{K}|}$, this corresponds to Case M. By Corollary 1 and the previously established fact that $\mathcal{C}(m, k') \cap \mathcal{C}(m, k'') = \emptyset$ for all $k'' \neq k'$ and each m , the family $\{\mathcal{C}(m_1, k)\}_k$ forms a partition of $\mathcal{C}$.

Hence, for each c , there exists a unique k such that $D_k(c) = m_1$. Denote this k as $k(c, m_1)$. Then,

$$p_m(m_1)p_{c|m,k}(c|m_1, k(c, m_1)) = 1,$$

which gives

$$p_{c|m,k}(c|m_1, k(c, m_1)) = \frac{1}{p_m(m_1)} = \frac{1}{|\mathcal{C}(m_1, k(c, m_1))|}.$$

Thus, $|\mathcal{C}(m_1, k(c, m_1))| \cdot |\mathcal{K}| > 1$, contradicting that $\{\mathcal{C}(m_1, k)\}_k$ is a partition of $\mathcal{C}$. $\square$

Proposition 8 (restated). For a message distribution p_m , there exists a continuous scheme E_c such that $\text{Adv}_{E_c, p_m, p_k}^{\text{mrs}} = p_{\max}$ for any key distribution p_k with $p_k^{\max} \leq \frac{1}{|\mathcal{M}|}$.

Proof of Proposition 8. This proposition corresponds only to Case M.

In that case, HES for uniform message distribution suffices. $\square$

Lemma S.1. For $f_a(x) = \max\{\frac{a}{x}, \frac{1-a}{1-x}\} + \max\{\frac{a}{1-x}, \frac{1-a}{x}\}$, it always holds

$$f_{a_1}(x) > f_{a_2}(x)$$

for $0 < a_1 < a_2 < 1/2$ and $0 < x < 1$.

Example 1 (restated). For $p_m = (\frac{1}{2}, \frac{1}{2})$ and $p_k = (p_1, p_2, p_3)$ with $\frac{1}{2} > p_1 \geq p_2 \geq p_3$:

1. If $p_2 \geq \frac{1}{3}$,

$$\inf_{EE \in \Omega_{EE}} \text{Adv}_{EE, p_m, p_k}^{\text{mrs}} = \min \left\{ \frac{1-2p_1}{2}, \frac{(2p_1-1)(1-3p_1)}{2(1-p_1)p_1}, \frac{1-3p_3}{4}, \frac{(1-2p_2)(p_2-p_3)}{2(1-p_2)p_2} \right\} + \frac{1}{2}.$$

2. If $p_2 \leq \frac{1}{3}$,

$$\inf_{EE \in \Omega_{EE}} \text{Adv}_{EE, p_m, p_k}^{\text{mrs}} = \min \left\{ \frac{1-2p_1}{2}, \frac{3p_1-1}{4} \right\} + \frac{1}{2}.$$

Then, $\inf_{EE \in \Omega_{EE}} \text{Adv}_{EE, p_m, p_k}^{\text{mrs}} \geq \frac{1}{2}$, where the equality holds if and only if $p_1 = p_2 = p_3 = \frac{1}{3}$.

Derivation of Example 1. In an EE construction, for each key k the ciphertext distribution is uniform over $\mathcal{C}$, and for every message m the size of the set $\mathcal{C}(m, k)$ is independent of k . Consequently, EE constructions can freely adjust the sizes of the equivalence classes

$$S_{(m_{i_1}, m_{i_2}, m_{i_3})}$$

while preserving the uniformity and constant size of each $\mathcal{C}(m_i, k)$.

For notational convenience, write

$$s_{i_1 i_2 i_3} = |S_{(m_{i_1}, m_{i_2}, m_{i_3})}|, \quad \alpha_i = |\mathcal{C}(m_i, k)|.$$

These variables satisfy

$$\begin{aligned} s_{111} + s_{112} + s_{121} + s_{122} &= \alpha_1, \\ s_{111} + s_{112} + s_{211} + s_{212} &= \alpha_1, \\ s_{111} + s_{121} + s_{211} + s_{221} &= \alpha_1, \\ \sum_{i_1, i_2, i_3=1}^2 s_{i_1 i_2 i_3} &= 1, \\ \alpha_1 + \alpha_2 &= 1, \\ s_{i_1 i_2 i_3} &\geq 0 \quad \forall i_1, i_2, i_3 \in \{1, 2\}. \end{aligned}$$

First, observe that setting

$$s_{111} = s_{222} = 0$$

cannot increase the adversary's advantage. Indeed, any positive mass on s_{111} or s_{222} contributes an advantage of 1, and one can redistribute this mass to other classes to lower the overall advantage.

Hence, without loss of generality we set $s_{111} = s_{222} = 0$ and treat α_1 , s_{112} , and s_{121} as free parameters. The remaining variables admit the closed-form expressions

$$\begin{aligned} s_{122} &= \alpha_1 - s_{112} - s_{121}, \\ s_{211} &= -1 + 3\alpha_1 - s_{112} - s_{121}, \\ s_{212} &= 1 - 2\alpha_1 + s_{121}, \\ s_{221} &= 1 - 2\alpha_1 + s_{112}, \\ \alpha_2 &= 1 - \alpha_1, \end{aligned}$$

subject to

$$\frac{1}{3} \leq \alpha_1 \leq \frac{2}{3}, \quad L \leq s_{112} \leq U, \quad L \leq s_{121} \leq U,$$

where

$$L = \max\{0, 2\alpha_1 - 1\}, \quad U = \min\{\alpha_1, 3\alpha_1 - 1\}.$$

For any $c \in S_{i_1 i_2 i_3}$,

$$\max_m \Pr[M = m, C = c] = \frac{1}{2} \max \left\{ \frac{\sum_{j:i_j=1} p_j}{\alpha_1}, \frac{\sum_{j:i_j=2} p_j}{\alpha_2} \right\}.$$

Therefore the overall advantage is

$$\mathbf{Adv}_{\text{EE}, p_m, p_k}^{\text{mrs}} = \sum_{i_1, i_2, i_3} \frac{s_{i_1 i_2 i_3}}{2} \max \left\{ \frac{\sum_{j:i_j=1} p_j}{\alpha_1}, \frac{\sum_{j:i_j=2} p_j}{\alpha_2} \right\}.$$

Fix α_1 , the feasible region for $(s_{i_1 i_2 i_3})_{i_1 i_2 i_3}$ is a triangle, and $\mathbf{Adv}_{\text{EE}, p_m, p_k}^{\text{mrs}}$ is affine in these two variables as

$$\frac{1}{2} s_{121} A_1 + \frac{1}{2} s_{112} A_2 + \frac{1}{2} A_3,$$

where

$$\begin{aligned} A_1 &= \max \left\{ \frac{p_2}{\alpha_1}, \frac{1-p_2}{1-\alpha_1} \right\} + \max \left\{ \frac{p_2}{1-\alpha_1}, \frac{1-p_2}{\alpha_1} \right\} \\ &\quad - \max \left\{ \frac{p_1}{\alpha_1}, \frac{1-p_1}{1-\alpha_1} \right\} - \max \left\{ \frac{p_1}{1-\alpha_1}, \frac{1-p_1}{\alpha_1} \right\}, \\ A_2 &= \max \left\{ \frac{p_3}{\alpha_1}, \frac{1-p_3}{1-\alpha_1} \right\} + \max \left\{ \frac{1-p_3}{\alpha_1}, \frac{p_3}{1-\alpha_1} \right\} \end{aligned}$$

$$\begin{aligned}
& -\max\left\{\frac{p_1}{\alpha_1}, \frac{1-p_1}{1-\alpha_1}\right\} - \max\left\{\frac{p_1}{1-\alpha_1}, \frac{1-p_1}{\alpha_1}\right\}, \\
A_3 = & \alpha_1 \max\left\{\frac{p_1}{\alpha_1}, \frac{1-p_1}{1-\alpha_1}\right\} + \max\left\{\frac{p_2}{\alpha_1}, \frac{1-p_2}{1-\alpha_1}\right\} + \max\left\{\frac{p_3}{\alpha_1}, \frac{1-p_3}{1-\alpha_1}\right\} \\
& + 3\alpha_1 \max\left\{\frac{p_1}{1-\alpha_1}, \frac{1-p_1}{\alpha_1}\right\} - 2\alpha_1 \max\left\{\frac{p_2}{\alpha_1}, \frac{1-p_2}{1-\alpha_1}\right\} \\
& - 2\alpha_1 \max\left\{\frac{p_3}{\alpha_1}, \frac{1-p_3}{1-\alpha_1}\right\} - \max\left\{\frac{p_1}{1-\alpha_1}, \frac{1-p_1}{\alpha_1}\right\}.
\end{aligned}$$

As $1 > p_1 \geq p_2 \geq p_3 \geq 0$, by Lemma S.1, $A_1 \geq 0$ and $A_2 \geq 0$. Therefore, $\mathbf{Adv}_{\mathbf{EE}, p_m, p_k}^{\text{mrs}}$ achieves its minimum value A_3 when

$$s_{112} = s_{121} = L.$$

At this vertex $s_{i_1 i_2 i_3}$ take the values

$$\begin{aligned}
s_{111} &= s_{222} = 0, \\
s_{122} &= \alpha_1 - 2L, \\
s_{211} &= 3\alpha_1 - 1 - 2L, \\
s_{212} &= 1 - 2\alpha_1 + L, \\
s_{221} &= 1 - 2\alpha_1 + L.
\end{aligned}$$

Naturally, we can partition the space $[\frac{1}{3}, \frac{2}{3}]$ of α_1 into several intervals such that, within each interval, each expression of the form $\max\{\cdot, \cdot\}$ becomes fixed. Then,

$$\mathbf{Adv}_{\mathbf{EE}, p_m, p_k}^{\text{mrs}} = A + \frac{B}{\alpha_1} + \frac{C}{1-\alpha_1},$$

where A, B, C are constants determined by (p_1, p_2, p_3) . We can compute the minimum of $\mathbf{Adv}_{\mathbf{EE}, p_m, p_k}^{\text{mrs}}$ over each interval and then determine the global minimum.

The interval boundaries consist of all switching points of the max functions, along with the endpoints of the interval $[\frac{1}{3}, \frac{2}{3}]$, forming the set

$$\left\{\frac{1}{3}, \frac{1}{2}, \frac{2}{3}, p_1, p_2, p_3, p_1+p_2, p_1+p_3, p_2+p_3\right\} \cap \left[\frac{1}{3}, \frac{2}{3}\right].$$

Considering the symmetry between m_1 and m_2 , we can construct an $\mathbf{EE}'$ with $s'_{i'_1 i'_2 i'_3} = s_{i_1 i_2 i_3}$, $i'_j = 3 - i_j$ and $\alpha'_i = 1 - \alpha_i$ such that

$$\mathbf{Adv}_{\mathbf{EE}', p_m, p_k}^{\text{mrs}} = \mathbf{Adv}_{\mathbf{EE}, p_m, p_k}^{\text{mrs}}.$$

Thus, $\mathbf{Adv}_{\mathbf{EE}, p_m, p_k}^{\text{mrs}}$ achieves the same value at α_1 and $1 - \alpha_1$. We only need to consider the minimum on the interval $[\frac{1}{3}, \frac{1}{2}]$ with the vertex satisfying

$$\begin{aligned}
s_{111} &= s_{222} = s_{112} = s_{121} = 0, \\
s_{122} &= \alpha_1, \\
s_{211} &= 3\alpha_1 - 1, \\
s_{212} &= s_{221} = 1 - 2\alpha_1,
\end{aligned}$$

and the boundary point set

$$P = \left\{\frac{1}{3}, \frac{1}{2}, p_1, p_2, p_3\right\} \cap \left[\frac{1}{3}, \frac{1}{2}\right].$$

In fact, for $0 < \alpha_1 < 1$, it always holds that $B \leq 0$ and $C \leq 0$ (see the following calculation). Therefore, $\mathbf{Adv}_{\mathbf{EE}, p_m, p_k}^{\text{mrs}}$ is a concave function of α_1 and its minimum occurs at one of the interval endpoints.

P varies in two cases: 1) $p_2 \geq \frac{1}{3}$ and 2) $p_2 \leq \frac{1}{3}$.

Case 1: If $p_2 \geq \frac{1}{3}$, then

$$p_3 = 1 - p_1 - p_2 \leq \frac{1}{3} \leq p_2 \leq p_1 < \frac{1}{2}.$$

Thus,

$$P = \left\{\frac{1}{3}, p_2, p_1, \frac{1}{2}\right\}.$$

On each resulting interval, $\mathbf{Adv}_{\mathbf{EE}, p_m, p_k}^{\text{mrs}}$ simplifies to one of the following expressions:

$$\begin{aligned} & \frac{3}{2} - \frac{1-p_3}{2(1-\alpha_1)} - \frac{p_3}{2\alpha_1}, \\ & \frac{5}{2} - \frac{1-p_1}{2\alpha_1} - \frac{p_1+1}{2(1-\alpha_1)}, \\ & 2 - \frac{1-p_1}{2\alpha_1} - \frac{p_1}{1-\alpha_1}. \end{aligned}$$

Thus, we compute the values on the endpoints:

$$\begin{aligned} & \frac{1}{4}(1-3p_3) + \frac{1}{2}, \\ & \frac{(2p_2-1)(p_1+2p_2-1)}{2(p_2-1)p_2} + \frac{1}{2}, \\ & \frac{(2p_1-1)(3p_1-1)}{2(p_1-1)p_1} + \frac{1}{2}, \\ & \frac{1}{2}(1-2p_1) + \frac{1}{2}. \end{aligned}$$

Finally, the global minimum is given by:

$$\min \left\{ \frac{1-2p_1}{2}, \frac{(2p_1-1)(1-3p_1)}{2(1-p_1)p_1}, \frac{1-3p_3}{4}, \frac{(1-2p_2)(p_2-p_3)}{2(1-p_2)p_2} \right\} + \frac{1}{2}.$$

Case 2: If $p_2 \leq \frac{1}{3}$, then

$$p_3 \leq p_2 \leq \frac{1}{3} \leq p_1 \leq \frac{1}{2}.$$

Therefore,

$$P = \left\{ \frac{1}{3}, p_1, \frac{1}{2} \right\}.$$

On the resulting intervals, the expression for $\mathbf{Adv}_{\mathbf{EE}, p_m, p_k}^{\text{mrs}}$ becomes:

$$\begin{aligned} & \frac{5}{2} - \frac{1-p_1}{2\alpha_1} - \frac{p_1+1}{2(1-\alpha_1)}, \\ & 2 - \frac{1-p_1}{2\alpha_1} - \frac{p_1}{1-\alpha_1}. \end{aligned}$$

We evaluate the expression at the endpoints:

$$\begin{aligned} & \frac{3p_1-1}{4} + \frac{1}{2}, \\ & \frac{(2p_1-1)(3p_1-1)}{2(p_1-1)p_1} + \frac{1}{2}, \\ & \frac{1-2p_1}{2} + \frac{1}{2}. \end{aligned}$$

Thus, the minimum value of $\mathbf{Adv}_{\mathbf{EE}, p_m, p_k}^{\text{mrs}}$ is:

$$\min \left\{ \frac{1-2p_1}{2}, \frac{3p_1-1}{4} \right\} + \frac{1}{2}.$$

□

Example 2 (restated). For $p_m = (\frac{1}{2}, \frac{1}{2})$ and $p_k = (p_1, p_2, p_3)$ with $\frac{1}{2} > p_1 \geq p_2 \geq p_3$:

$$\inf_{\mathbf{HE} \in \Omega_{\mathbf{HE}}} \mathbf{Adv}_{\mathbf{HE}, p_m, p_k}^{\text{mrs}} = \frac{1-2p_1}{2} + \frac{1}{2} > \frac{1}{2}.$$

Derivation of Example 2. By the derivation of Example 1, setting $\alpha_1 = \frac{1}{2}$ naturally implies this bound.

□

Theorem 16 (restated). For any message and key distributions (p_m, p_k) , HES satisfies

$$I(K; C) = 0 \quad \text{and} \quad \text{Adv}_{\text{HES}, p_m, p_k}^{\text{mrs}} \leq p_k^{\max} + \frac{1 - p_k^{\max}}{\lfloor 1/p_m^{\max} \rfloor} \mathbf{1}_{1/p_m^{\max} < |\mathcal{K}|} \leq 2p_{\max}.$$

Proof of Theorem 16. Without loss of generality, assume $p_{\max} < \frac{1}{2}$ and $1/p_m^{\max} < |\mathcal{K}|$, since otherwise the claim is trivial (see the proof of Theorem 12).

Let $t = \lfloor 1/p_m(m_1) \rfloor$ and partition the key space $\mathcal{K}$ into $\lceil N_k/t \rceil$ groups, where the i -th group is $\mathcal{K}_i = \{k_{ti+1}, \dots, k_{t(i+1)}\}$ (the last group ends with k_{N_k}).

By construction of HES, there are no collisions within any group $\mathcal{K}_i$: for any $k_1, k_2 \in \mathcal{K}_i$ and $c \in \mathcal{C}$, we have $D_{k_1}(c) \neq D_{k_2}(c)$. Since HES is a HE scheme, $p_{c|m,k}(c|m, k) = 1/p_m(m)$. Hence,

$$\begin{aligned} p_{m,c}(m, c) &= \sum_{k: D_k(c)=m} p_k(k) p_m(m) p_{c|m,k}(c|m, k) \\ &= \sum_{k: D_k(c)=m} p_k(k) \leq \sum_{i=0}^{\lceil N_k/t \rceil - 1} p_k(k_{ti+1}). \end{aligned}$$

Therefore,

$$\begin{aligned} \text{Adv}_{\text{HES}, p_m, p_k}^{\text{mrs}} &\leq \sum_{i=0}^{\lceil N_k/t \rceil - 1} p_k(k_{ti+1}) \\ &\leq p_k(k_1) + \sum_{i=1}^{\lceil N_k/t \rceil - 1} \frac{1}{t} \sum_{j=0}^{t-1} p_k(k_{ti+1-j}) \\ &\leq p_k(k_1) + \frac{1}{t} \sum_{i=2}^{N_k} p_k(k_i) \\ &= p_k(k_1) + \frac{1 - p_k(k_1)}{t}. \end{aligned}$$

Since $t = \lfloor 1/p_m(m_1) \rfloor \geq 1/p_m(m_1) - 1$, we get

$$\begin{aligned} p_k(k_1) + \frac{1 - p_k(k_1)}{t} &\leq p_k(k_1) + \frac{1 - p_k(k_1)}{1/p_m(m_1) - 1} \\ &= \frac{p_k(k_1) + p_m(m_1) - 2p_k(k_1)p_m(m_1)}{1 - p_m(m_1)} \\ &\leq \frac{p_{\max} + (1 - 2p_{\max})p_{\min}}{1 - p_{\max}} \\ &\leq \frac{p_{\max} + (1 - 2p_{\max})p_{\max}}{1 - p_{\max}} = 2p_{\max}, \end{aligned}$$

where $p_{\max} = \max\{p_k(k_1), p_m(m_1)\}$ and $p_{\min} = \min\{p_k(k_1), p_m(m_1)\}$. $\square$

Proposition 10 (restated). For any EE construction $\text{EE}^{(l)}$ in the random oracle or ideal cipher models and the associated continuous scheme EE_c , as $l \rightarrow \infty$,

$$\mathbb{E}[I_{\text{EE}^{(l)}}(K; C^{(l)})] \rightarrow I_{\text{EE}_c}(K; C) \quad \text{and} \quad \mathbb{E}[\text{Adv}_{\text{EE}^{(l)}, p_m, p_k}^{\text{mrs}}] \rightarrow \text{Adv}_{\text{EE}_c, p_m, p_k}^{\text{mrs}}.$$

Proof of Proposition 10. In an EE construction, $p_{c|m,k}(c|m, k) = \frac{1}{2^l p_d^{(l)}(m)}$, so $p_{c|k}(c|k)$, $p_c(c)$, and $p_{m,c}(m, c)$ are constant for all $c \in S_m$. Therefore,

$$\begin{aligned} I(K; C) &= \sum_k p_k(k) \sum_c p_{c|k}(c|k) \log \frac{p_{c|k}(c|k)}{p_c(c)} \\ &= \sum_m |S_m| \sum_k p_k(k) p_{c|k}(c|k) \log \frac{p_{c|k}(c|k)}{p_c(c)}. \\ \text{Adv}_{\text{EE}^{(l)}, p_m, p_k}^{\text{mrs}} &= \sum_c \max_m p_{m,c}(m, c) = \sum_m |S_m| \max_m p_{m,c}(m, c). \end{aligned}$$

It suffices to compute $\mathbb{E}[|S_{\mathbf{m}}|]$ in the RO/IC setting. Let I_c be the indicator that $D_{k_i}(c) = m_{j_i}$ for all i . Then

$$\mathbb{E}[I_c] = \Pr[D_{k_i}(c) = m_{j_i}, \forall i] = \Pr[\text{decode}(c \oplus H(k_i)) = m_{j_i}, \forall i].$$

Since $c \oplus H(k_i)$ is independently uniform over $\{0, 1\}^l$ in the RO/IC models,

$$\mathbb{E}[|S_{\mathbf{m}}|] = \sum_c \mathbb{E}[I_c] = \sum_c \Pr[\text{decode}(c \oplus H(k_i)) = m_{j_i}, \forall i] = 2^l \prod_i p_d^{(l)}(m_{j_i}).$$

As $p_d^{(l)}(m_{j_i}) \rightarrow p_d(m_{j_i})$, we set $|S_{\mathbf{m}}| = \prod_i p_d(m_{j_i})$ and $p_{c|m,k}(c|m, k) = 1/p_d(m)$ in the continuous scheme EE_c , which yields the claimed convergence. $\square$

Example 3 (restated). For $p_m = (\frac{1}{2}, \frac{1}{2})$ and $p_k = (p_1, p_2, p_3)$ with $\frac{1}{2} > p_1 \geq p_2 \geq p_3$, in the random-oracle or ideal-cipher model, the continuous HE scheme HE_c (with $p_d = p_m$) and EE constructions EE_c (with general p_d) satisfy the bounds:

$$\text{Adv}_{\text{HE}_c, p_m, p_k}^{\text{mrs}} = \frac{3}{4} \quad \text{and} \quad \inf_{\text{EE}_c \in \Omega_{\text{EE}}} \text{Adv}_{\text{EE}_c, p_m, p_k}^{\text{mrs}} = \frac{1}{2} + p_1(1 - p_1) \geq \frac{13}{18}.$$

Derivation of Example 3. We borrow the notation from the derivation of Example 1. By Proposition 10, we have $s_{i_1 i_2 i_3} = \alpha_{i_1} \alpha_{i_2} \alpha_{i_3}$ where $i_1, i_2, i_3 \in \{1, 2\}$.

For the HE scheme, we have $\alpha_1 = \alpha_2 = \frac{1}{2}$, so $s_{i_1 i_2 i_3} = \frac{1}{8}$, and:

$$\begin{aligned} \text{Adv}_{\text{HE}_c, p_m, p_k}^{\text{mrs}} &= \sum_{i_1, i_2, i_3} \frac{s_{i_1 i_2 i_3}}{2} \max \left\{ \frac{\sum_{j: i_j=1} p_j}{\alpha_1}, \frac{\sum_{j: i_j=2} p_j}{\alpha_2} \right\} \\ &= \frac{1}{8} \left(\max\{1, 0\} + \max\{1 - p_3, p_3\} + \max\{1 - p_2, p_2\} + \max\{1 - p_1, p_1\} \right. \\ &\quad \left. + \max\{p_1, 1 - p_1\} + \max\{p_2, 1 - p_2\} + \max\{p_3, 1 - p_3\} + \max\{0, 1\} \right) \\ &= \frac{1}{8} (1 + 1 - p_3 + 1 - p_2 + 1 - p_1 + 1 - p_1 + 1 - p_2 + 1 - p_3 + 1) \\ &= \frac{3}{4}. \end{aligned}$$

For EE constructions, suppose $\alpha_1 = \alpha, \alpha_2 = 1 - \alpha$ and $0 < \alpha \leq \frac{1}{2}$. We have:

$$\begin{aligned} \text{Adv}_{\text{EE}_c, p_m, p_k}^{\text{mrs}} &= \sum_{i_1, i_2, i_3} \frac{s_{i_1 i_2 i_3}}{2} \max \left\{ \frac{\sum_{j: i_j=1} p_j}{\alpha}, \frac{\sum_{j: i_j=2} p_j}{1 - \alpha} \right\} \\ &= \frac{\alpha^3}{2} \max \left\{ \frac{1}{\alpha}, 0 \right\} \\ &\quad + \frac{\alpha^2(1 - \alpha)}{2} \left(\max \left\{ \frac{1 - p_3}{\alpha}, \frac{p_3}{1 - \alpha} \right\} + \max \left\{ \frac{1 - p_2}{\alpha}, \frac{p_2}{1 - \alpha} \right\} \right. \\ &\quad \left. + \max \left\{ \frac{1 - p_1}{\alpha}, \frac{p_1}{1 - \alpha} \right\} \right) \\ &\quad + \frac{\alpha(1 - \alpha)^2}{2} \left(\max \left\{ \frac{p_1}{\alpha}, \frac{1 - p_1}{1 - \alpha} \right\} + \max \left\{ \frac{p_2}{\alpha}, \frac{1 - p_2}{1 - \alpha} \right\} \right. \\ &\quad \left. + \max \left\{ \frac{p_3}{\alpha}, \frac{1 - p_3}{1 - \alpha} \right\} \right) + \frac{(1 - \alpha)^3}{2} \max \left\{ 0, \frac{1}{1 - \alpha} \right\} \\ &= \frac{\alpha^2}{2} + \frac{\alpha^2(1 - \alpha)}{2} \left(\frac{1 - p_3}{\alpha} + \frac{1 - p_2}{\alpha} + \frac{1 - p_1}{\alpha} \right) \\ &\quad + \frac{1 - \alpha}{2} (\max\{p_1, \alpha\} - p_1\alpha + \max\{p_2, \alpha\} - p_2\alpha + \max\{p_3, \alpha\} - p_3\alpha) \\ &\quad + \frac{(1 - \alpha)^2}{2} \\ &= \frac{1}{2} + \frac{1 - \alpha}{2} (\max\{p_1, \alpha\} + \max\{p_2, \alpha\} + \max\{p_3, \alpha\} - \alpha) \end{aligned}$$

$$= \begin{cases} \frac{\alpha^2}{2} - \alpha + 1, & 0 < \alpha \leq p_3 \\ \frac{1+(1-\alpha)(1-p_3)}{2}, & p_3 < \alpha \leq p_2 \\ \frac{1+(1-\alpha)(p_1+\alpha)}{2}, & p_2 < \alpha \leq p_1 \\ \frac{1}{2} + \alpha(1-\alpha), & p_1 < \alpha \leq \frac{1}{2} \end{cases}.$$

By simple analysis of the above piecewise function of α , the minimum value is achieved when $\alpha = p_1$, therefore:

$$\inf_{E \in \Omega_{EE}} \mathbf{Adv}_{E, p_m, p_k}^{\text{mrs}} = \frac{1}{2} + p_1(1 - p_1).$$

Moreover, the above results show that the optimal encoder is $p_d = (p_1, 1 - p_1)$ but not $(\frac{1}{2}, \frac{1}{2})$ as in the HE scheme. $\square$

Theorem 15 (restated). *There exist distributions p_m and p_k such that no continuous scheme E_c can satisfy both $I(K; C) = 0$ and $\mathbf{Adv}_{E_c, p_m, p_k}^{\text{mrs}} = p_{\max}$.*

Proof of Theorem 15. Similar to the analysis of key confidentiality (KC) for OE in Section 5.6, we leverage the necessary conditions for KC and MRS to prove the theorem.

Let p_m be the uniform distribution over $\{m_1, m_2\}$, and let p_k be the uniform distribution over $\{k_1, k_2, k_3\}$. This setting corresponds to Case M, since $p_m(m) = \frac{1}{2} > \frac{1}{3} = p_k(k)$.

Assume that such an encryption scheme exists and denote it by E . By Theorem 9, we have $p_{c|k}(c|k) = p_c(c)$. Since E is piecewise uniform, it can be stretched to make p_c uniform over the entire ciphertext space.

By Theorem 11, both m_1 and m_2 must maximize

$$p_{m,c}(m, c) = \sum_{k: D_k(c)=m} p_m(m) \cdot p_k(k) \cdot p_{c|k}(c|k) = \frac{1}{6} \cdot |\{k | D_k(c) = m\}|$$

for each c . However, since $\sum_m |\{k | D_k(c) = m\}| = 3$, it is impossible for both $|\{k | D_k(c) = m_1\}|$ and $|\{k | D_k(c) = m_2\}|$ to equal $\frac{3}{2}$. Therefore, no such encryption scheme E exists, which completes the proof. $\square$

S.10. Sufficiency and Necessity for Optimal Security

Based on the analysis in Sections 4 and 5, we summarize the necessary and sufficient conditions for achieving optimal security. By categorizing different spaces of encryption schemes, we clarify these conditions: sufficient conditions are determined by their feasibility within these spaces, while necessary conditions are identified by whether optimally secure schemes exist within these specific spaces.

For KC, let $I_{p_m, p_k}(\Omega)$ represent the infimum of $I_E(C, K)$ across all encryption schemes $E \in \Omega$ for the given p_m and p_k . For message confidentiality, denote $\mathbf{Adv}_{\Omega, p_m, p_k}^{\text{mrs}}$ as the infimum of the advantage $\mathbf{Adv}_{E, p_m, p_k}^{\text{mrs}}$ over all schemes $E \in \Omega$. For a family $\mathcal{E}$ of encryption schemes, where each scheme is parameterized by l as $E^{(l)} = (\mathcal{M}, \mathcal{K}, \mathcal{C}^{(l)}, \mathcal{E}^{(l)}, \mathcal{D}^{(l)})$, the infima of $I_{E^{(l)}}(C, K)$ and $\mathbf{Adv}_{E^{(l)}, p_m, p_k}^{\text{mrs}}$ are respectively denoted as $I_E(C, K)$ and $\mathbf{Adv}_{E, p_m, p_k}^{\text{mrs}}$.

With the previous analysis in Sections 4 and 5, on the spaces Ω_E of symmetric encryption, for arbitrary message and key distributions (p_m, p_k) , we have the optimal security with $I_{p_m, p_k}(\Omega_E) = 0$ and $\mathbf{Adv}_{\Omega, p_m, p_k}^{\text{mrs}} = p_{\max}$.

We consider the spaces of deterministic encryption, probabilistic encryption, EE constructions and HE schemes, denoted as Ω_{DE} , Ω_{PE} , Ω_{EE} , and Ω_{HE} , respectively.

Moreover, we analyze probabilistic encryption schemes characterized by different attributes: 1) ciphertext uniformity (U), 2) ciphertext cardinality dependence or independence from the key (KD or KID), and 3) ciphertext cardinality dependence or independence from the message (MD or MID). We denote these spaces as Ω_{PE}^P for attributes P , where $P = U, KD, KID, MD, MID$, or combinations thereof.

We also discuss the spaces of symmetric encryption schemes achieving optimal security for MC and KC, denoted as Ω_E^{MC} and Ω_E^{KC} , respectively.

Delving into the insights from Section 4, we have:

1. For general non-uniform message distributions p_m , it typically holds that $I_{p_m, p_k}(\Omega_{DE}) > 0$ and $I_{p_m, p_k}(\Omega_{PE}^{\text{U, MID}}) > 0$. In other words, it generally follows that $\Omega_E^{\text{KC}} \cap \Omega_{PE}^{\text{U}} \subseteq \Omega_{PE}^{\text{U, MD}}$. This implies that uniform probabilistic encryption with optimal KC must be message-dependent. Without the constraint of uniformity, this proposition does not hold, as some ciphertexts can be assigned negligible probabilities to equalize the ciphertext numbers across messages.
2. $\Omega_{EE} = \Omega_{PE}^{\text{U, MD, KID}} \cup \Omega_{DE}$. This follows from Proposition 9: every uniform, message-dependent, and key-independent probabilistic encryption scheme can be represented as an EE construction.
3. $\Omega_E^{\text{KC}} \subseteq \Omega_E^{\text{M-IND}}$. This means that encryption schemes achieving optimal security must be M-IND.
4. $I_{p_m, p_k}(\Omega_{HE}) = I_{p_m, p_k}(\Omega_{PE}^{\text{U}}) = I_{p_m, p_k}(\Omega_E) = 0$. This follows trivially, as a HE scheme with an MS-IND encoder achieves optimal security.

Refined from the insights in Section 5, we have:

1. For general non-uniform message and key distributions, it typically holds that $\mathbf{Adv}_{\Omega_{EE}, p_m, p_k}^{\text{mrs}} > p_{\max}$. Moreover, it generally holds that $\Omega_E^{\text{MC}} \cap \Omega_{\text{PE}}^{\text{U}} \subseteq \Omega_{\text{PE}}^{\text{U, KD}}$. This implies that uniform probabilistic encryption with optimal message confidentiality must usually be key-dependent. Similar to KC, without the constraint of uniformity, this proposition does not hold.
2. $\mathbf{Adv}_{\Omega_{\text{PE}}, p_m, p_k}^{\text{mrs}} = p_{\max}$, as our OE $\in \Omega_{\text{PE}}^{\text{U}}$ achieves the optimal security.